BUSINESS INTELLIGENCE

Assoc. Prof. Nguyễn Trí Thành, PhD

UNIVERSITY OF ENGINEERING AND TECHNOLOGY

FACULTY OF INFORMATION TECHNOLOGY

DEPARTMENT OF INFORMATION SYSTEMS

ntthanh@vnu.edu.vn

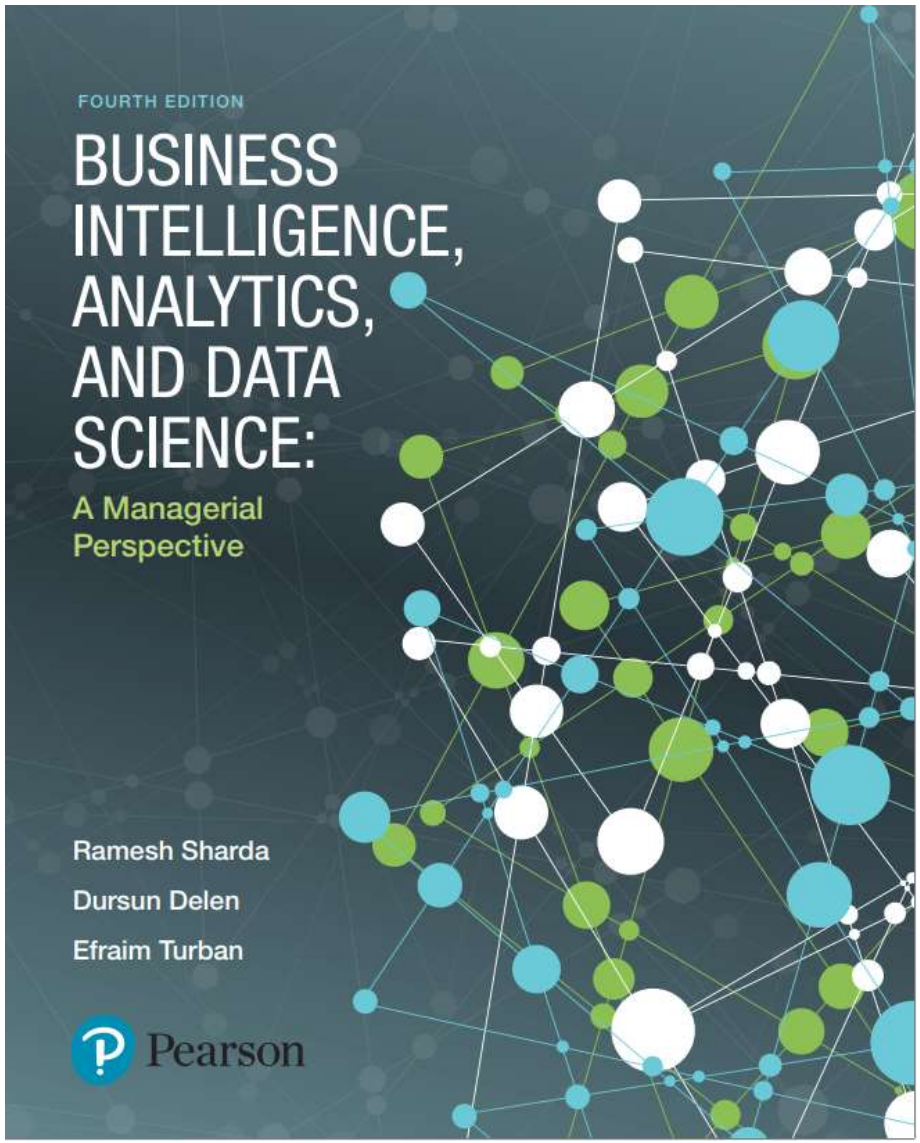
Business Intelligence, Analytics, and Data Science

A managerial perspective

Outline

- Data monetization
- Descriptive analytics I
- Descriptive analytics II

References



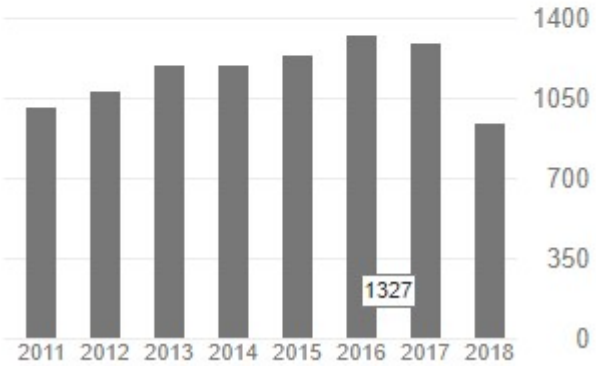


Ramesh Sharda

Oklahoma State University
Verified email at okstate.edu - Homepage
Data Science Decision Support Systems IT analytics data mining

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4th edition, 2017

Book introduction

- Provide real examples of companies with applied DS for success
- Systematic introduction about data science
- Can be useful for beginners especially managers
- Can be used as the dictionary for references for: tools, technologies, problems

DATA MONETIZATION

Data - the asset

What is the background / motivation of data science/ business analytics?

Data - the asset

What is the background / motivation of data science/ business analytics?

- Data is the asset of the company
- Data is the oil of the 21st Century – Analytics is the engine
  - *"Data … the next natural resource"* (Virginia Rometty, CEO, IBM, 2013)
- Monetize information, monetize data, data monetization
- Economics of information - Infornomics

Data – the resource

Oil – the commodity that got us here

Current generations might be forgiven for being blasé about the mega role of oil in our daily lives. For most of us, it has always been a given, therefore we seldom take a moment, if ever, to try and imagine a life without it.

From bitumen for waterproofing boats thousands of years ago, to a preservative for corpses before burial, to fuel for lighting homes and streets, to powering transport and even warships and aviation, oil has been and continues to be the major source of energy globally.



Data – the next natural resource

Much like the vast, untapped lakes of oil of earlier millennia, we are now sitting on massive seas of facts, figures and detail. In addition, our technology continues to stream ever-increasing rivers of new data into this pool every second of every day.

Yet, unlike fossil fuels, data isn't depleted when you use it and has a miniscule environmental footprint.

In fact, knowledge derived from data has enabled us to develop clean, renewable energy sources, which are inexhaustible and significantly healthier for us all.

Data analytics

Harness the power of data analytics

In the same way as engines are required to convert oil into energy, so analytics are required to extract the value form data.

Only then can insights follow, which, when applied, can be used to produce business benefits such as:

- **Eliminating the cost of waste or downtime** e.g. identifying when resources/teams/individuals are under-utilised. Why is one more productive than another? Where is this occurring?
- **Identifying new revenue opportunities** by examining sales margins on products/services or by salesperson. Why do we have more market share in one area than another? What are the demographics and motivations of customers? Which customers are likely to buy again if
  we offer something to them?
- **Reducing risk** by identifying types of repeated safety incidents, areas of non-compliance with processes, anomalies, and unusual activity.

ALL enterprises are

- **Collecting data** in marketing, finance, sales, entertainment, human resources, payroll, risk, operations etc. Some of this even happens automatically e.g. Google Analytics on a company's web site, items in your Inbox, or click-throughs on a newsletter.
- **Storing data**. Much of a company's data is held in database systems, ERP systems, spreadsheets, disparate cloud systems, text files etc.

Every enterprise collects and stores data.



Data analytics

SMART enterprises also:

- **Connect their data together** – so that hits on a website can be used to predict sales of a product, or demographics and customer sales data can show market share and sales opportunities.
- **Analyse their data** – to gain insights into what's actually happening in their businesses.
- **Take action** using the insights and knowledge gained from up-to-the-moment data to make better decisions about optimising activities, and to make more accurate predictions about future performance.

This provides them with a distinct edge over companies that aren't doing this.

Seize the future

Oil was the economic driver of the last century. It created massive opportunities and wealth for economies worldwide.

Data and analytics have the same potential.

They are providing smart companies with incredibly valuable sources of competitive advantage, and creating new opportunities for smart businesses to create wealth and benefit in the 21st century.

Don't get left behind.

Start analysing and using your data today!

Another introduction

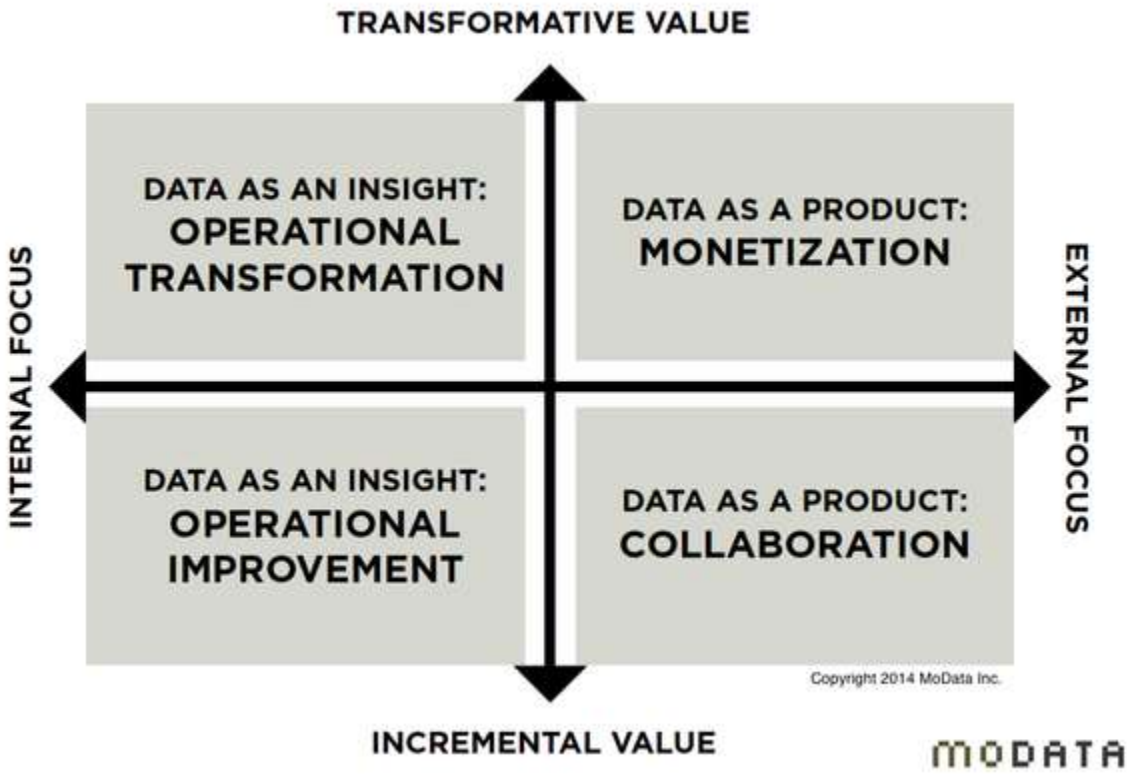
Khoa học dữ liệu, tiền hóa dữ liệu và khả năng thực thi tại Việt Nam

Hà Quang Thụy và Nguyễn Trí Thành,
Phòng Thí nghiệm Khoa học dữ liệu và Công nghệ Tri thức,
Trường Đại học Công nghệ, Đại học Quốc gia Hà Nội

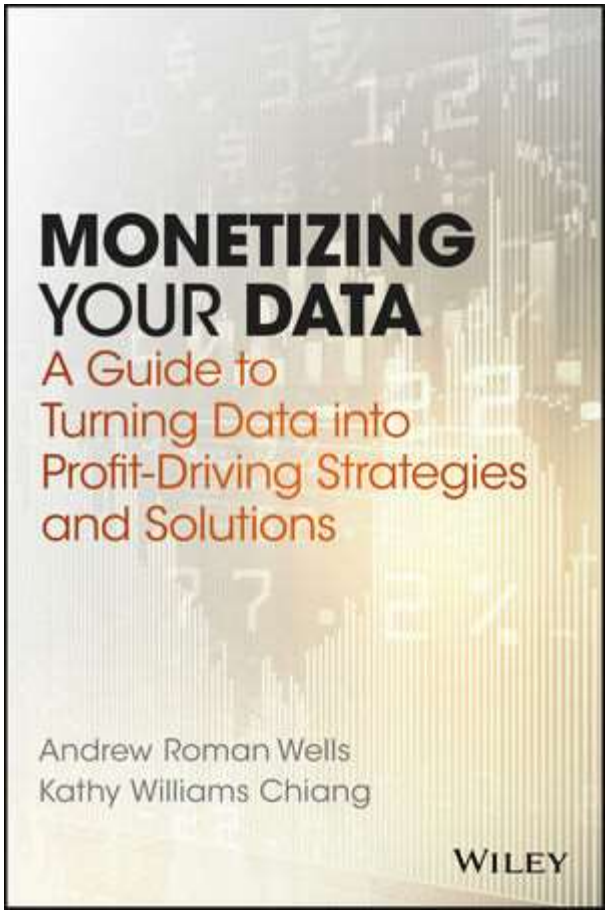
Tóm tắt: Tại các nền kinh tế phát triển, dữ liệu đã trở thành một tài sản then chốt được ví như "dầu mỏ trong Thế kỷ 21" và khoa học dữ liệu là một phương tiện chuyển hóa nguồn "dầu mỏ" dữ liệu thành tiền. Tuy nhiên, khoa học dữ liệu và chuyển hóa dữ liệu thành tiền gặp một số thách thức không nhỏ. Bài viết này cung cấp một số nội dung cơ bản về tài sản dữ liệu, tiền hóa dữ liệu, khoa học dữ liệu và một số liên hệ với Việt Nam. Một chính sách khoa học dữ liệu phù hợp có khả năng đưa doanh nghiệp Việt Nam tham gia xứng đáng vào chuỗi giá trị toàn cầu.

Từ khóa: khai phá dữ liệu, khoa học dữ liệu, kinh tế học thông tin, tài sản dữ liệu, tiền hóa dữ liệu

The value from data



Books


MONETIZING
YOUR DATA
A Guide to
Turning Data into
Profit-Driving Strategies
and Solutions
Andrew Roman Wells
Kathy Williams Chiang
WILEY


How will Big Data companies monetize data in 2018
Kdnuggets.com | Author: Yash Mehta


Data Monetization
Strategies
Information
Builders

Textbook structure

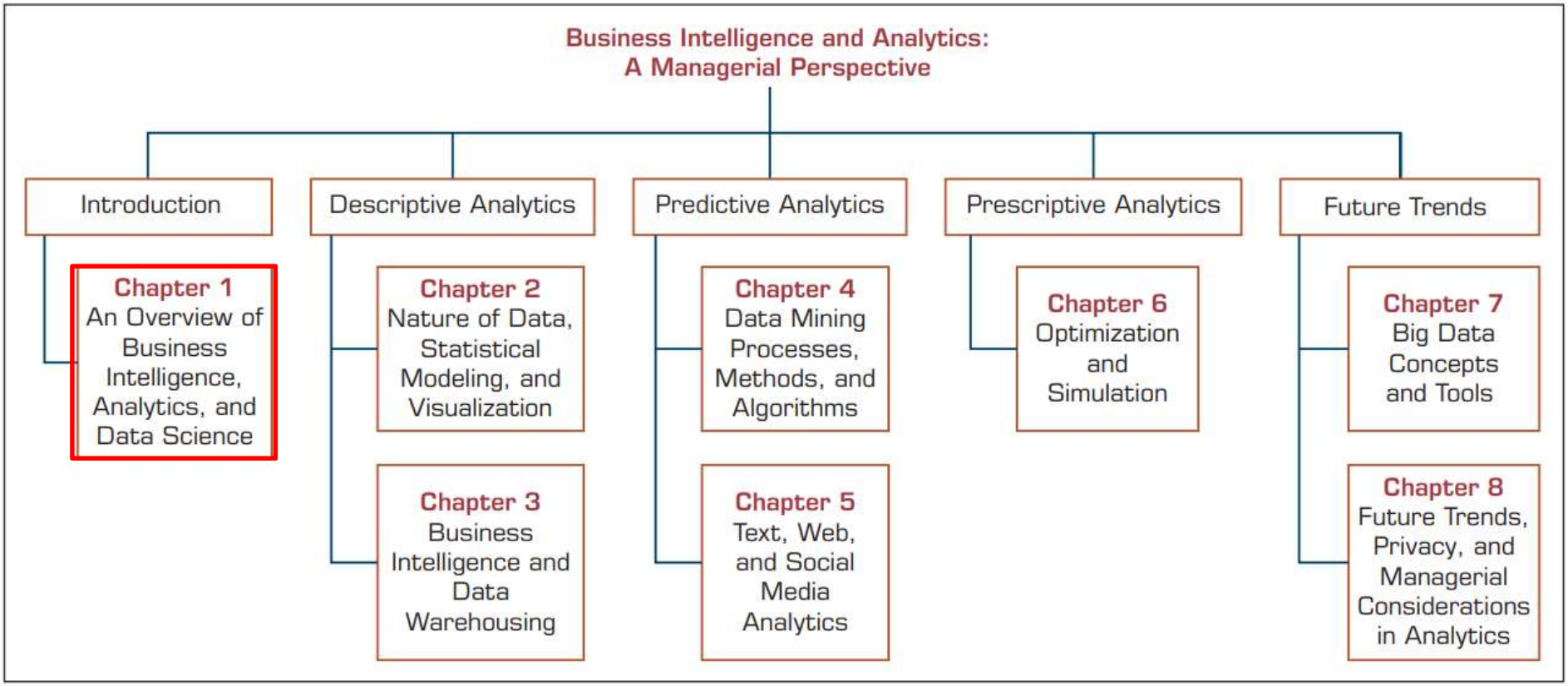


FIGURE 1.15 Plan of the Book.

Business Intelligence, Analytics, and Data Science

A managerial perspective

OVERVIEW

Analytics tasks

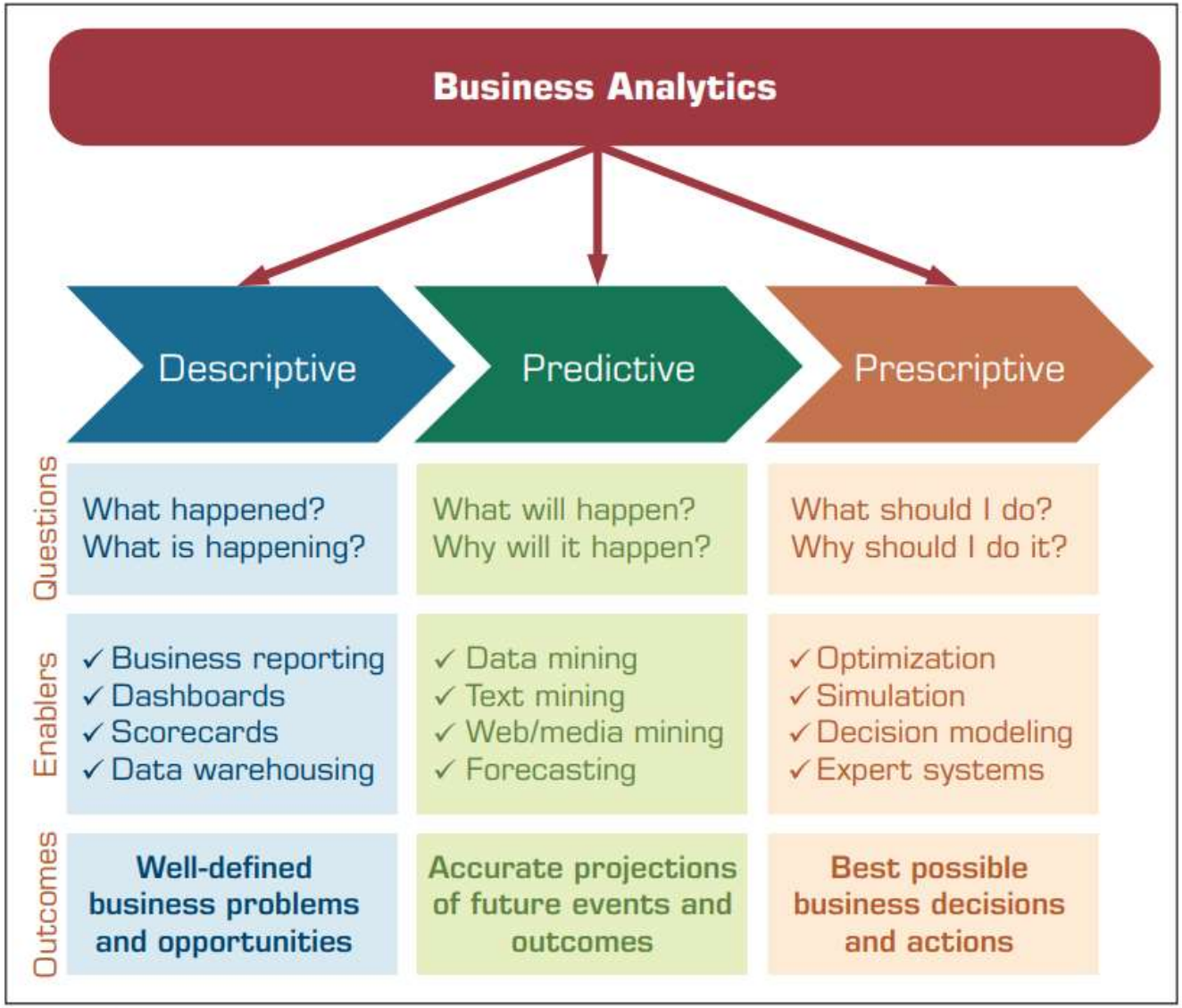


FIGURE 1.11 Three Types of Analytics.

Concepts

Analytics or Data Science?

Even as the concept of analytics is receiving more attention in industry and academic circles, another term has already been introduced and is becoming popular. The new term is *data science*. Thus, the practitioners of data science are data scientists. D. J. Patil of LinkedIn is sometimes credited with creating the term *data science*. There have been some attempts to describe the differences between data analysts and data scientists (e.g., see emc.com/collateral/about/news/emc-data-science-study-wp.pdf). One view is that *data analyst* is just another term for professionals who were doing BI in the form of data compilation, cleaning, reporting, and perhaps some visualization. Their skill sets included Excel, some SQL knowledge, and reporting. You would recognize those capabilities as descriptive or reporting analytics. In contrast, a data scientist is responsible for predictive analysis, statistical analysis, and more advanced analytical tools and algorithms. They may have a deeper knowledge of algorithms and may recognize them under various labels—data mining, knowledge discovery, or machine learning. Some of these professionals may also need deeper programming knowledge to be able to write code for data cleaning/analysis in current Web-oriented languages such as Java or Python and statistical languages such as R. Many analytics professionals also need to build significant expertise in statistical modeling, experimentation, and analysis. Again, our readers should recognize that these fall under the predictive and prescriptive analytics umbrella. However, prescriptive analytics also includes more significant expertise in OR including optimization, simulation, decision analysis, and so on. Those who cover these fields are more likely to be called data scientists than analytics professionals.

Concepts

Our view is that the distinction between analytics and data scientist is more of a degree of technical knowledge and skill sets than functions. It may also be more of a distinction across disciplines. Computer science, statistics, and applied mathematics programs appear to prefer the data science label, reserving the analytics label for more business-oriented professionals. As another example of this, applied physics professionals have proposed using *network science* as the term for describing analytics that relate to groups of people—social networks, supply chain networks, and so forth. See http://barabasi.com/networksciencebook/ for an evolving textbook on this topic.

Aside from a clear difference in the skill s ssionals who only have to do descriptive/reporting analytics versus those who engage in all three types of analytics, the distinction is fuzzy between the two labels, at best. We observe that graduates of our analytics programs tend to be responsible for tasks which are more in line with data science professionals (as defined by some circles) than just reporting analytics. This book is clearly

BI architecture

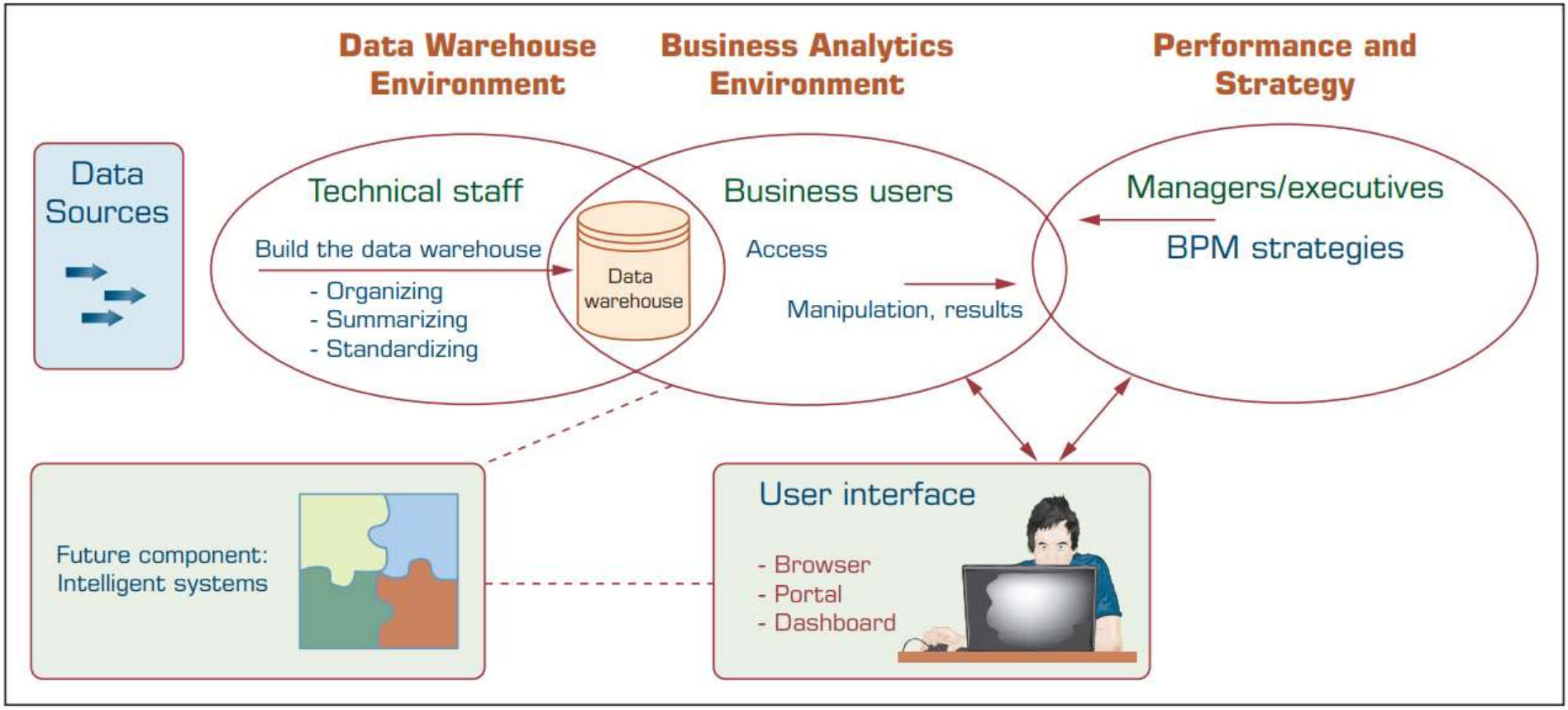


FIGURE 1.10 A High-Level Architecture of BI. (*Source:* Based on W. Eckerson, *Smart Companies in the 21st Century: The Secrets of Creating Successful Business Intelligent Solutions.* The Data Warehousing Institute, Seattle, WA, 2003, p. 32, Illustration 5.)

Business Intelligence, Analytics, and Data Science

A managerial perspective

DESCRIPTIVE ANALYTICS II-BI AND DATA WAREHOUSING

Content

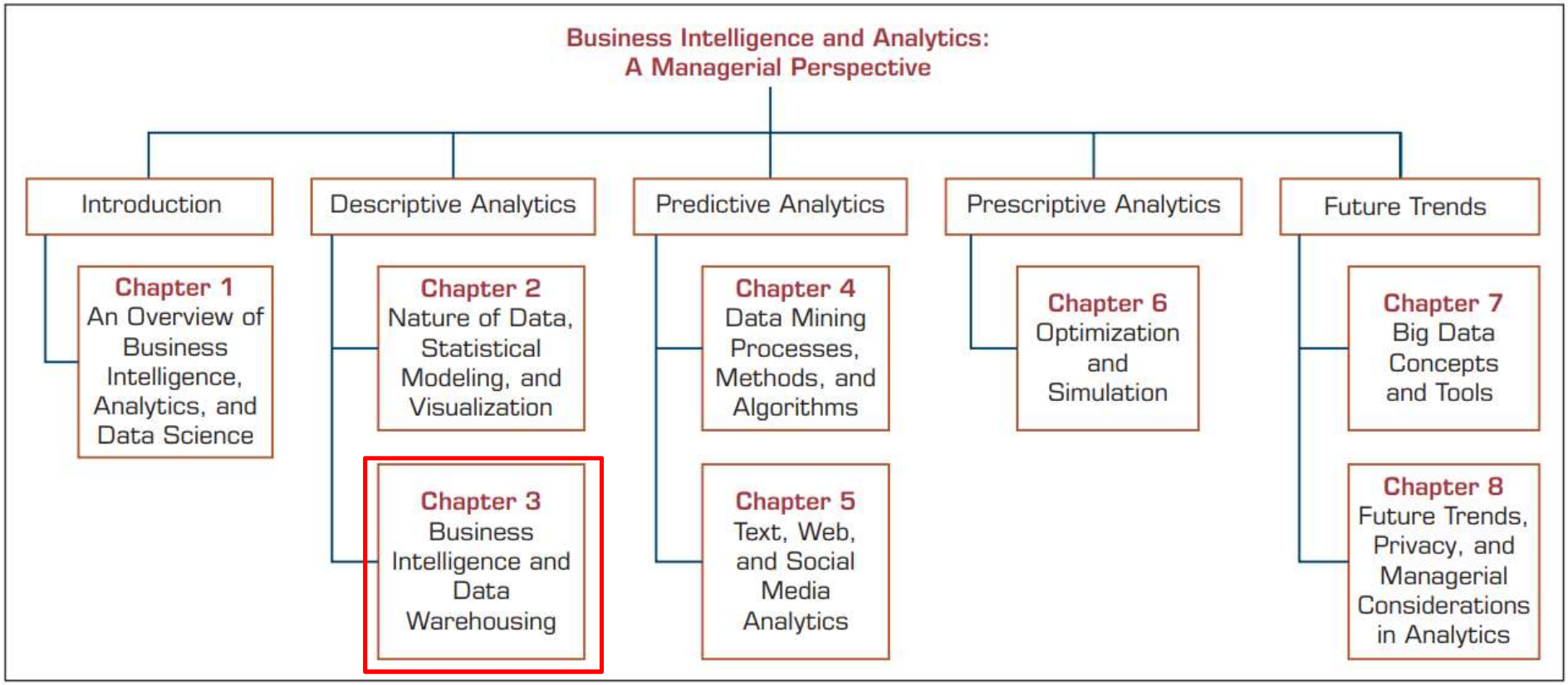


FIGURE 1.15 Plan of the Book.

Data sources for analytics

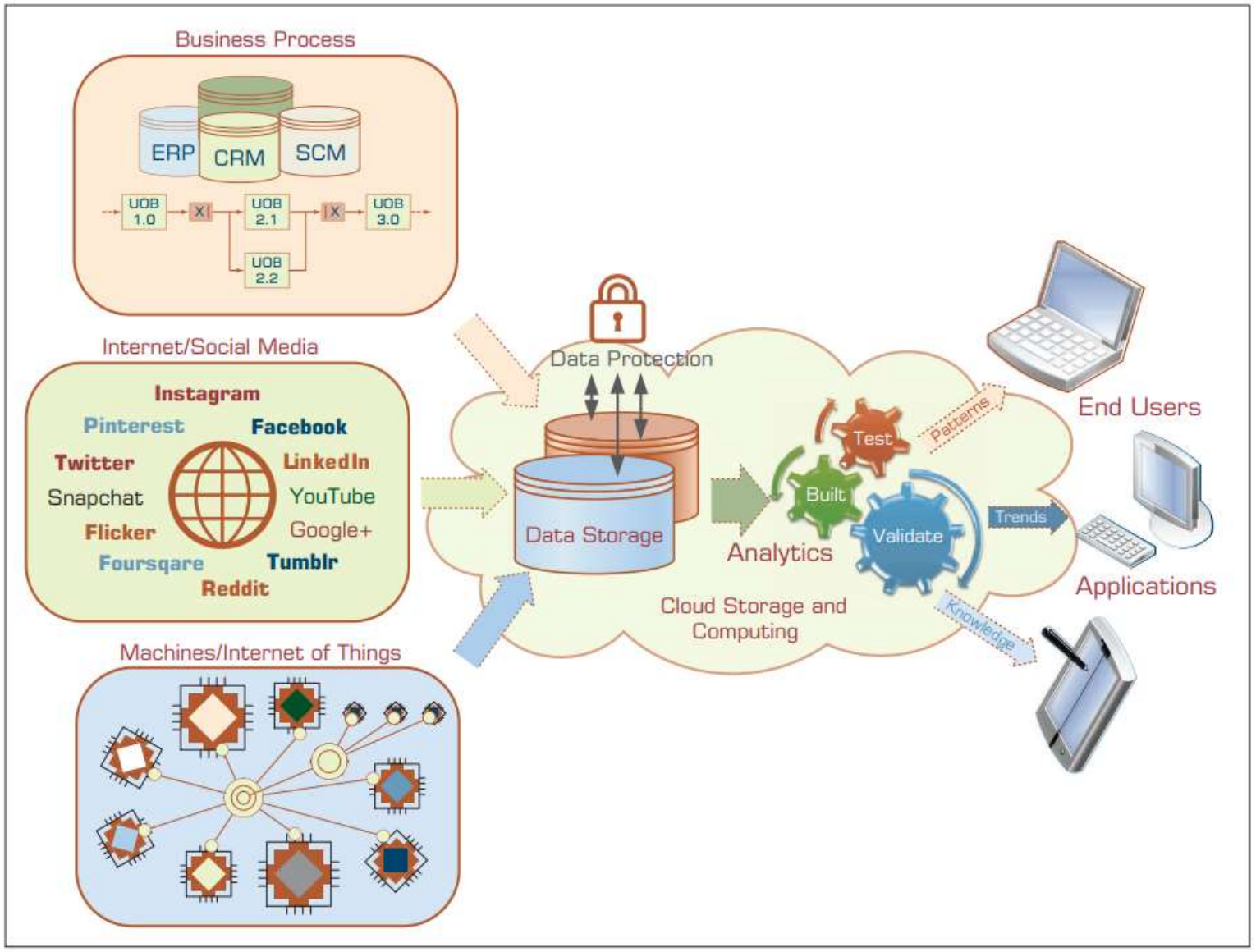


FIGURE 2.1 A Data to Knowledge Continuum.

Data processing

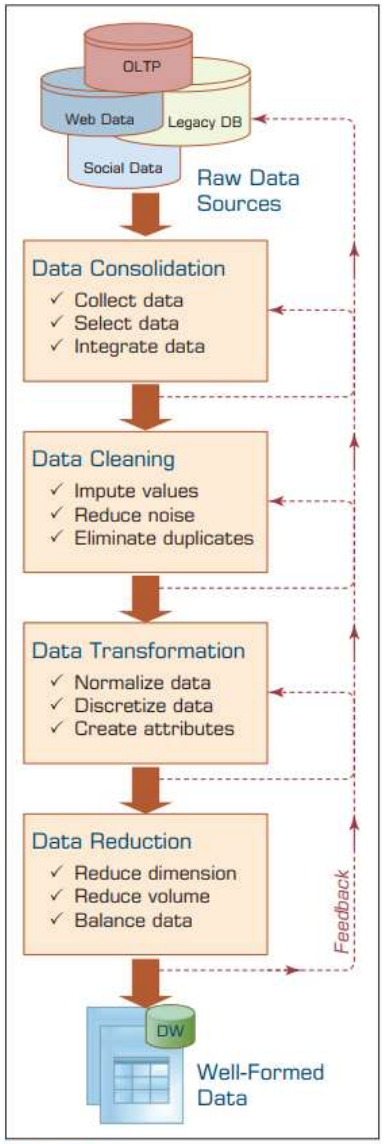
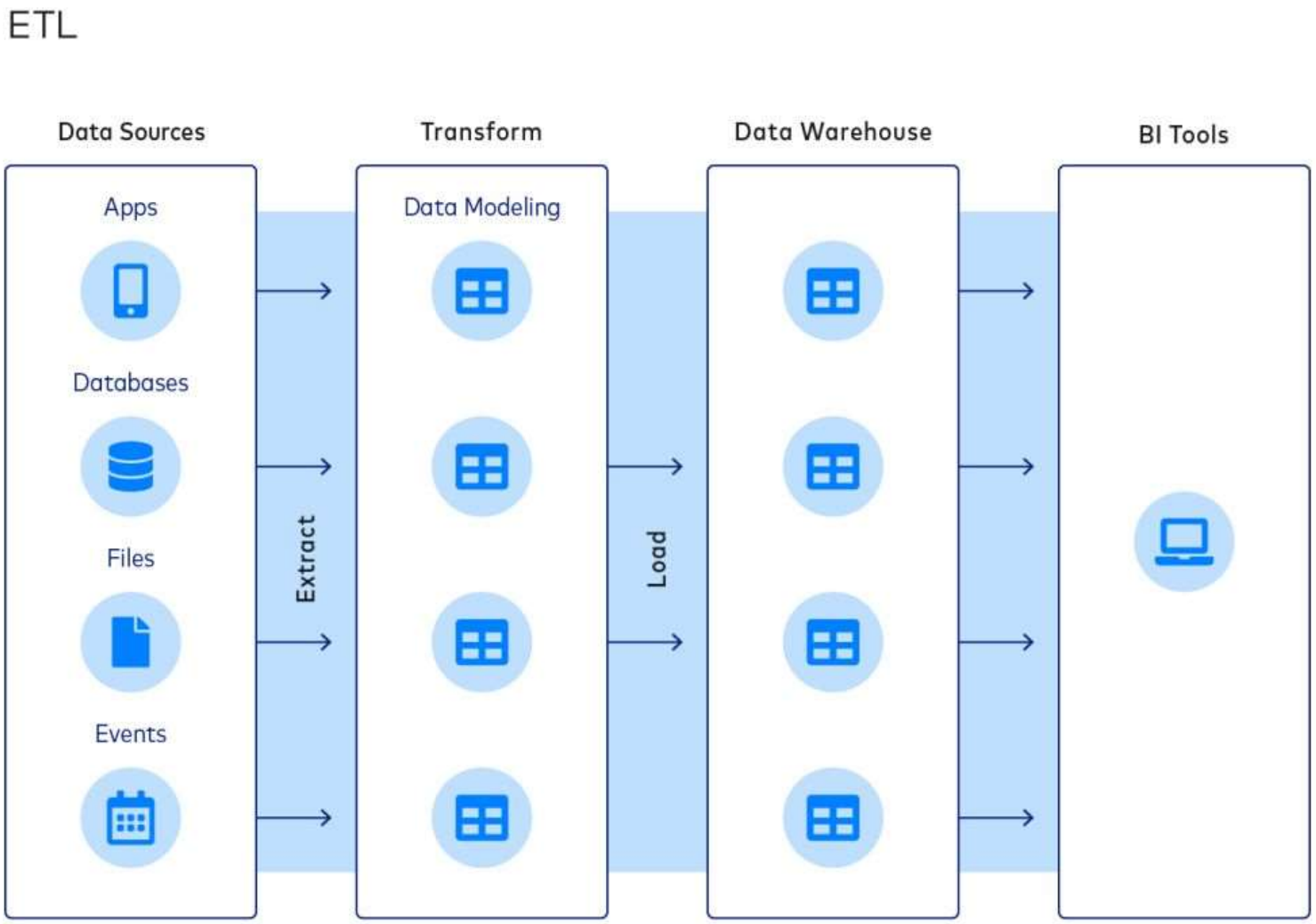


FIGURE 2.3 Data Preprocessing Steps.

ETL


ETL
Data Sources
Transform
Data Warehouse
BI Tools
Apps
Databases
Files
Events
Extract
Data Modeling
Load

ETL tools

ORACLE DATA INTEGRATOR

Microsoft SQL Server Integration Services

TALEND FOR DATA INTEGRATION & BIG DATA

pentaho

AWS Glue

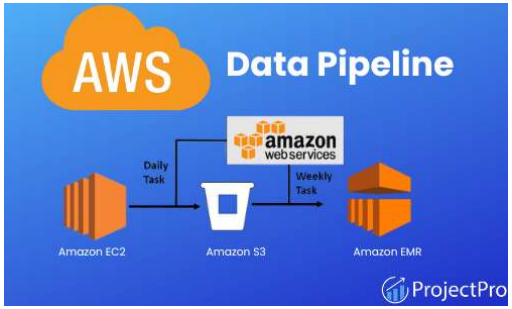
Informatica PowerCenter - ETL Tools



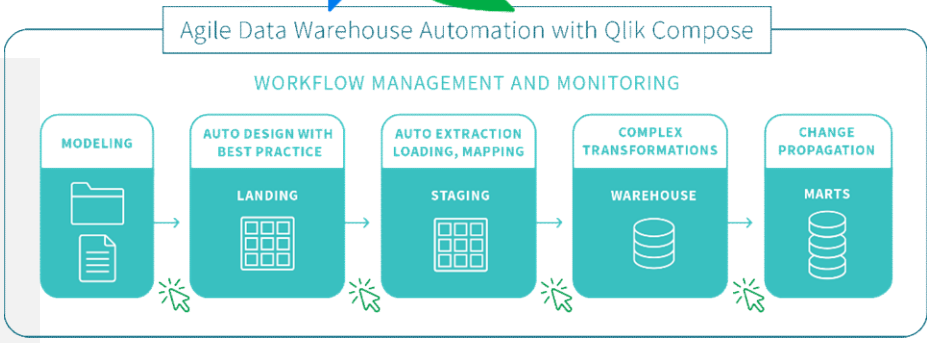
APACHE Spark Structured Streaming

IBM InfoSphere DataStage

Apache Airflow



Azure Data Factory



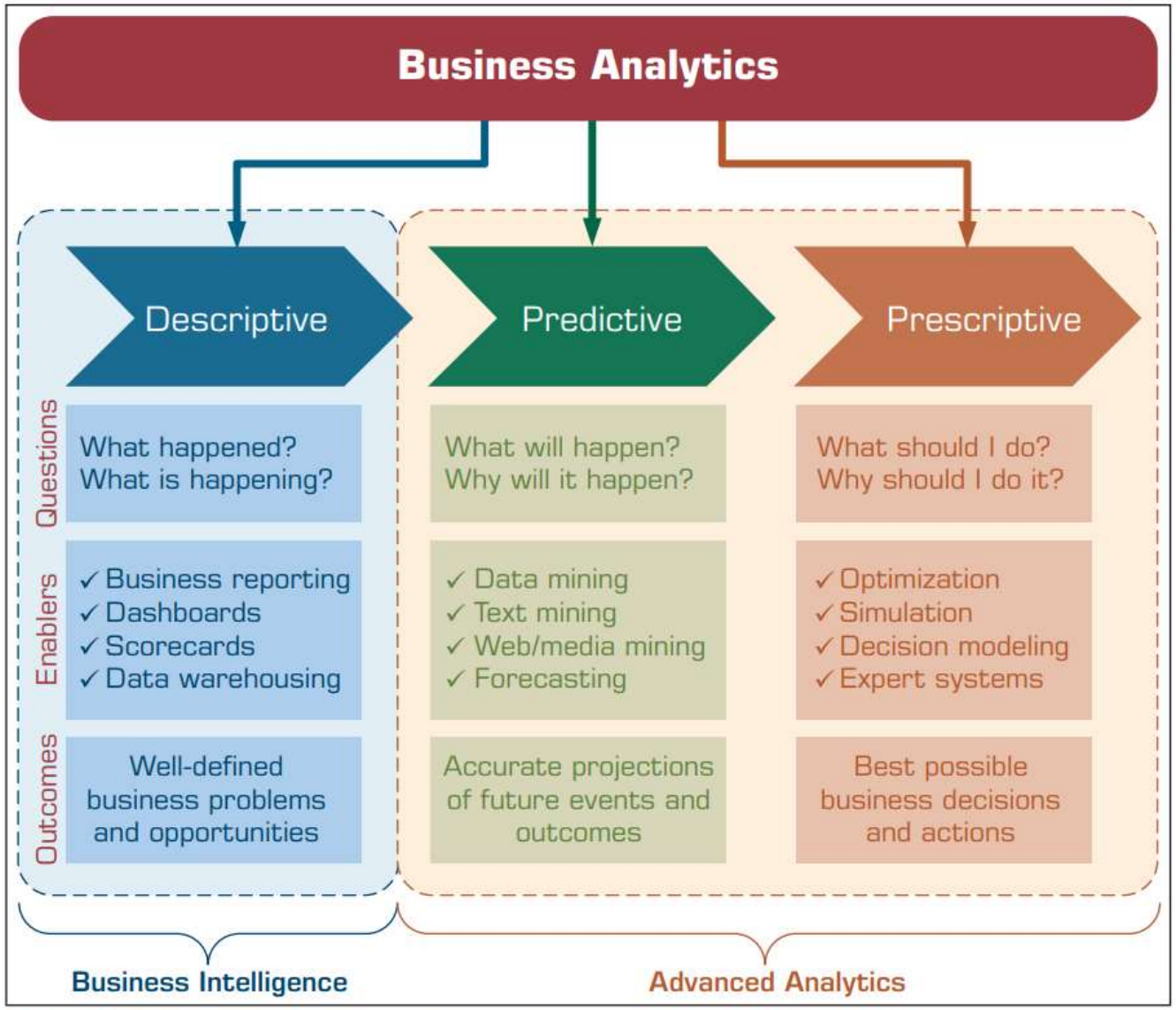


FIGURE 3.1 Relationship between Business Analytics and BI, and BI and Data Warehousing.

Data warehousing

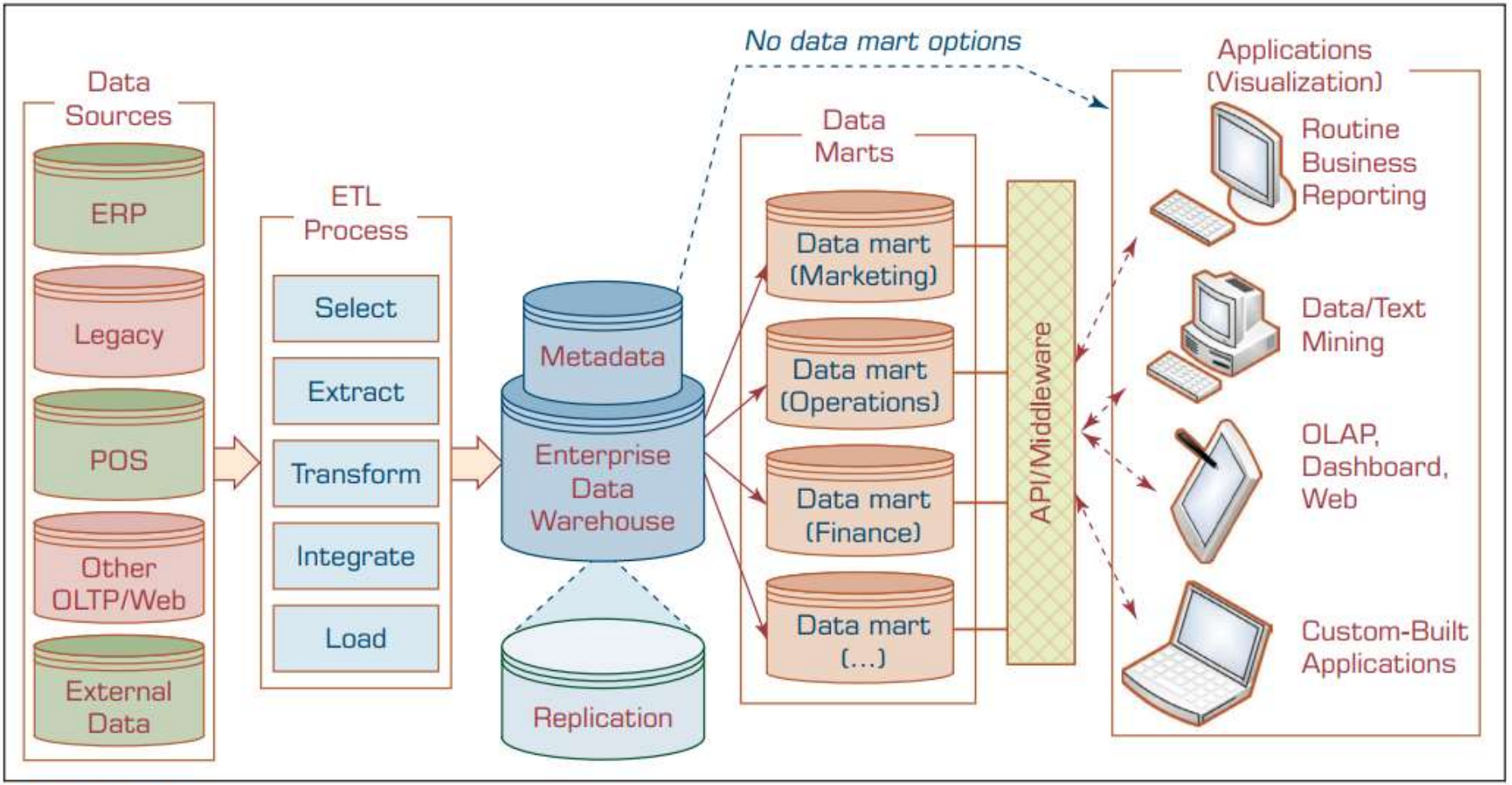


FIGURE 3.4 A Data Warehouse Framework and Views.

Data warehouse development

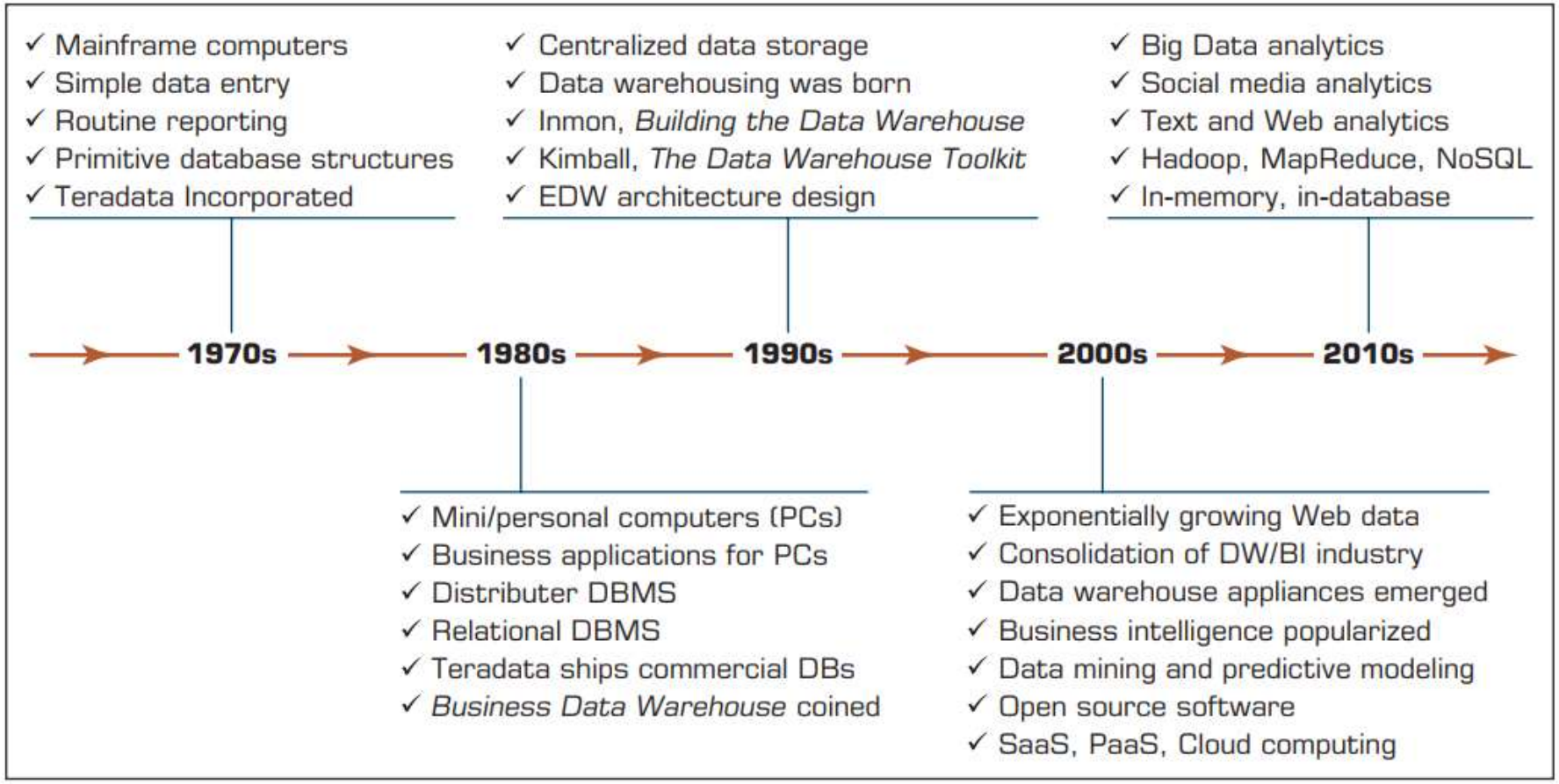


FIGURE 3.2 A List of Events That Led to Data Warehousing Development.

What is Data Warehouse?

- Defined in many different ways, but not rigorously.
  - A decision support database that is maintained separately from the organization's operational database
  - Support information processing by providing a solid platform of consolidated, historical data for analysis.
- "A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process."—W. H. Inmon
- Data warehousing:
  - The process of constructing and using data warehouses

Data Warehouse—Subject-Oriented

- Organized around major subjects, such as customer, product, sales
- Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing
- Provide a simple and concise view around particular subject issues by excluding data that are not useful in the decision support process

Data Warehouse—Integrated

- Constructed by integrating multiple, heterogeneous data sources
  - relational databases, flat files, on-line transaction records
- Data cleaning and data integration techniques are applied.
  - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
    - E.g., Hotel price: currency, tax, breakfast covered, etc.
  - When data is moved to the warehouse, it is converted.

Data Warehouse—Time Variant

- The time horizon for the data warehouse is significantly longer than that of operational systems
  - Operational database: current value data
  - Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the data warehouse
  - Contains an element of time, explicitly or implicitly
  - But the key of operational data may or may not contain "time element"

Data Warehouse—Nonvolatile

- A physically separate store of data transformed from the operational environment
- Operational update of data does not occur in the data warehouse environment
  - Does not require transaction processing, recovery, and concurrency control mechanisms
  - Requires only two operations in data accessing:
    - *initial loading of data* and *access of data*

Data Warehouse vs. Heterogeneous DBMS

- Traditional heterogeneous DB integration: A query driven approach
  - Build wrappers/mediators on top of heterogeneous databases
  - When a query is posed to a client site, a meta-dictionary is used to translate the query into queries appropriate for individual heterogeneous sites involved, and the results are integrated into a global answer set
  - Complex information filtering, compete for resources
- Data warehouse: update-driven, high performance
  - Information from heterogeneous sources is integrated in advance and stored in warehouses for direct query and analysis

Data Warehouse vs. Operational DBMS

- OLTP (on-line transaction processing)
  - Major task of traditional relational DBMS
  - Day-to-day operations: purchasing, inventory, banking, manufacturing, payroll, registration, accounting, etc.
- OLAP (on-line analytical processing)
  - Major task of data warehouse system
  - Data analysis and decision making
- Distinct features (OLTP vs. OLAP):
  - User and system orientation: customer vs. market
  - Data contents: current, detailed vs. historical, consolidated
  - Database design: ER + application vs. star + subject
  - View: current, local vs. evolutionary, integrated
  - Access patterns: update vs. read-only but complex queries

OLTP vs. OLAP

	OLTP	**OLAP**
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date detailed, flat relational isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions
#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response

Characteristics of Data Warehousing

Characteristics of Data Warehousing

A common way to introduce data warehousing is to refer to its fundamental characteristics (see Inmon, 2005):

- ***Subject oriented.*** Data are organized by detailed subject, such as sales, products, or customers, containing only information relevant for decision support. Subject orientation enables users to determine not only how their business is performing, but also why. A data warehouse differs from an operational database in that most operational databases have a product orientation and are tuned to handle transactions that update the database. Subject orientation provides a more comprehensive view of the organization.

Characteristics of Data Warehousing

- ***Integrated.*** Integration is closely related to subject orientation. Data warehouses must place data from different sources into a consistent format. To do so, they must deal with naming conflicts and discrepancies among units of measure. A data warehouse is presumed to be totally integrated.
- ***Time variant (time series).*** A warehouse maintains historical data. The data do not necessarily provide current status (except in real-time systems). They detect trends, deviations, and long-term relationships for forecasting and comparisons, leading to decision making. Every data warehouse has a temporal quality. Time is the one important dimension that all data warehouses must support. Data for analysis from multiple sources contain multiple time points (e.g., daily, weekly, monthly views).
- ***Nonvolatile.*** After data are entered into a data warehouse, users cannot change or update the data. Obsolete data are discarded, and changes are recorded as new data.

These characteristics enable data warehouses to be tuned almost exclusively for data access. Some additional characteristics may include the following:

DWH providers

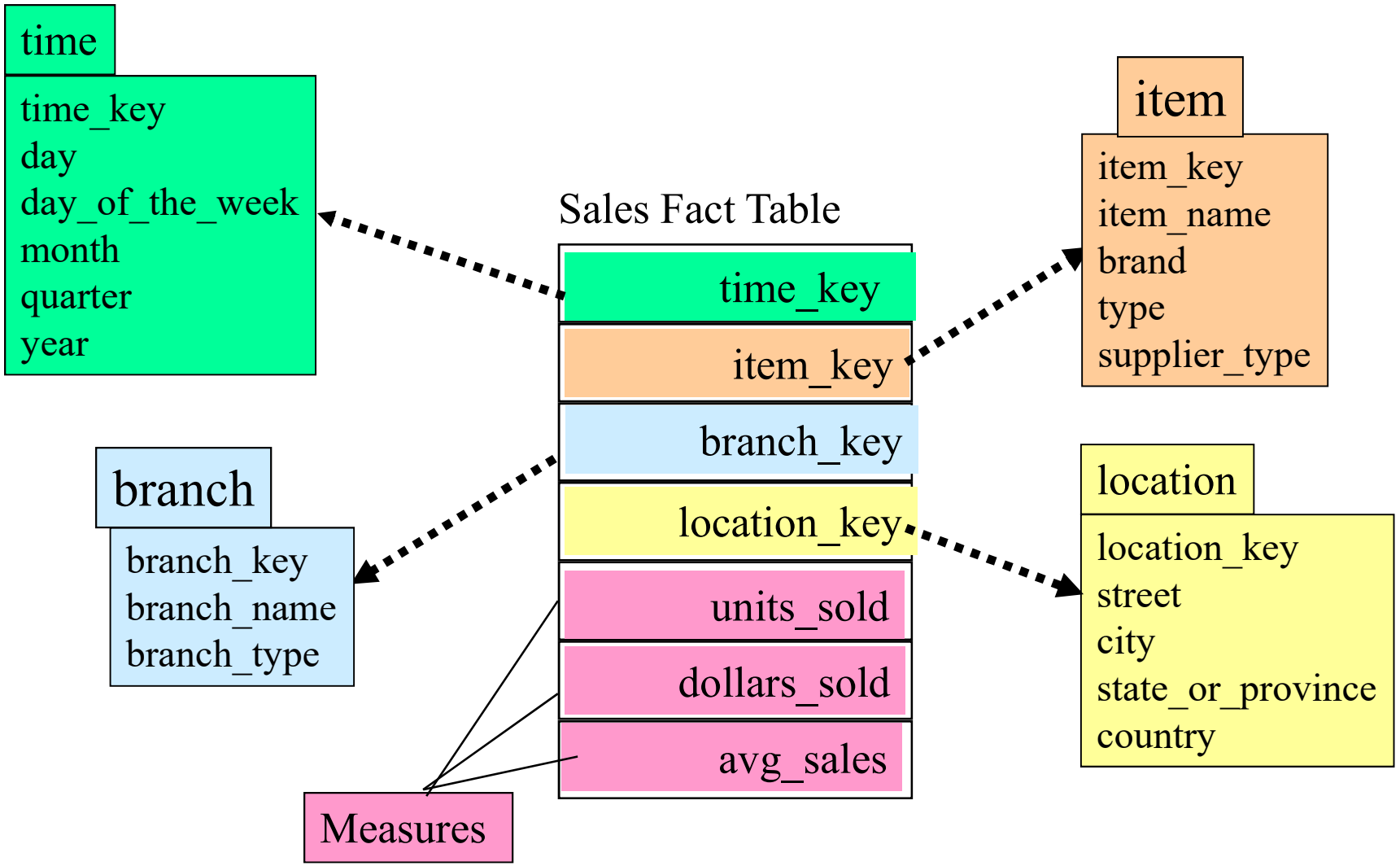
TABLE 3.2 Sample List of Data Warehousing Vendors

Vendor	Product Offerings
Business Objects (businessobjects.com)	A comprehensive set of BI and data visualization software (now owned by SAP)
Computer Associates (cai.com)	Comprehensive set of data warehouse (DW) tools and products
DataMirror (datamirror.com)	DW administration, management, and performance products
Data Advantage Group (dataadvantagegroup.com)	Metadata software
Dell (dell.com)	DW servers
Embarcadero Technologies (embarcadero.com)	DW administration, management, and performance products
Greenplum (greenplum.com)	Data warehousing and data appliance solution provider (now owned by EMC)
Harte-Hanks (harte-hanks.com)	Customer relationship management (CRM) products and services
HP (hp.com)	DW servers
Hummingbird Ltd. (hummingbird.com)	DW engines and exploration warehouses
Hyperion Solutions (hyperion.com)	Comprehensive set of DW tools, products, and applications
IBM InfoSphere (www-01.ibm.com/software/data/infosphere)	Data integration, DW, master data management, Big Data products
Informatica (informatica.com)	DW administration, management, and performance products
Microsoft (microsoft.com)	DW tools and products
Netezza	DW software and hardware (DW appliance) provider (now owned by IBM)
Oracle (including PeopleSoft and Siebel; oracle.com)	DW, ERP, and CRM tools, products, and applications
SAS Institute (sas.com)	DW tools, products, and applications
Siemens (siemens.com)	DW servers
Sybase (sybase.com)	Comprehensive set of DW tools and applications
Teradata (teradata.com)	DW tools, DW appliances, DW consultancy, and applications

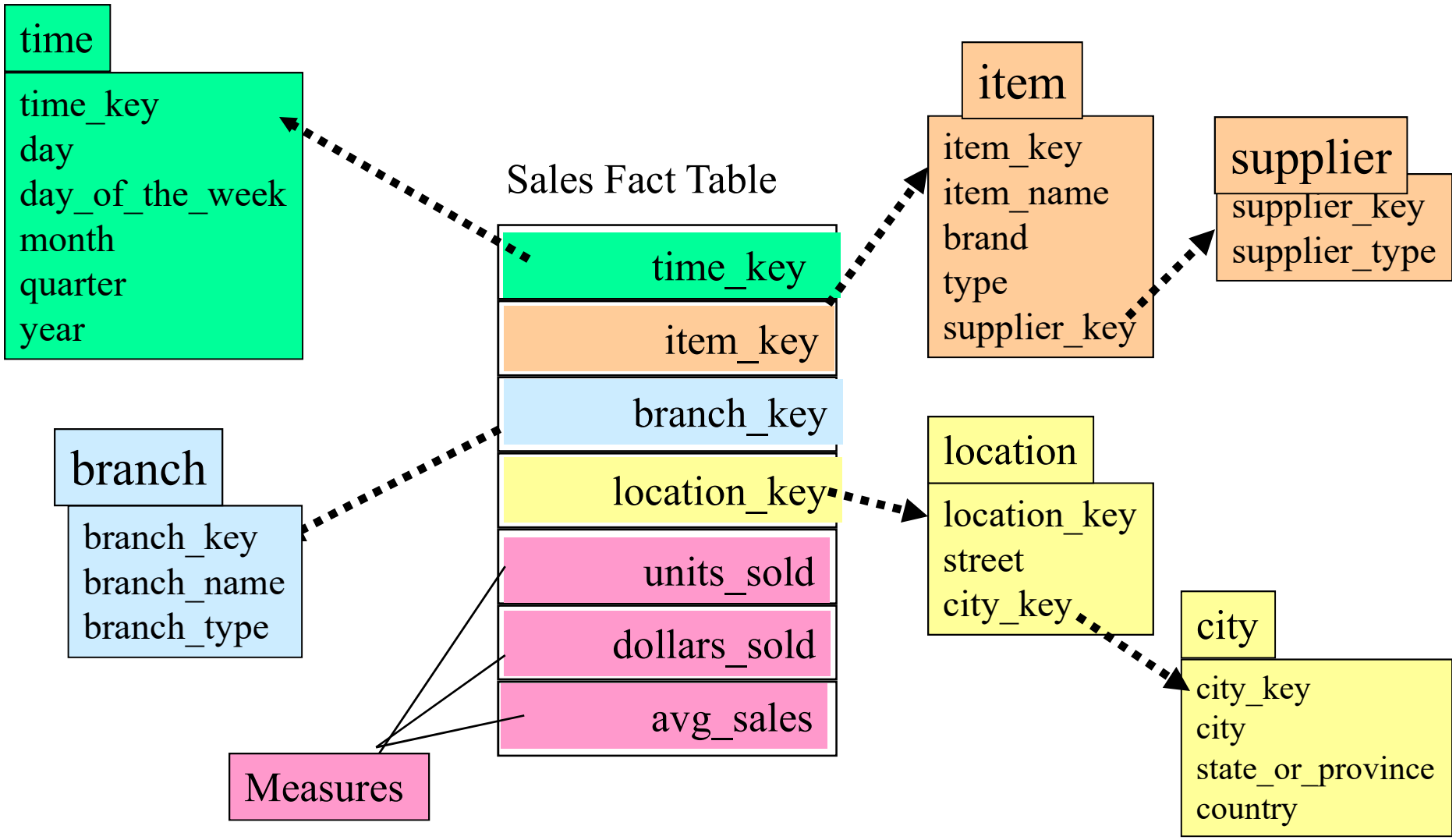
Modeling of Data Warehouses

- Modeling data warehouses: dimensions & measures
  - Star schema: A fact table in the middle connected to a set of dimension tables
  - Snowflake schema: A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake
  - Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

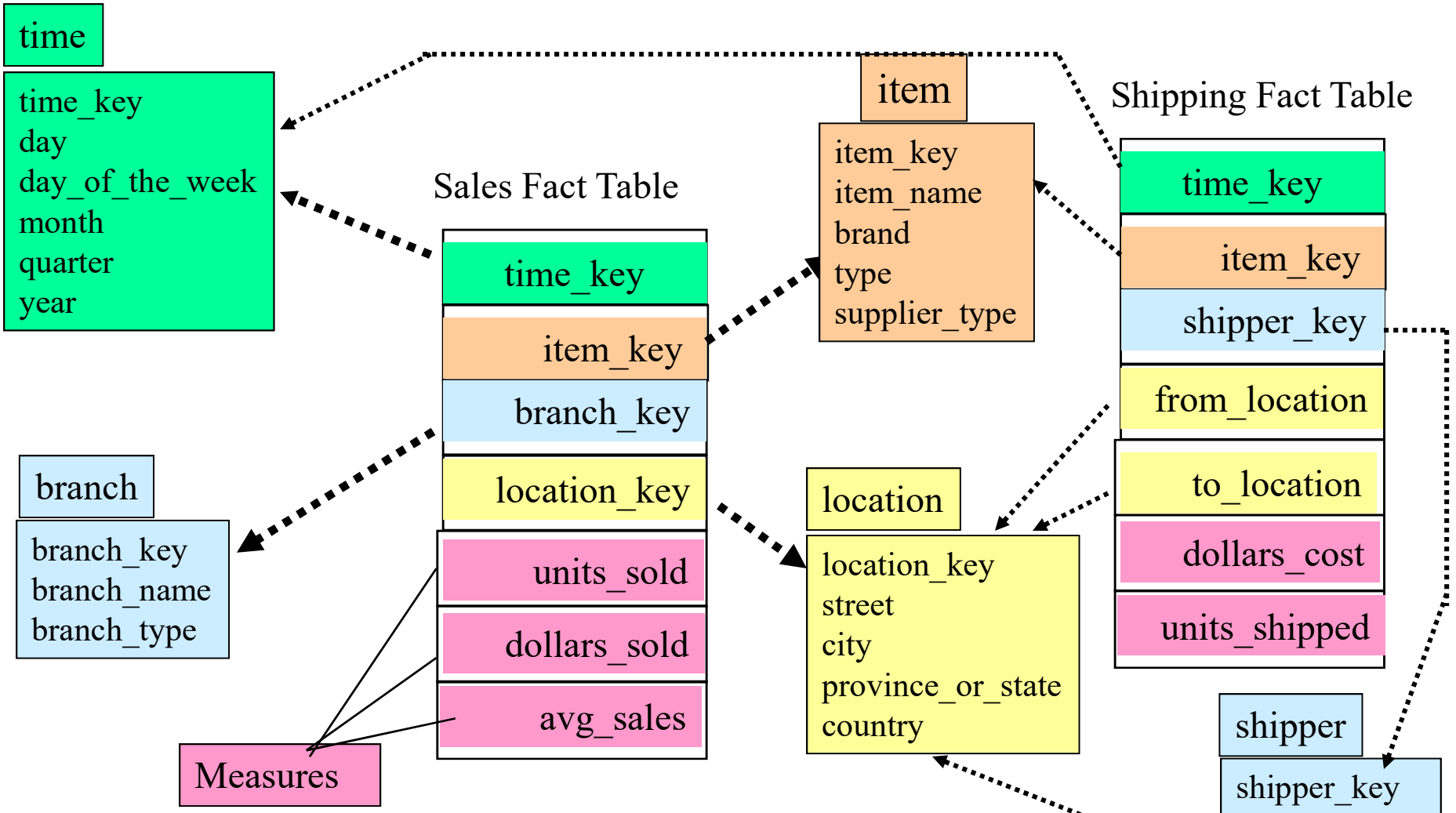
Example of Star Schema



Example of Snowflake Schema

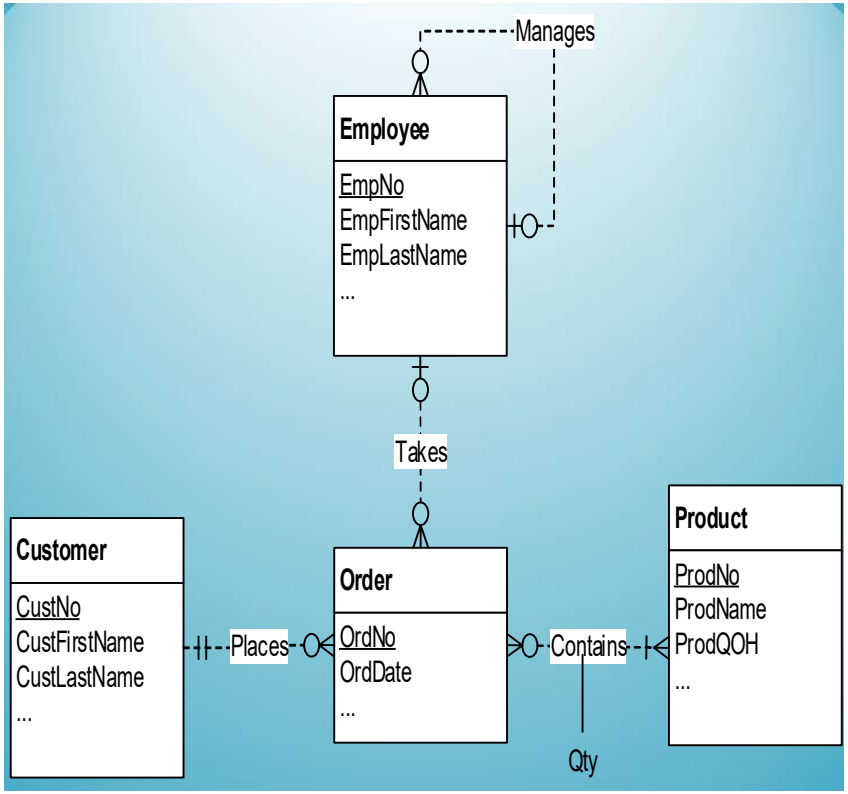


Example of Fact onstellation

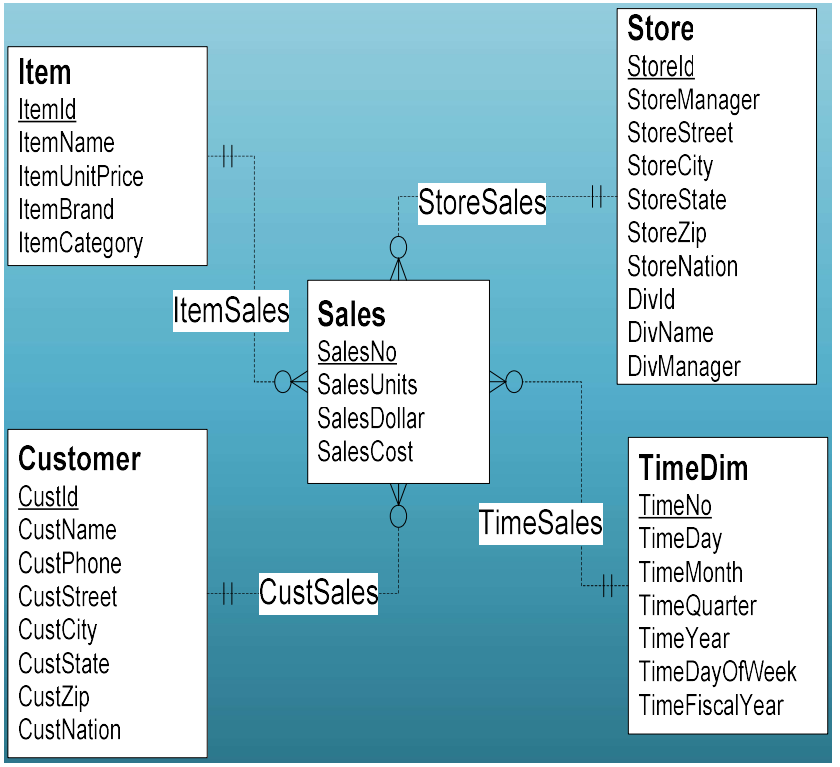


Schema Comparison

Operational database



Data warehouse



OLAP Operations

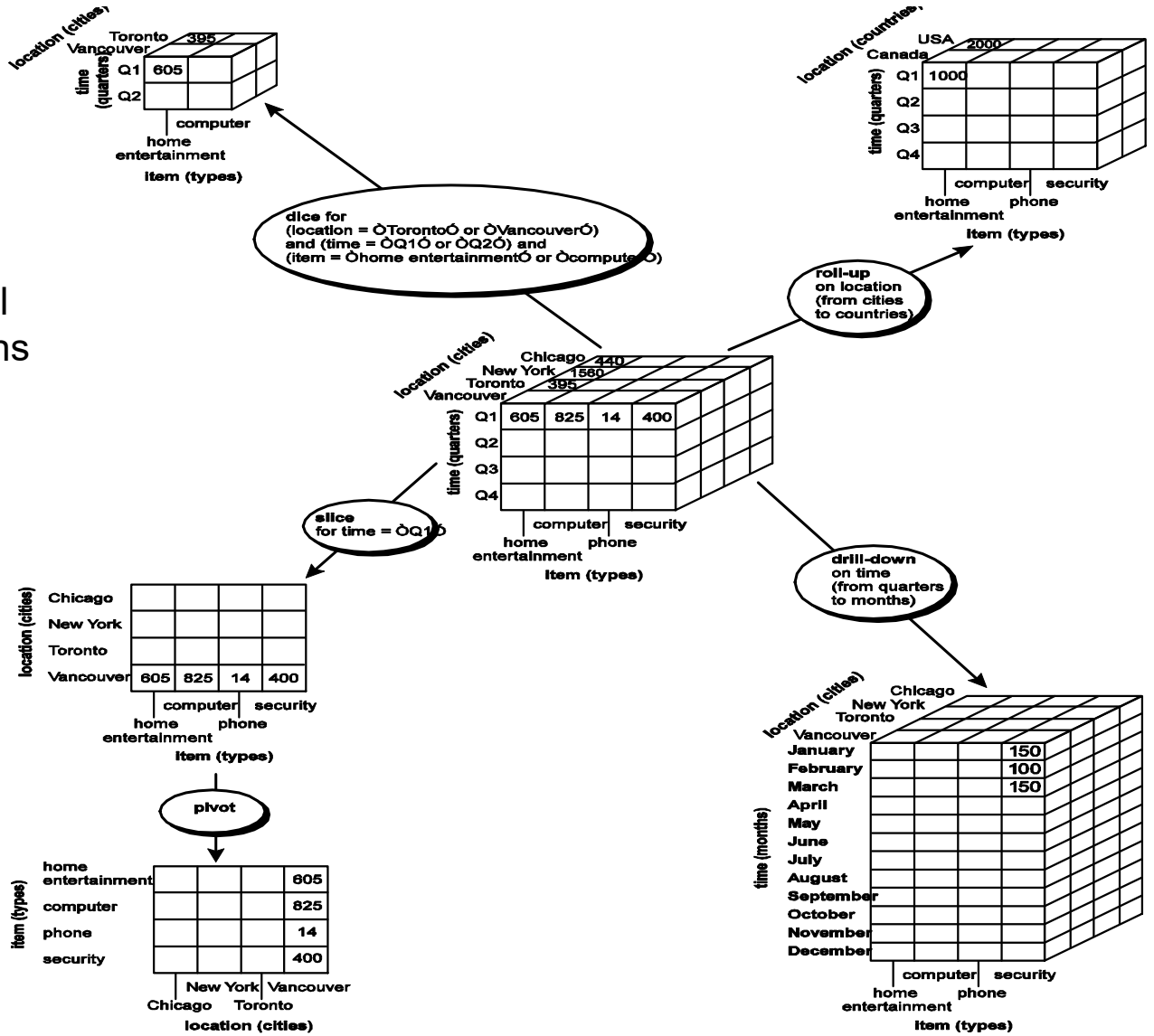


Fig. 3.10 Typical OLAP Operations

Business Intelligence, Analytics, and Data Science

A managerial perspective

DESCRIPTIVE ANALYTICS I-NATURE OF DATA, STATISTICAL MODELING, AND VISUALIZATION

Content

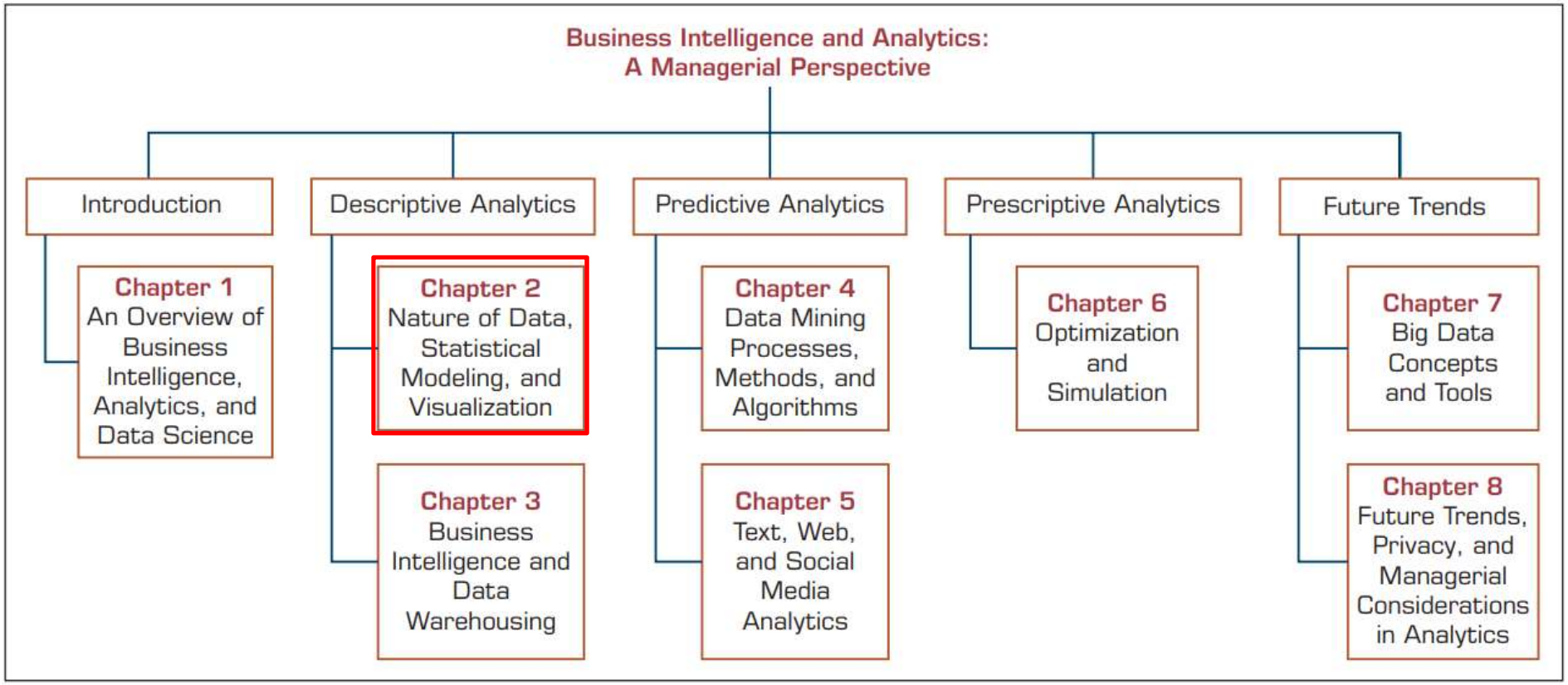


FIGURE 1.15 Plan of the Book.

Data requirements

- **Data source reliability** refers to the originality and appropriateness of the storage medium where the data is obtained—answering the question of "Do we have the right confidence and belief in this data source?" If it all possible, one should always look for the original source/creator of the data to eliminate/mitigate the possibilities of data misrepresentation and data transformation caused by the mishandling of the data as it moved from the source to destination through one or more steps and stops along the way. Every move of the data creates a chance to unintentionally drop or reformat data items, which limits the integrity and perhaps true accuracy of the data set.

Data requirements

- **Data content accuracy** means that data are correct and are a good match for the analytics problem—answering the question of "Do we have the right data for the job?" The data should represent what was intended or defined by the original source of the data. For example, the customer's contact information recorded in a record within a database should be the same as what the patient said it was. Data accuracy will be covered in more detail in the following subsection.

- **Data accessibility** means that the data are easily and readily obtainable—answering the question of "Can we easily get to the data when we need to?" Access to data may be tricky, especially if the data is stored in more than one location and storage medium and need to be merged/transformed while accessing and obtaining it. As the traditional relational database management systems leave their place (or coexist with) a new generation of data storage mediums like data lakes and Hadoop infra-structure, the importance/criticality of data accessibility is also increasing.

Data requirements

- **Data security and data privacy** means that the data is secured to only allow those people who have the authority and the need to access it and to prevent anyone else from reaching it. Increasing popularity in educational degrees and certificate programs for Information Assurance is an evidence to the criticality and the increasing urgency of this data quality metric. Any organization that maintains health records for individual patients must have systems in place that not only safeguard the data from unauthorized access (which is mandated by federal laws like Health Insurance Portability and Accountability Act [HIPPA]) but also accurately identifies each patient to allow proper and timely access to records by authorized users (Annas, 2003).

Data requirements

- **Data richness** means that all the required data elements are included in the data set. In essence, richness (or comprehensiveness) means that the available variables portray a rich enough dimensionality of the underlying subject matter for an accurate and worthy analytics study. It also means that the information content is complete (or near complete) to build a predictive and/or prescriptive analytics model.

- **Data consistency** means that the data are accurately collected and combined/merged. Consistent data represent the dimensional information (variables of interest) coming from potentially disparate sources but pertaining to the same subject. If the data integration/merging is not done properly, some of the variables of different subjects may find themselves in the same record—having two different patient records mixed up—for instance, it may happen while merging the demographic and clinical test result data records.

Data requirements

- **Data currency/data timeliness** means that the data should be up-to-date (or as recent/new as it needs to be) for a given analytics model. It also means that the data is recorded at or near the time of the event or observation so that the time-delay-related misrepresentation (incorrectly remembering and encoding) of the data is prevented. Because accurate analytics rely on accurate and timely data, an essential characteristic of analytics-ready data is the timeliness of the creation and access to data elements.

Data requirements

- **Data granularity** requires that the variables and data values be defined at the lowest (or as low as required) level of detail for the intended use of the data. If the data is aggregated, it may not contain the level of detail needed for an analytics algorithm to learn how to discern different records/cases from one another. For example, in a medical setting, numerical values for laboratory results should be recorded to the appropriate decimal place as required for the meaningful interpretation of test results and proper use of those values within an analytics algorithm. Similarly, in the collection of demographic data, data elements should be defined at a granular level to determine the differences in outcomes of care among various subpopulations. One thing to remember is that the data that is aggregated cannot be disaggregated (without access to the original source), but it can easily be aggregated from its granular representation.

Data requirements

Data validity is the term used to describe a match/mismatch between the actual and expected data values of a given variable. As part of data definition, the acceptable values or value ranges for each data element must be defined. For example, a valid data definition related to gender would include three values: male, female, and unknown.

Data relevancy means that the variables in the data set are all relevant to the study being conducted. Relevancy is not a dichotomous measure (whether a variable is relevant or not); rather, it has a spectrum of relevancy from least relevant to most relevant. Based on the analytics algorithms being used, one may choose to include only the most relevant information (i.e., variables) or if the algorithm is capable enough to sort them out, may choose to include all the relevant ones, regardless of their relevancy level. One thing that analytics studies should avoid is to include totally irrelevant data into the model building, as this may contaminate the information for the algorithm, resulting in inaccurate and misleading results.

Data types

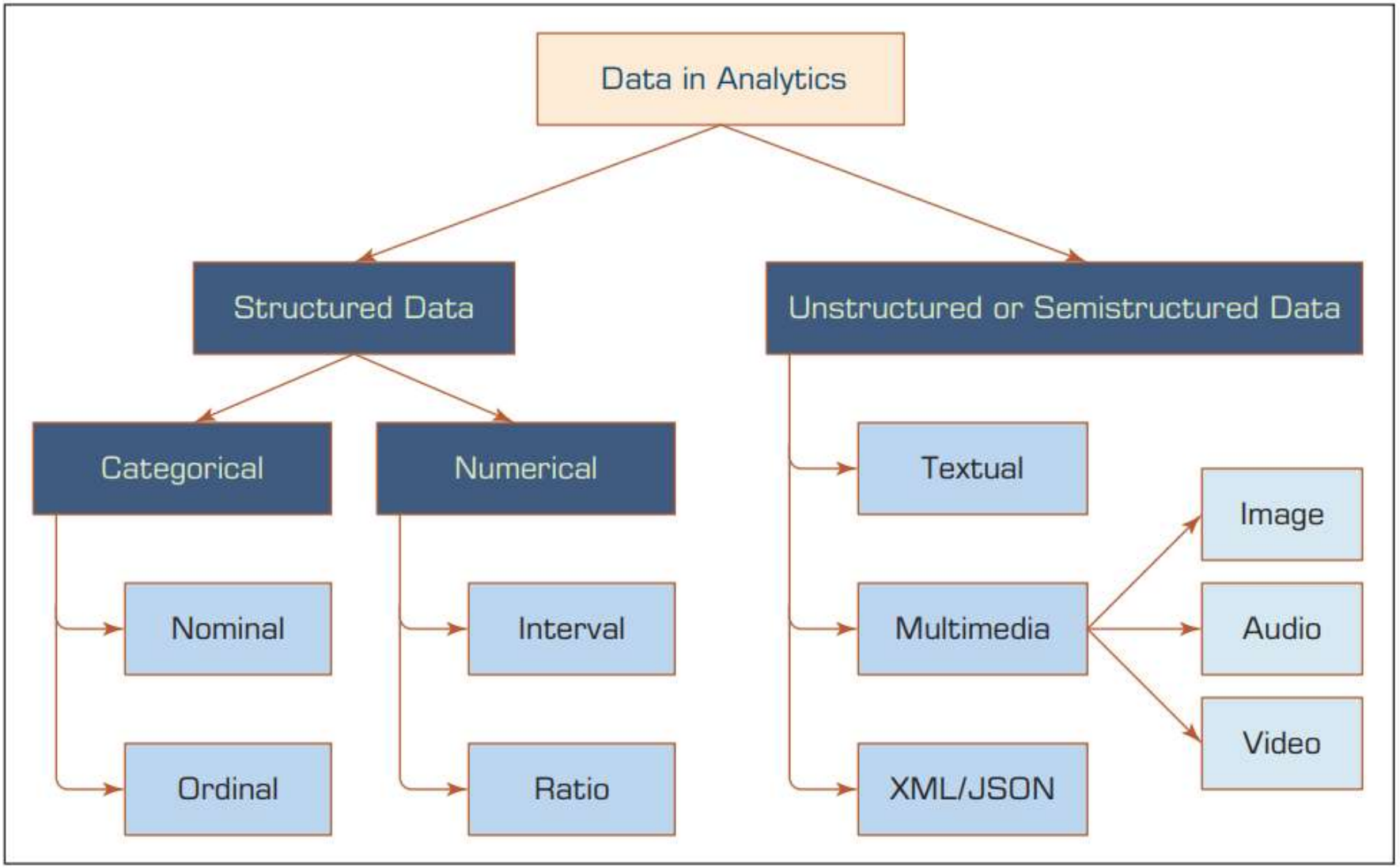


FIGURE 2.2 A Simple Taxonomy of Data.

Descriptive statistics

Descriptive Statistics for Descriptive Analytics

Descriptive statistics, as the name implies, describes the basic characteristics of the data at hand, often one variable at a time. Using formulas and numerical aggregations, descriptive statistics summarizes the data in such a way that often meaningful and easily understandable patterns emerge from the study. Although it is very useful in data analytics and very popular among the statistical methods, descriptive statistics does not allow making conclusions (or inferences) beyond the sample of the data being analyzed. That is, it is simply a nice way to characterize and describe the data on hand, without making conclusions (inferences or extrapolations) regarding the population of related hypotheses we might have in mind.

Descriptive statistics

In business analytics, descriptive statistics plays a critical role—it allows us to understand and explain/present our data in a meaningful manner using aggregated numbers, data tables, or charts/graphs. In essence, descriptive statistics helps us convert our numbers and symbols into meaningful representations for anyone to understand and use. Such an understanding not only helps business users in their decision-making processes, but also helps analytics professionals and data scientists to characterize and validate the data for other more sophisticated analytics tasks. Descriptive statistics allows analysts to identify data concertation, unusually large or small values (i.e., outliers), and unexpectedly distributed data values for numeric variables. Therefore, the methods in descriptive statistics can be classified as either measures for central tendency or measures of dispersion. In the following section we will use a simple description and mathematical formulation/representation of these measures. In mathematical representation, we will use $x_1, x_2, \ldots, x_n$ to represent individual values (observations) of the variable (measure) that we are interested in characterizing.

Descriptive statistics

Measures of Centrality Tendency (May Also Be Called Measures of Location or Centrality)

Measures of centrality are the mathematical methods by which we estimate or describe central positioning of a given variable of interest. A measure of central tendency is a single numerical value that aims to describe a set of data by simply identifying or estimating the central position within the data. The mean (often called the arithmetic mean or the simple average) is the most commonly used measure of central tendency. In addition to mean, you could also see median or mode being used to describe the centrality of a given variable. Although, the mean, median, and mode are all valid measures of central tendency, under different circumstances, one of these measures of centrality becomes more appropriate than the others. What follows are short descriptions of these measures, including how to calculate them mathematically and pointers on the circumstances in which they are the most appropriate measure to use.

Descriptive statistics

Arithmetic Mean

The **arithmetic mean** (or simply *mean* or *average*) is the sum of all the values/observations divided by the number of observations in the data set. It is by far the most popular and most commonly used measure of central tendency. It is used with continuous or discrete numeric data. For a given variable x, if we happen to have n values/observations ($x_1, x_2, \ldots, x_n$), we can write the arithmetic mean of the data sample ($\bar{x}$, pronounced as x-bar) as follows:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

or

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n}$$

The mean has several unique characteristics. For instance, the sum of the absolute deviations (differences between the mean and the observations) above the mean are the same as the sum of the deviations below the mean, balancing the values on either side of it. That said, it does not suggest, however, that half the observations are above and the other half are below the mean (a common misconception among those who do not know basic statistics). Also, the mean is unique for every data set and is meaningful and calculable for both interval- and ratio-type numeric data. One major downside is that the mean can be affected by outliers (observations that are considerably larger or smaller than the rest of the data points). Outliers can pull the mean toward their direction and, hence, bias the centrality representation. Therefore, if there are outliers or if the data is erratically

Descriptive statistic

Median

The **median** is the measure of center value in a given data set. It is the number in the middle of a given set of data that has been arranged/sorted in order of magnitude (either ascending or descending). If the number of observation is an odd number, identifying the median is very easy—just sort the observations based on their values and pick the value right in the middle. If the number of observations is an even number, then identify the two middle values, and then take the simple average of these two values. The median is meaningful and calculable for ratio, interval, and ordinal data types. Once determined, half the data points in the data are above and the other half are below the median. In contrary to the mean, the median is not affected by outliers or skewed data.

Descriptive statistic

Mode

The **mode** is the observation that occurs most frequently (the most frequent value in our data set). On a histogram it represents the highest bar in a bar chart, and hence, it may be considered as being the most popular option/value. The mode is most useful for data sets that contain a relatively small number of unique values. That is, it may be useless if the data have too many unique values (as is the case in many engineering measurements that capture high precision with a large number of decimal places), rendering each value having either one or a very small number representing its frequency. Although it is a useful measure (especially for nominal data), mode is not a very good representation of centrality, and therefore, it should not be used as the only measure of central tendency for a given data set.

Dispersion

Measures of Dispersion (May Also Be Called Measures of Spread or Decentrality)

Measures of **dispersion** are the mathematical methods used to estimate or describe the degree of variation in a given variable of interest. They are a representation of the numerical spread (compactness or lack thereof) of a given data set. To describe this dispersion, a number of statistical measures are developed; the most notable ones are range, variance, and standard deviation (and also quartiles and absolute deviation). One of the main reasons why the measures of dispersion/spread of data values are important is the fact that it gives us a framework within which we can judge the central tendency—gives us the indication of how well the mean (or other centrality measures) represents the sample data. If the dispersion of values in the data set is large, the mean is not deemed to be a very good representation of the data. This is because a large dispersion measure indicates large differences between individual scores. Also, in research, it is often perceived as a positive sign to see a small variation within each data sample, as it may indicate homogeneity, similarity, and robustness within the collected data.

Dispersion

Variance

A more comprehensive and sophisticated measure of dispersion is the **variance**. It is a method used to calculate the deviation of all data points in a given data set from the mean. The larger the variance, the more the data are spread out from the mean and the more variability one can observe in the data sample. To prevent the offsetting of negative and positive differences, the variance takes into account the square of the distances from the mean. The formula for a data sample can be written as

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \overline{x})^2}{n - 1}$$

where n is the number of samples, $\overline{x}$ is the mean of the sample and x_i is the ith value in the data set. The larger values of variance indicate more dispersion, whereas smaller values indicate compression in the overall data set. Because the differences are squared, larger deviations from the mean contribute significantly to the value of variance. Again, because the differences are squared, the numbers that represent deviation/variance become somewhat meaningless (as opposed to a dollar difference, herein you are given a squared dollar difference). Therefore, instead of variance, in many business applications we use a more meaningful dispersion measure, called standard deviation.

Dispersion

Standard Deviation

The **standard deviation** is also a measure of the spread of values within a set of data. The standard deviation is calculated by simply taking the square root of the variations. The following formula shows the calculation of standard deviation from a given sample of data points.

$$s = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n - 1}}$$

Dispersion

Mean Absolute Deviation

In addition to variance and standard deviation, sometimes we also use **mean absolute deviation** to measure dispersion in a data set. It is a simpler way to calculate the overall deviation from the mean. Specifically, it is calculated by measuring the absolute values of the differences between each data point and the mean and summing them. It provides a measure of spread without being specific about the data point being lower or higher than the mean. The following formula shows the calculation of the mean absolute deviation:

$$MAD = \frac{\sum_{i=1}^{n} |x_i - \bar{x}|}{n}$$

Centrality & Dispersion

Box-and-Whiskers Plot

The **box-and-whiskers plot** (or simply a **box plot**) is a graphical illustration of several descriptive statistics about a given data set. They can be either horizontal or vertical, but vertical is the most common representation, especially in modern-day analytics software products. It is known to be first created and presented by John W. Tukey in 1969. Box plot is often used to illustrate both centrality and dispersion of a given data set (i.e., the distribution of the sample data) in an easy-to-understand graphical notation. Figure 2.8 shows a couple of box plots side by side, sharing the same *y*-axis. As shown therein, a single

Centrality & Dispersion

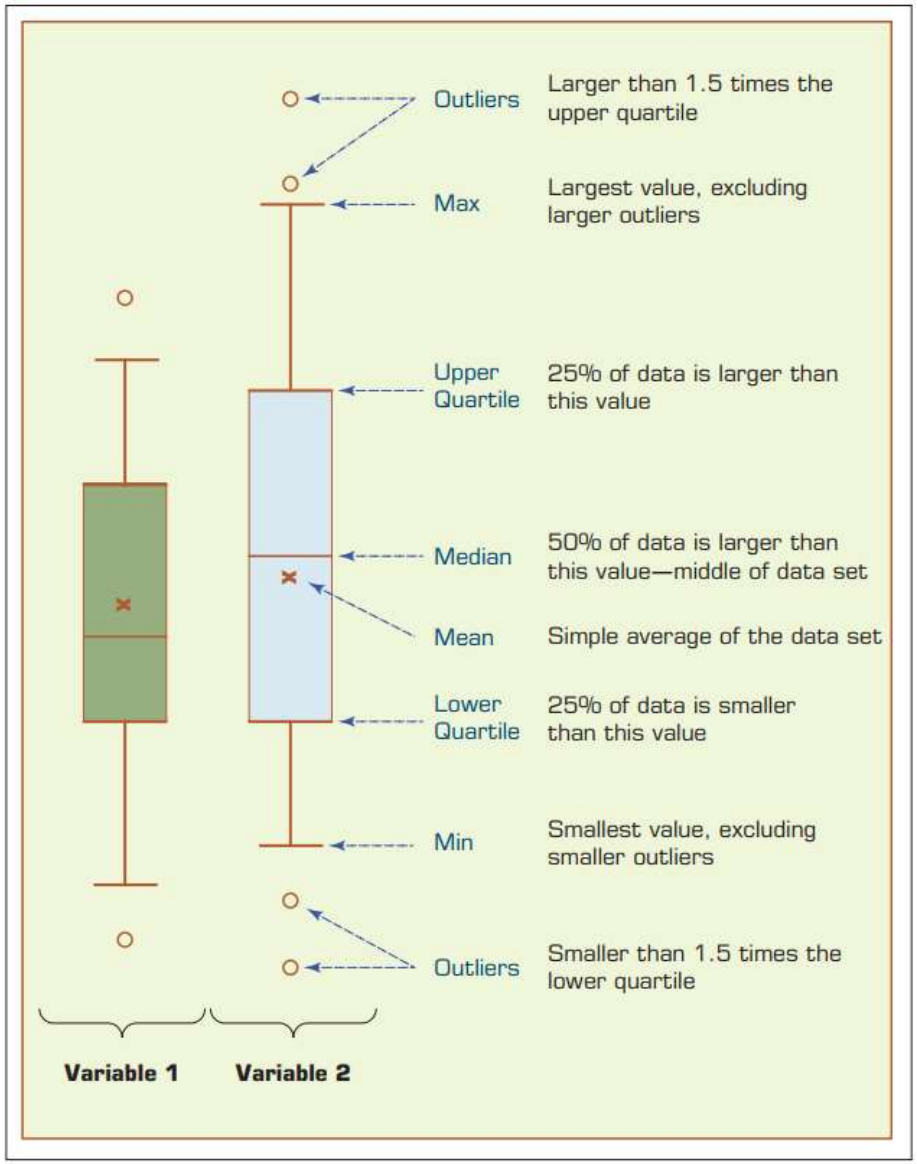


FIGURE 2.8 Understanding the Specifics about Box-and-Whiskers Plots.

Shape of distribution

Skewness is a measure of asymmetry (sway) in a distribution of the data that portrays a unimodal structure—only one peak exists in the distribution of the data. Because normal distribution is a perfectly symmetric unimodal distribution, it does not have skewness, that is, its skewness measure (i.e., the value of the coefficient of skewness) is equal to zero. The skewness measure/value can be either positive or negative. If the distribution sways left (i.e., tail is on the right side and the mean is smaller than median), then it produces a positive skewness measure, and if the distribution sways right (i.e., the tail is on the left side and the mean is larger than median), then it produces a negative skewness measure. In Figure 2.9, (c) represents a positively skewed distribution, whereas (d) represents a negatively skewed distribution. In the same figure, both (a) and (b) represent perfect symmetry and hence zero measure for skewness.

$$Skewness = S = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^3}{(n-1)s^3}$$

where s is the standard deviation and n is the number of samples.

Shape of distribution

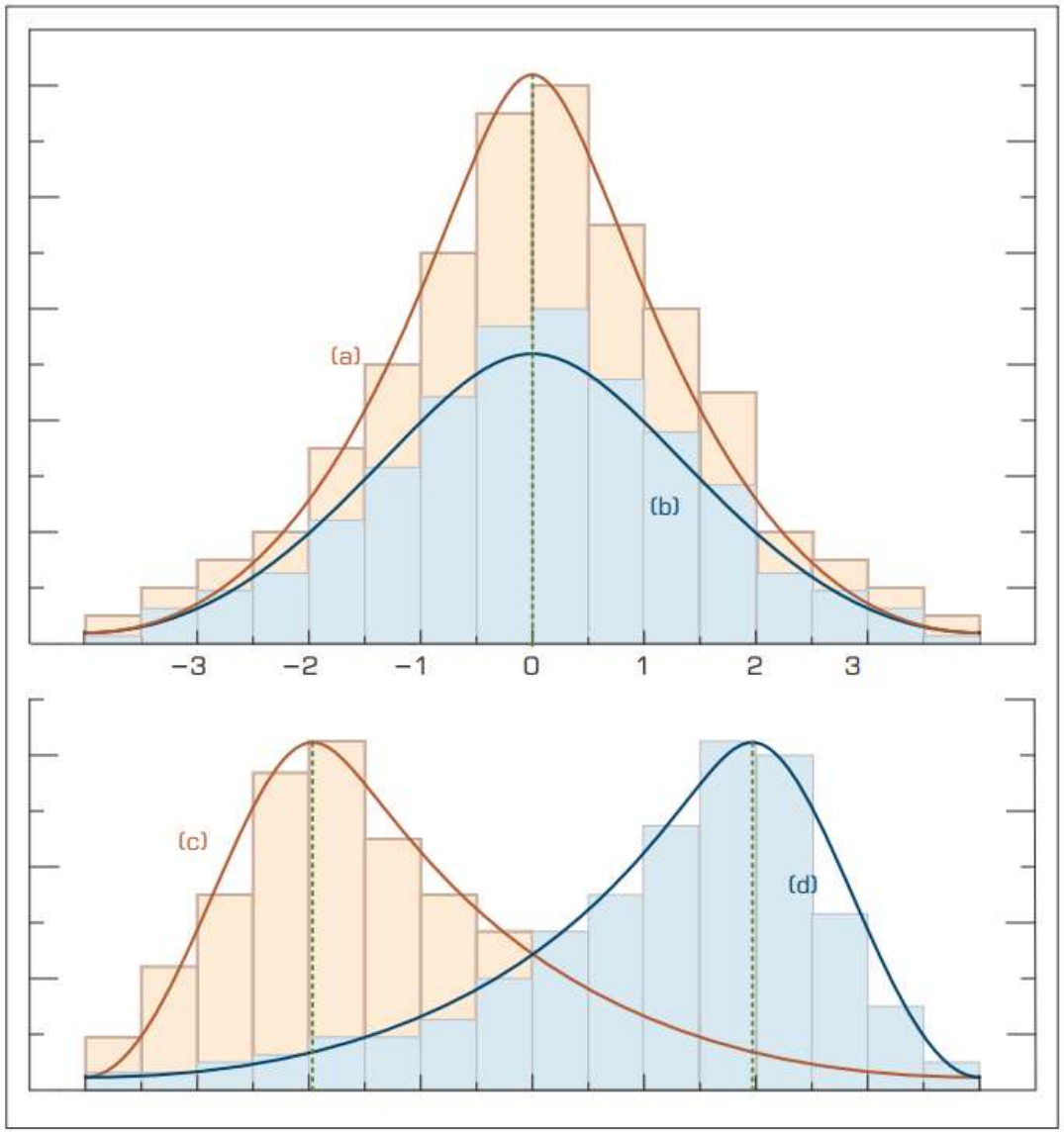
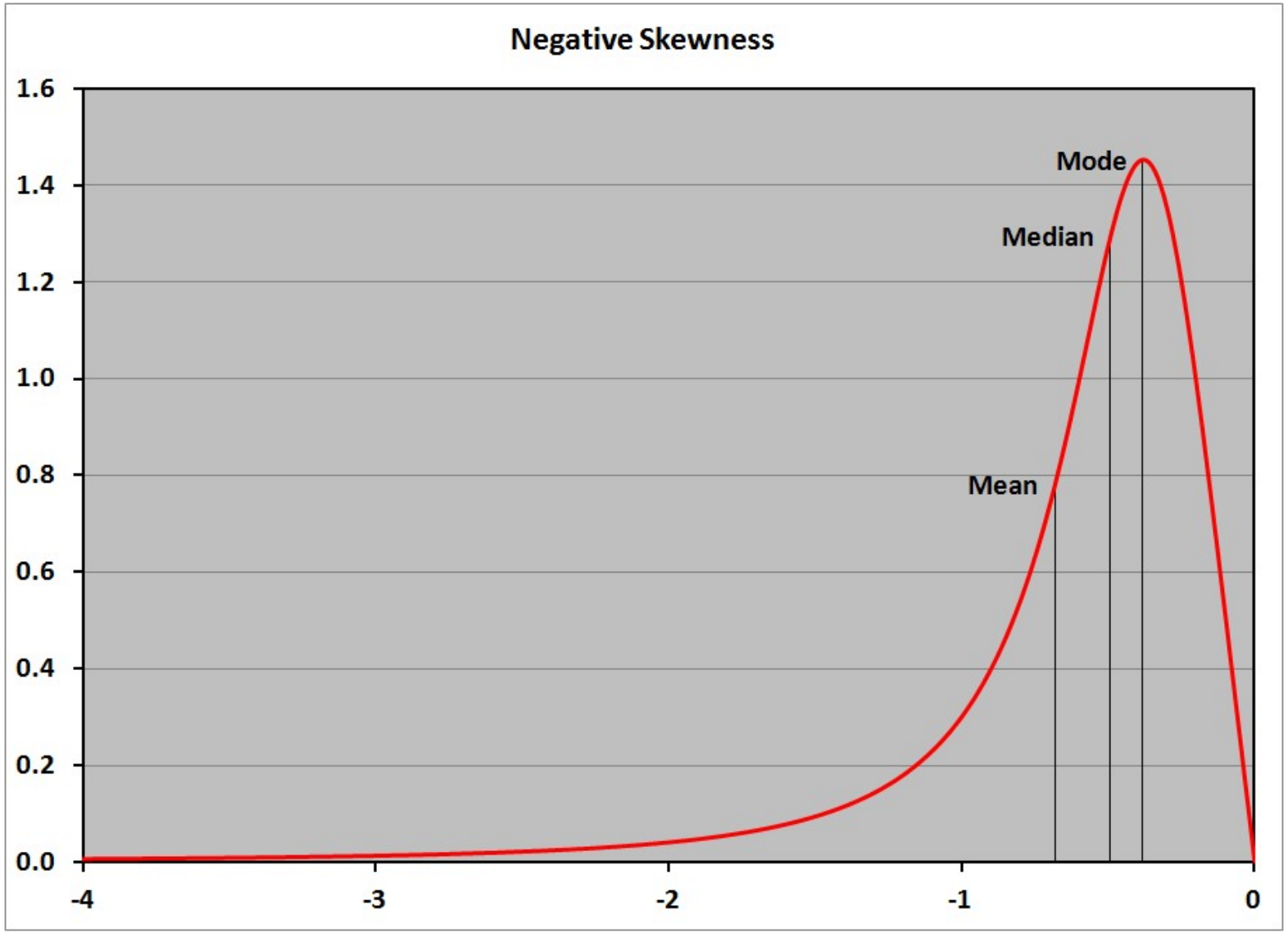
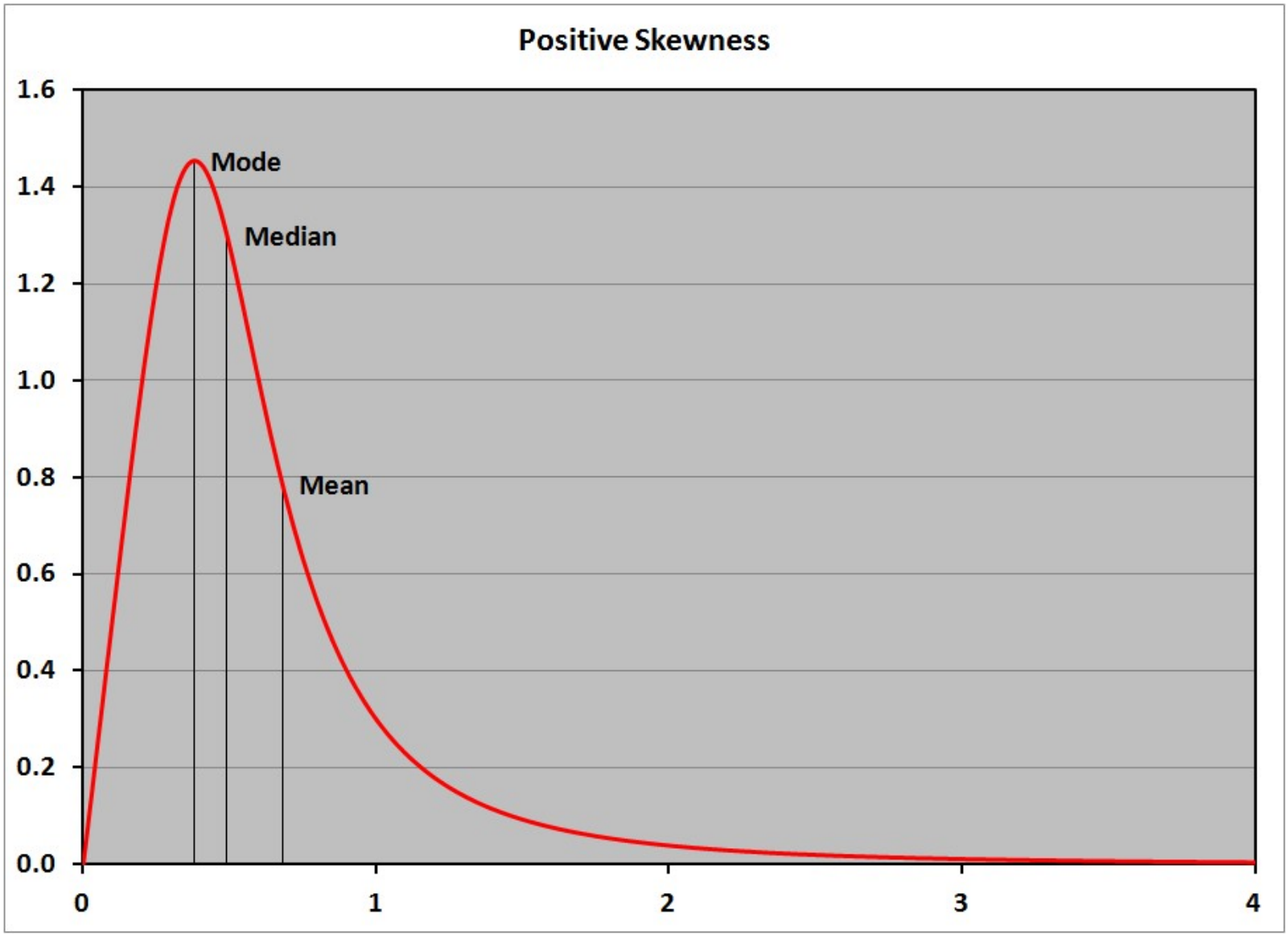


FIGURE 2.9 Relationship between Dispersion and Shape Properties.

Shape of distribution



Shape of distribution



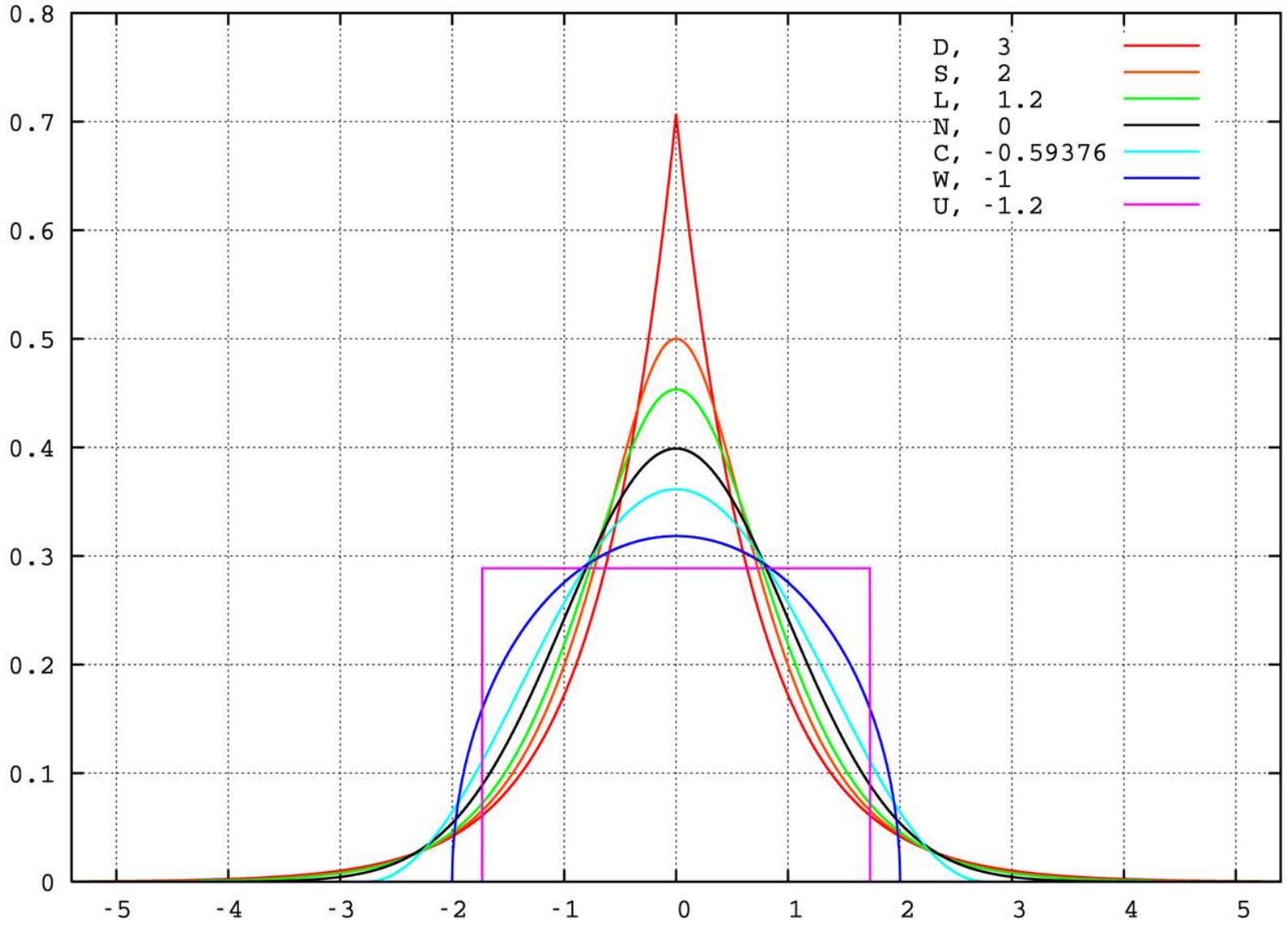
Shape of distribution

Kurtosis is another measure to use in characterizing the shape of a unimodal distribution. As opposed to the sway in shape, kurtosis is more interested in characterizing the peak/tall/skinny nature of the distribution. Specifically, kurtosis measures the degree to which a distribution is more or less peaked than a normal distribution. Whereas a positive kurtosis indicates a relatively peaked/tall distribution, a negative kurtosis indicates a relatively flat/short distribution. As a reference point, a normal distribution has a kurtosis of 3. The formula for kurtosis can be written as

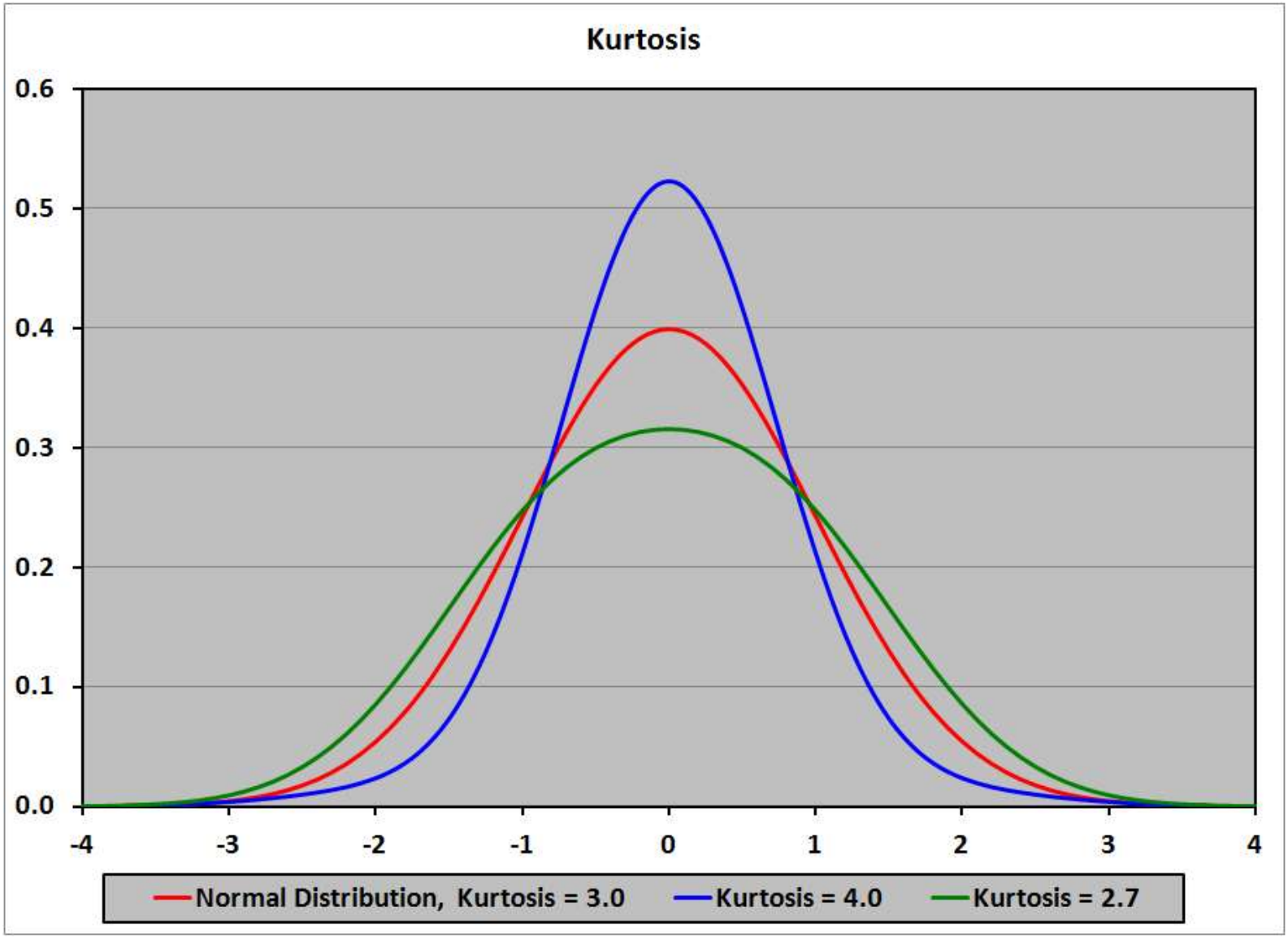
$$Kurtosis = K = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^4}{ns^4} - 3$$

Descriptive statistics (as well as inferential statistics) can easily be calculated using commercially viable statistical software packages (e.g., SAS, SPSS, Minitab, JMP, Statistica) or free/open source tools (e.g., R). Perhaps the most convenient way to calculate descriptive and some of the inferential statistics is to use Excel. Technology Insights 2.1 describes in detail how to use Microsoft Excel to calculate descriptive statistics.

Shape of distribution



Shape of distribution



Regression

2.6 Regression Modeling for Inferential Statistics

Regression, especially linear regression, is perhaps the most widely known and used analytics technique in statistics. Historically speaking, the roots of regression date back to the 1920s and 1930s, to the earlier work on inherited characteristics of sweet peas by Sir Francis Galton and subsequently by Karl Pearson. Since then regression has become the statistical technique for characterization of relationships between explanatory (input) variable(s) and response (output) variable(s).

As popular as it is, essentially, regression is a relatively simple statistical technique to model the dependence of a variable (response or output variable) on one (or more) explanatory (input) variables. Once identified, this relationship between the variables can be formally represented as a linear/additive function/equation. As is the case with many other modeling techniques, regression aims to capture the functional relationship between and among the characteristics of the real world and describe this relationship with a mathematical model, which may then be used to discover and understand the complexities of reality—explore and explain relationships or forecast future occurrences.

Regression

Regression can be used for one of two purposes: hypothesis testing—investigating potential relationships between different variables, and prediction/forecasting—estimating values of a response variables based on one or more explanatory variables. These two uses are not mutually exclusive. The explanatory power of regression is also the foundation of its prediction ability. In hypothesis testing (theory building), regression analysis can reveal the existence/strength and the directions of relationships between a number of explanatory variables (often represented with x_i) and the response variable (often represented with y). In prediction, regression identifies additive mathematical relationships (in the form of an equation) between one or more explanatory variables and a response variable. Once determined, this equation can be used to forecast the values of the response variable for a given set of values of the explanatory variables.

Regression

SIMPLE VERSUS MULTIPLE REGRESSION If the regression equation is built between one response variable and one explanatory variable, then it is called simple regression. For instance, the regression equation built to predict/explain the relationship between a height of a person (explanatory variable) and the weight of a person (response variable) is a good example of simple regression. Multiple regression is the extension of simple regression where the explanatory variables are more than one. For instance, in the previous example, if we were to include not only the height of the person but also other personal characteristics (e.g., BMI, gender, ethnicity) to predict the weight of a person, then we would be performing multiple regression analysis. In both cases, the relationship between the response variable and the explanatory variable(s) are linear and additive in nature. If the relationships are not linear, then we may want to use one of many other nonlinear regression methods to better capture the relationships between the input and output variables.

Approach

How Do We Develop the Linear Regression Model?

To understand the relationship between two variables, the simplest thing that one can do is to draw a scatter plot, where the *y*-axis represents the values of the response variable and the *x*-axis represents the values of the explanatory variable (see Figure 2.13). A scatter catter plot would show the changes in the response variable as a function of the changes in the explanatory variable. In the case shown in Figure 2.13, there seems to be a positive relationship between the two; as the explanatory variable values increase, so does the response variable.

Simple regression analysis aims to find a mathematical representation of this relationship. In reality, it tries to find the signature of a straight line passing through right between the plotted dots (representing the observation/historical data) in such a way that it minimizes the distance between the dots and the line (the predicted values on the

Approach

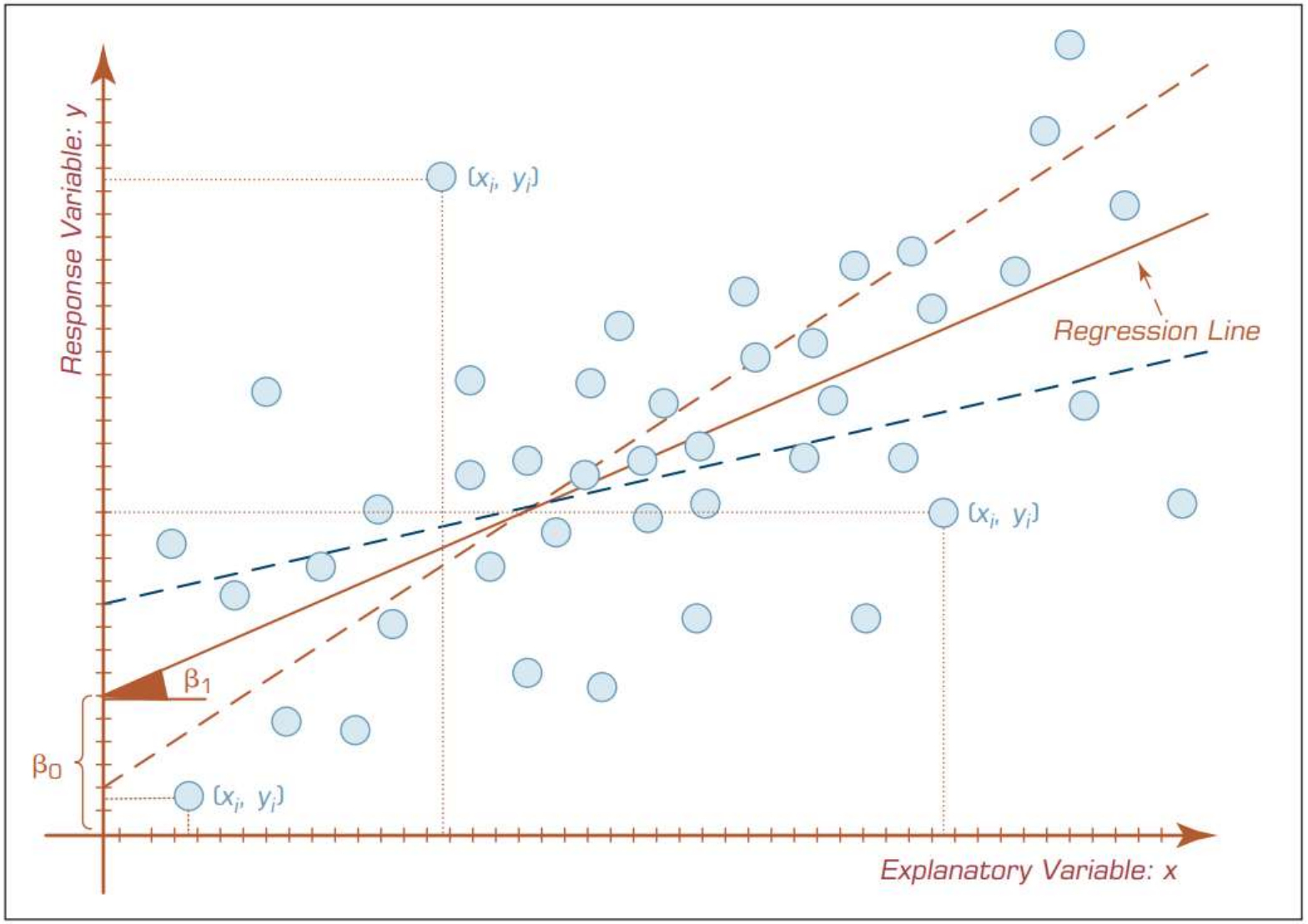


FIGURE 2.13 A Scatter Plot and a Linear Regression Line.

Approach

theoretical regression line). Even though there are several methods/algorithms proposed to identify the regression line, the one that is most commonly used is called the **ordinary least squares** (**OLS**) method. The OLS method aims to minimize the sum of squared residuals (squared vertical distances between the observation and the regression point) and leads to a mathematical expression for the estimated value of the regression line (which are known as β parameters). For simple **linear regression**, the aforementioned relationship between the response variable (y) and the explanatory variable(s) (x) can be shown as a simple equation as follows:

$$y = \beta_0 + \beta_1 x$$

In this equation, β_0 is called the intercept, and β_1 is called the slope. Once OLS determines the values of these two coefficients, the simple equation can be used to forecast the values of y for given values of x. The sign and the value of β_1 also reveal the direction and the strengths of relationship between the two variables.

Approach

In this equation, β_0 is called the intercept, and β_1 is called the slope. Once OLS determines the values of these two coefficients, the simple equation can be used to forecast the values of y for given values of x. The sign and the value of β_1 also reveal the direction and the strengths of relationship between the two variables.

If the model is of a multiple linear regression type, then there would be more coefficients to be determined, one for each additional explanatory variable. As the following formula shows, the additional explanatory variable would be multiplied with the new β_i coefficients and summed together to establish a linear additive representation of the response variable.

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \ldots + \beta_n x_n$$

Important assumptions

What Are the Most Important Assumptions in Linear Regression?

Even though they are still the choice of many for data analyses (both for explanatory as well as for predictive modeling purposes), linear regression models suffer from several highly restrictive assumptions. The validity of the linear model built depends on its ability to comply with these assumptions. Here are the most commonly pronounced assumptions:

1. **Linearity.** This assumption states that the relationship between the response variable and the explanatory variables are linear. That is, the expected value of the response variable is a straight-line function of each explanatory variable, while holding all other explanatory variables fixed. Also, the slope of the line does not depend on the values of the other variables. It also implies that the effects of different explanatory variables on the expected value of the response variable are additive in nature.
2. **Independence** (of errors). This assumption states that the errors of the response variable are uncorrelated with each other. This independence of the errors is weaker than actual statistical independence, which is a stronger condition and is often not needed for linear regression analysis.

Important assumptions

3. **Normality** (of errors). This assumption states that the errors of the response variable are normally distributed. That is, they are supposed to be totally random and should not represent any nonrandom patterns.
4. **Constant variance** (of errors). This assumption, also called homoscedasticity, states that the response variables have the same variance in their error, regardless of the values of the explanatory variables. In practice this assumption is invalid if the response variable varies over a wide enough range/scale.
5. **Multicollinearity.** This assumption states that the explanatory variables are not correlated (i.e., do not replicate the same but provide a different perspective of the information needed for the model). Multicollinearity can be triggered by having two or more perfectly correlated explanatory variables presented to the model (e.g., if the same explanatory variable is mistakenly included in the model twice, one with a slight transformation of the same variable). A correlation-based data assessment usually catches this error.

There are statistical techniques developed to identify the violation of these assumptions and techniques to mitigate them. The most important part for a modeler is to be aware of their existence and to put in place the means to assess the models to make sure that the models are compliant with the assumptions they are built on.

Logistic regression

Logistic Regression

Logistic regression is a very popular, statistically sound, probability-based classification algorithm that employs supervised **learning**. It was developed in the 1940s as a complement to linear regression and linear discriminant analysis methods. It has been used extensively in numerous disciplines, including the medical and social sciences fields. Logistic regression is similar to linear regression in that it also aims to regress to a mathematical function that explains the relationship between the response variable and the explanatory variables using a sample of past observations (training data). It differs from linear regression with one major point: its output (response variable) is a class as opposed to a numerical variable. That is, whereas linear regression is used to estimate a continuous numerical variable, logistic regression is used to classify a categorical variable. Even though the original form of logistic regression was developed for a binary output variable (e.g., 1/0, yes/no, pass/fail, accept/reject), the present-day modified version is capable of predicting multiclass output variables (i.e., multinomial logistic regression). If there is only one predictor variable and one predicted variable, the method is called simple logistic regression (similar to calling linear regression models with only one independent variable as simple linear regression).

Logistic regression

In predictive analytics, logistic regression models are used to develop probabilistic models between one or more explanatory/predictor variables (which may be a mix of both continuous and categorical in nature) and a class/response variable (which may be binomial/binary or multinomial/multiclass). Unlike ordinary linear regression, logistic regression is used for predicting categorical (often binary) outcomes of the response variable—treating the response variable as the outcome of a Bernoulli trial. Therefore, logistic regression takes the natural logarithm of the odds of the response variable to create a continuous criterion as a transformed version of the response variable. Thus the logit transformation is referred to as the link function in logistic regression—even though the response variable in logistic regression is categorical or binomial, the logit is the continuous criterion on which linear regression is conducted. Figure 2.15 shows a logistic regression function where the odds are represented in the x-axis (a linear function of the independent variables), whereas the probabilistic outcome is shown in the y-axis (i.e., response variable values change between 0 and 1).

Logistic regression

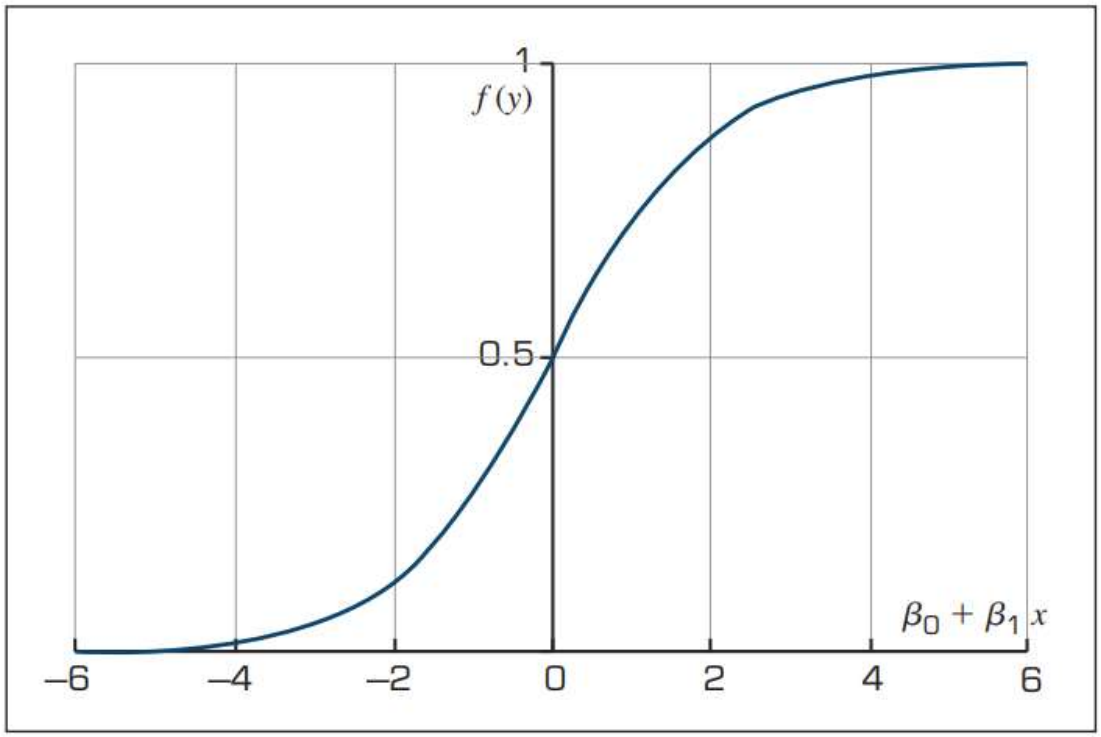


FIGURE 2.15 The Logistic Function.

The logistic function, $f(y)$ in Figure 2.15, is the core of logistic regression, which can only take values between 0 and 1. The following equation is a simple mathematical representation of this function:

$$f(y) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Experiment

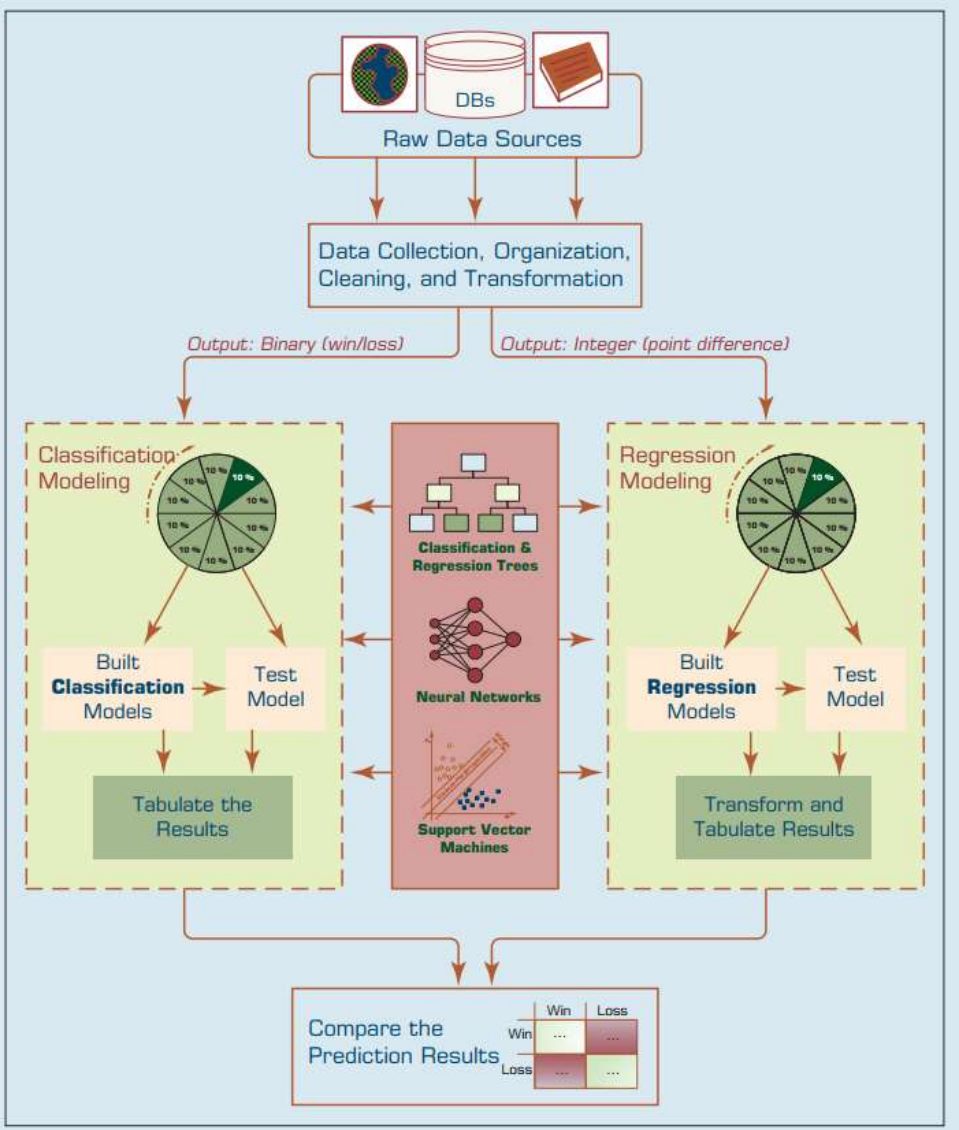


FIGURE 2.16 The Graphical Illustration of the Methodology Employed in the Study.

Time series forecasting

Time Series Forecasting

Sometimes the variable that we are interested in (i.e., the response variable) may not have distinctly identifiable explanatory variables, or there may be too many of them in a highly complex relationship. In such cases, if the data is available in a desired format, a prediction model, the so-called time series, can be developed. A time series is a sequence of data points of the variable of interest, measured and represented at successive points in time spaced at uniform time intervals. Examples of time series include monthly rain volumes in a geographic area, the daily closing value of the stock market indexes, daily sales totals for a grocery store. Often, time series are visualized using a line chart. Figure 2.17 shows an example time series of sales volumes for the years 2008 through 2012 in a quarterly basis.

Time series forecasting is the use of mathematical modeling to predict future values of the variable of interest based on previously observed values. The time series plots/charts look and feel very similar to simple linear regression in that as was the case in simple linear regression, in time series there are two variables: the response variable and the time variable presented in a scatter plot. Beyond this look similarity, there is hardly any other commonality between the two. Although regression analysis is often employed in testing theories to see if current values of one or more explanatory variables explain (and hence predict) the response variable, the time series models are focused on extrapolating on their time-varying behavior to estimate the future values.

Time series forecasting

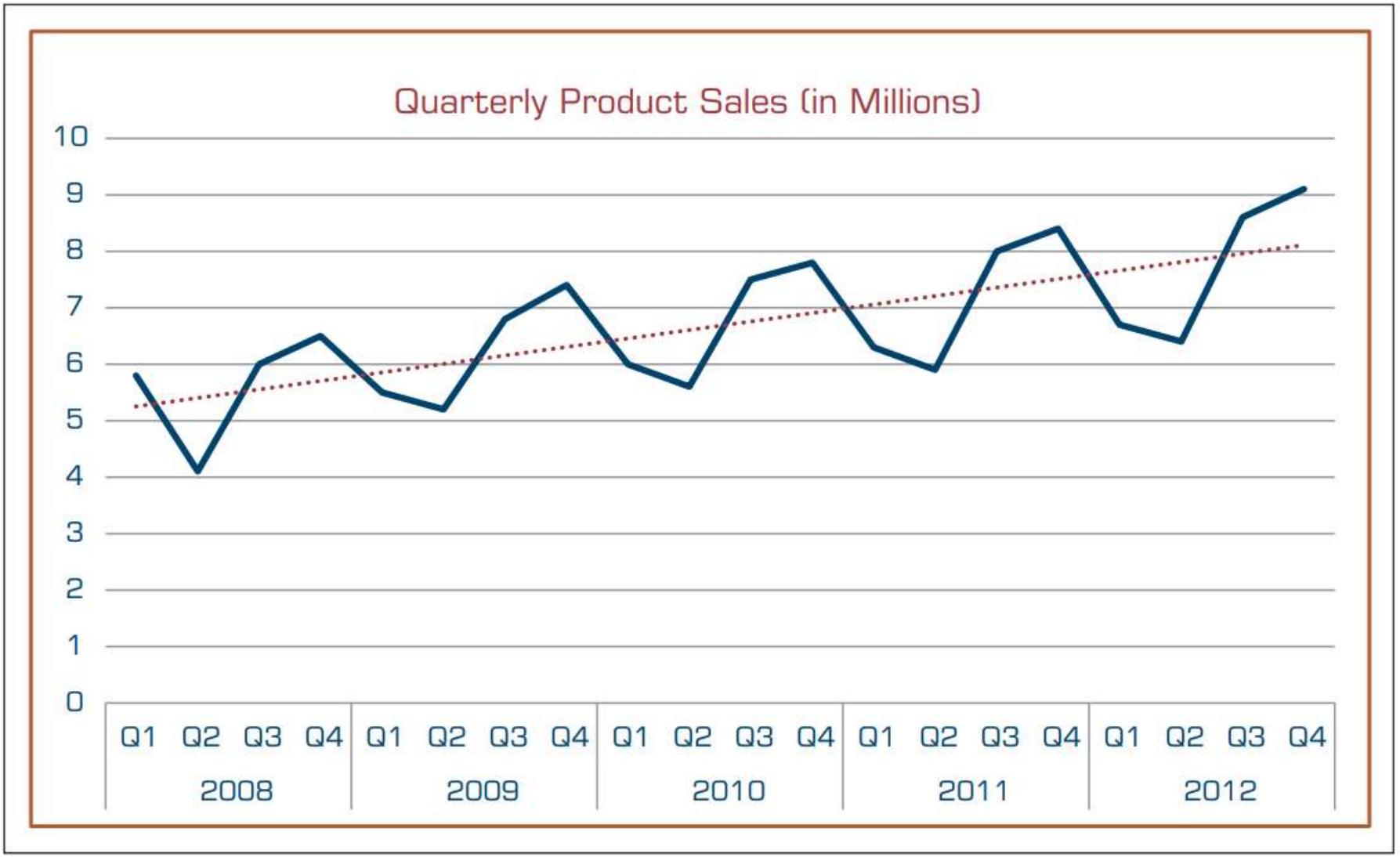


FIGURE 2.17 A Sample Time Series of Data on Quarterly Sales Volumes.

Reports

2.7 Business Reporting

Decision makers are in need of information to make accurate and timely decisions. Information is essentially the contextualization of data. In addition to statistical means that were explained in the previous section, information (descriptive analytics) can also be obtained using online analytics processing [OLTP] systems (see the simple taxonomy of descriptive analytics in Figure 2.7). The information is usually provided to the decision makers in the form of a written report (digital or on paper), although it can also be provided orally. Simply put, a **report** is any communication artifact prepared with the specific intention of conveying information in a digestible form to whoever needs it, whenever and wherever they may need it. It is usually a document that contains information (usually driven from data) organized in a narrative, graphic, and/or tabular form, prepared periodically (recurring) or on an as-needed (ad hoc) basis, referring to specific time periods, events, occurrences, or subjects. Business reports can fulfill many different (but often related) functions. Here are a few of the most prevailing ones:

- To ensure that all departments are functioning properly
- To provide information
- To provide the results of an analysis
- To persuade others to act
- To create an organizational memory (as part of a knowledge management system)

Reports

Business reporting (also called OLAP or BI) is an essential part of the larger drive toward improved, evidence-based, optimal managerial decision making. The foundation of these **business reports** is various sources of data coming from both inside and outside the organization (online transaction processing [OLTP] systems). Creation of these reports involves ETL (extract, transform, and load) procedures in coordination with a data warehouse and then using one or more reporting tools (see Chapter 3 for a detailed description of these concepts).

Due to the rapid expansion of information technology coupled with the need for improved competitiveness in business, there has been an increase in the use of computing power to produce unified reports that join different views of the enterprise in one place. Usually, this reporting process involves querying structured data sources, most of which were created using different logical data models and data dictionaries, to produce a human-readable, easily digestible report. These types of business reports allow managers and coworkers to stay informed and involved, review options and alternatives, and make informed decisions. Figure 2.18 shows the continuous cycle of data acquisition → information generation → decision making → business process management. Perhaps the most critical task in this cyclical process is the reporting (i.e., information generation)—converting data from different sources into actionable information.

Key to any successful report are clarity, brevity, completeness, and correctness. The nature of the report and the level of importance of these success factors change significantly

Reports

METRIC MANAGEMENT REPORTS In many organizations, business performance is managed through outcome-oriented metrics. For external groups, these are service-level agreements. For internal management, they are **key performance indicators (KPIs)**. Typically, there are enterprise-wide agreed targets to be tracked against over a period of time. They may be used as part of other management strategies such as Six Sigma or Total Quality Management.

DASHBOARD-TYPE REPORTS A popular idea in business reporting in recent years has been to present a range of different performance indicators on one page, like a dashboard in a car. Typically, dashboard vendors would provide a set of predefined reports with static elements and fixed structure, but also allow for customization of the dashboard widgets, views, and set targets for various metrics. It's common to have color-coded traffic lights defined for performance (red, orange, green) to draw management's attention to particular areas. A more detailed description of dashboards can be found in later part of this chapter.

BALANCED SCORECARD–TYPE REPORTS This is a method developed by Kaplan and Norton that attempts to present an integrated view of success in an organization. In addition to financial performance, balanced scorecard–type reports also include customer, business process, and learning and growth perspectives. More details on balanced scorecards are provided later in this chapter.

Reports

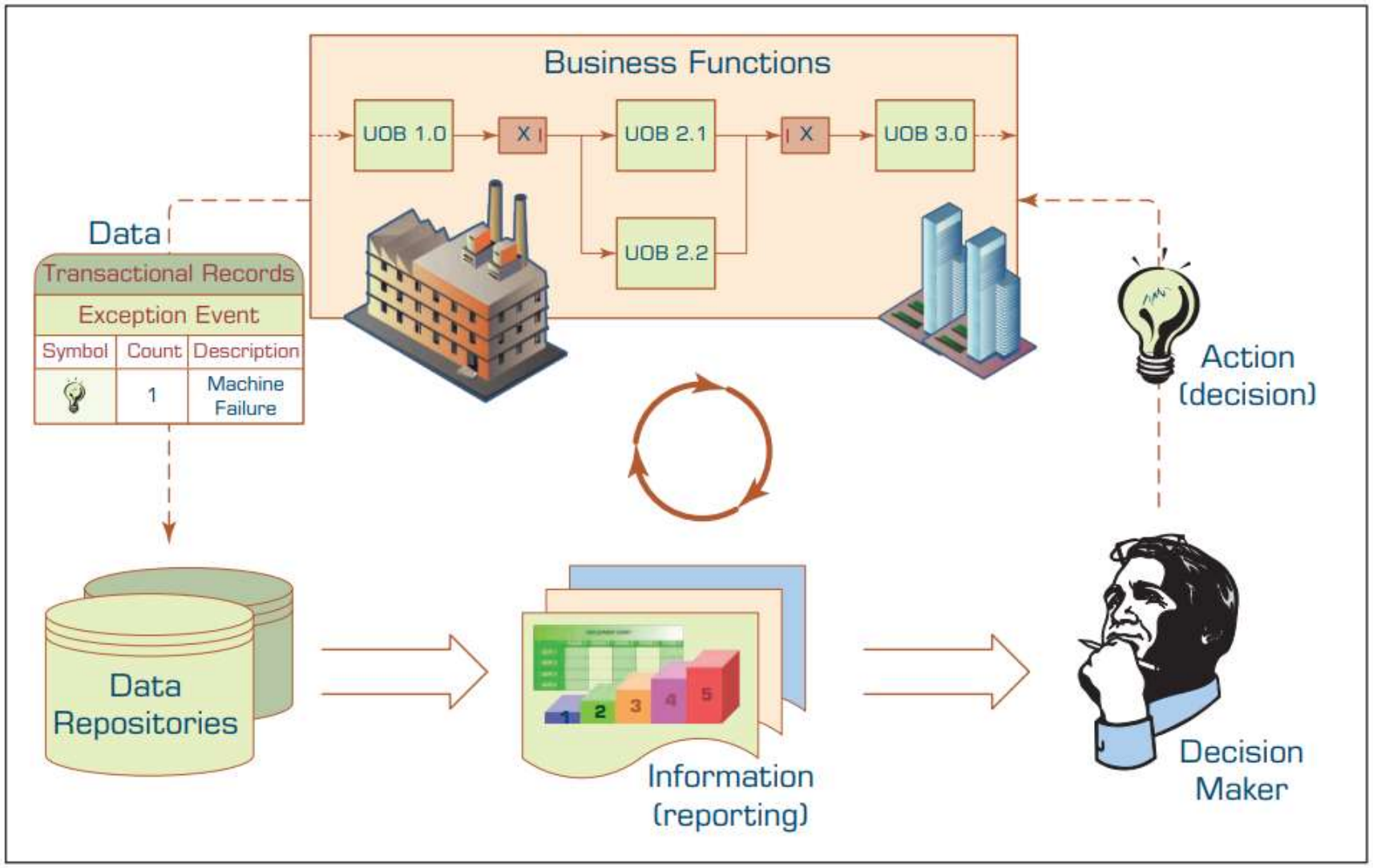


FIGURE 2.18 The Role of Information Reporting in Managerial Decision Making.

Data visualization-Dashboards

2.8 Data Visualization

Data visualization (or more appropriately, information visualization) has been defined as "the use of visual representations to explore, make sense of, and communicate data" (Few, 2007). Although the name that is commonly used is *data visualization*, usually what is meant by this is information visualization. Because information is the aggregation, summarization, and contextualization of data (raw facts), what is portrayed in visualizations is the information and not the data. However, because the two terms *data visualization* and *information visualization* are used interchangeably and synonymously, in this chapter we will follow suit.

Data visualization is closely related to the fields of information graphics, information visualization, scientific visualization, and statistical graphics. Until recently, the major forms of data visualization available in both BI applications have included charts and graphs, as well as the other types of visual elements used to create scorecards and dashboards.

To better understand the current and future trends in the field of data visualization, it helps to begin with some historical context.

Data visualization-Dashboards





Chart guideline

2.9 Different Types of Charts and Graphs

Often end users of business analytics systems are not sure what type of chart or graph to use for a specific purpose. Some charts or graphs are better at answering certain types of questions. Some look better than others. Some are simple; some are rather complex and crowded. What follows is a short description of the types of charts and/or graphs commonly found in most business analytics tools and what types of questions they are better at answering/analyzing. This material is compiled from several published articles and other literature (Abela, 2008; Hardin et al., 2012; SAS, 2014).

Basic Charts and Graphs

What follows are the basic charts and graphs that are commonly used for information visualization.

Chart guideline

LINE CHART Line charts are perhaps the most frequently used graphical visuals for time series data. Line charts (or a line graphs) show the relationship between two variables; they are most often used to track changes or trends over time (having one of the variables set to time on the *x*-axis). Line charts sequentially connect individual data points to help infer changing trends over a period of time. Line charts are often used to show time-dependent changes in the values of some measure, such as changes on a specific stock price over a 5-year period or changes on the number of daily customer service calls over a month.



Chart guideline

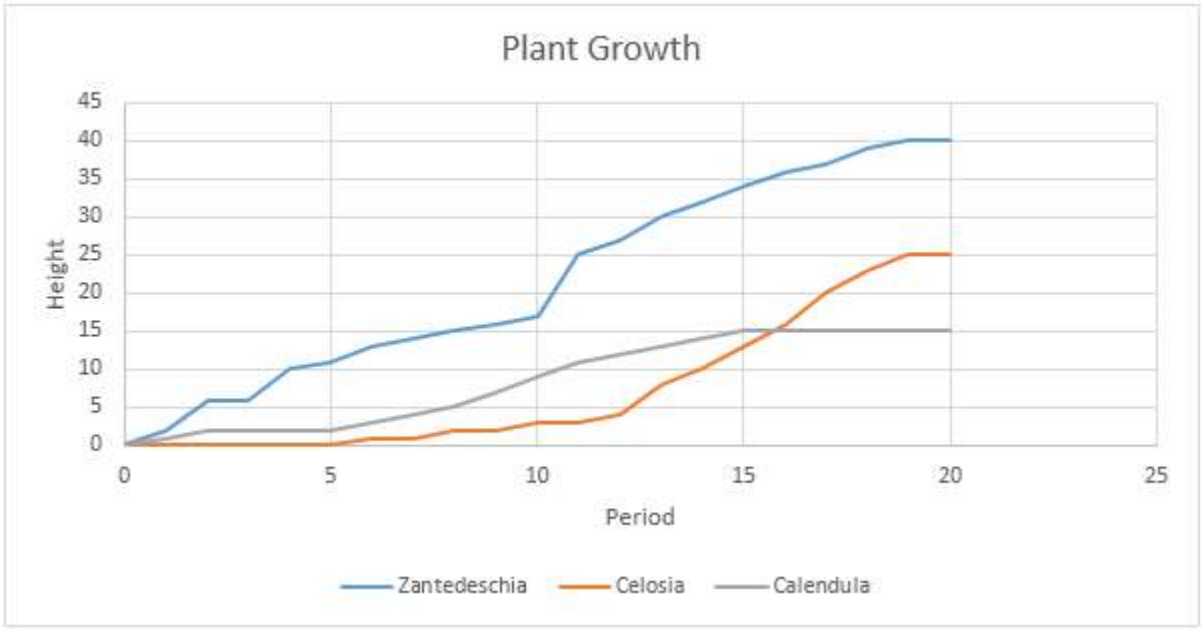
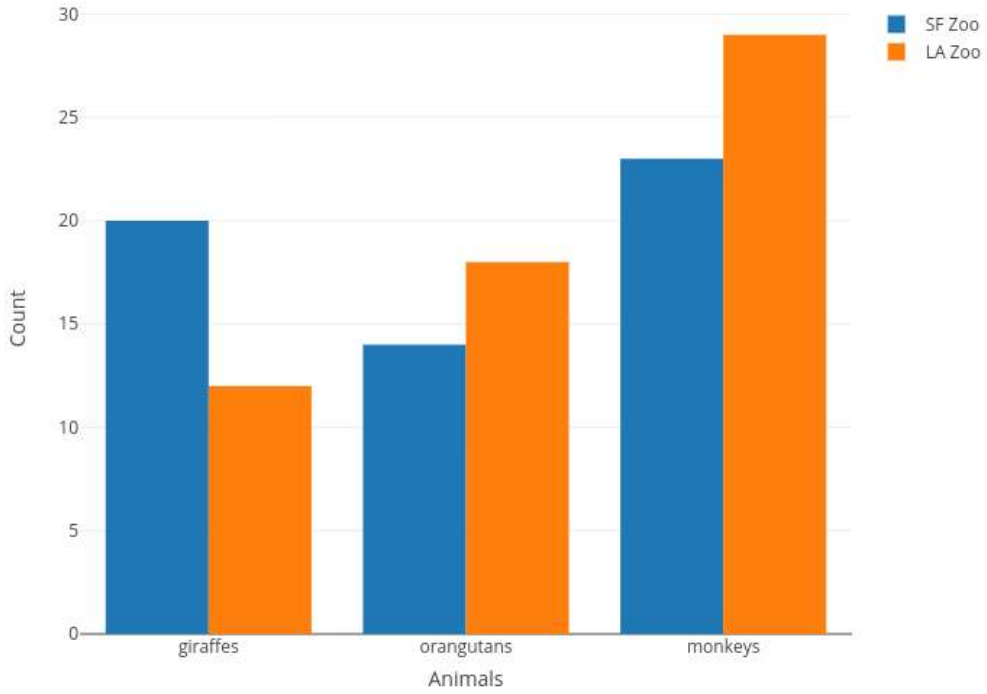


Chart guideline

BAR CHART Bar charts are among the most basic visuals used for data representation. Bar charts are effective when you have nominal data or numerical data that splits nicely into different categories so you can quickly see comparative results and trends within your data. Bar charts are often used to compare data across multiple categories such as percent of advertising spending by departments or by product categories. Bar charts can be vertically or horizontally oriented. They can also be stacked on top of each other to show multiple dimensions in a single chart.



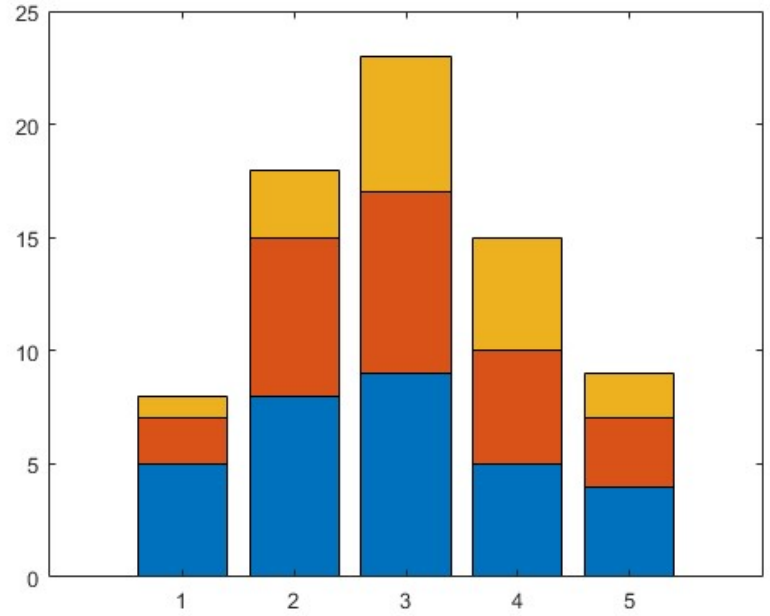


Chart guideline

PIE CHART **Pie charts** are visually appealing, as the name implies, pie-looking charts. Because they are so visually attractive, they are often incorrectly used. Pie charts should only be used to illustrate relative proportions of a specific measure. For instance, they can be used to show the relative percentage of an advertising budget spent on different product lines, or they can show relative proportions of majors declared by college students in their sophomore year. If the number of categories to show is more than just a few (say more than four), one should seriously consider using a bar chart instead of a pie chart.

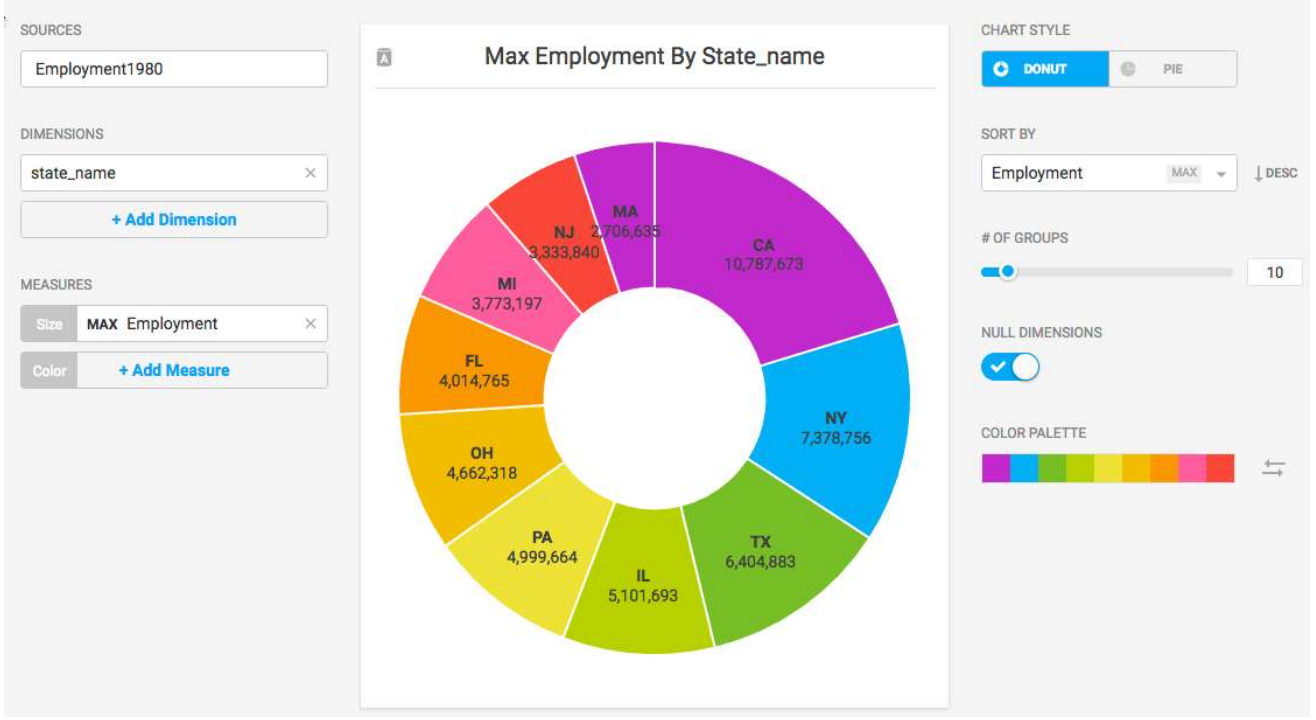
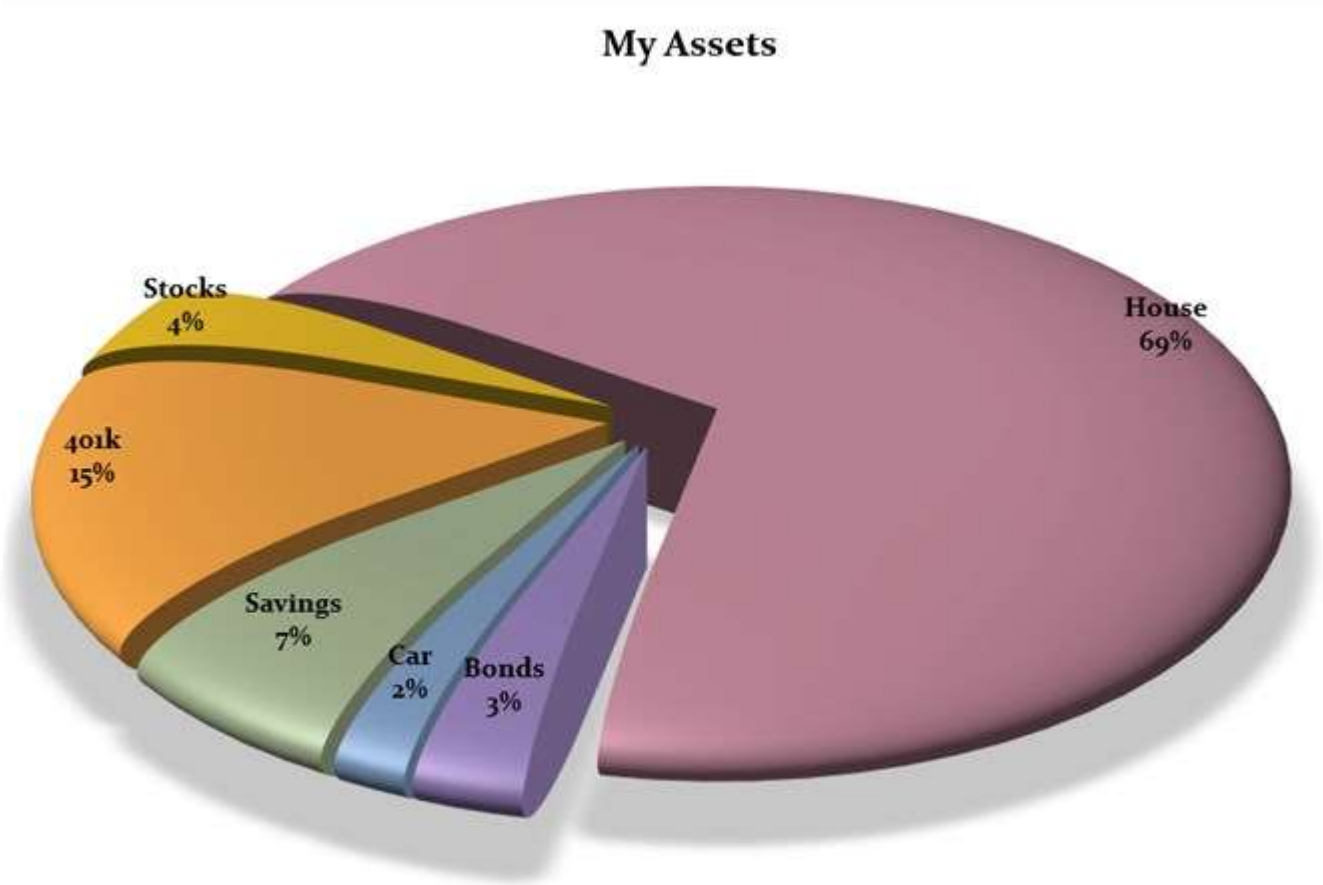


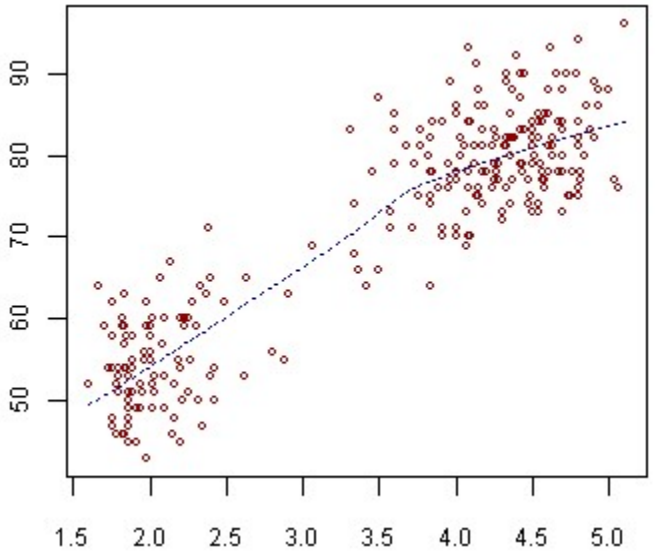
Chart guideline

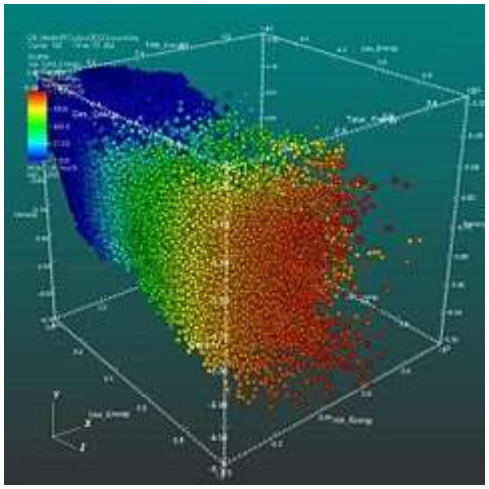


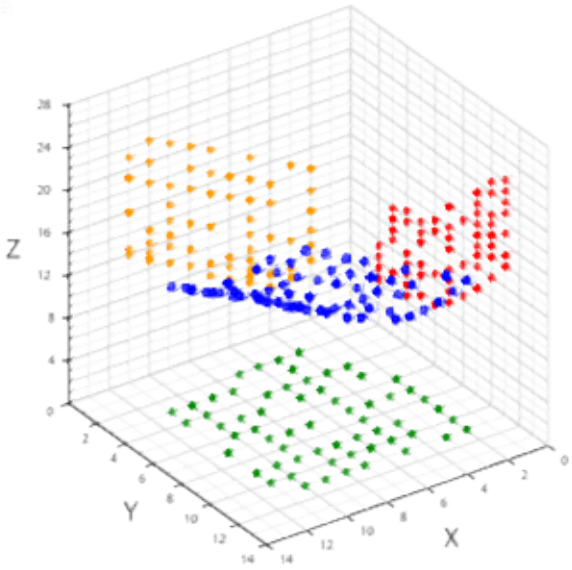
Scatter plot

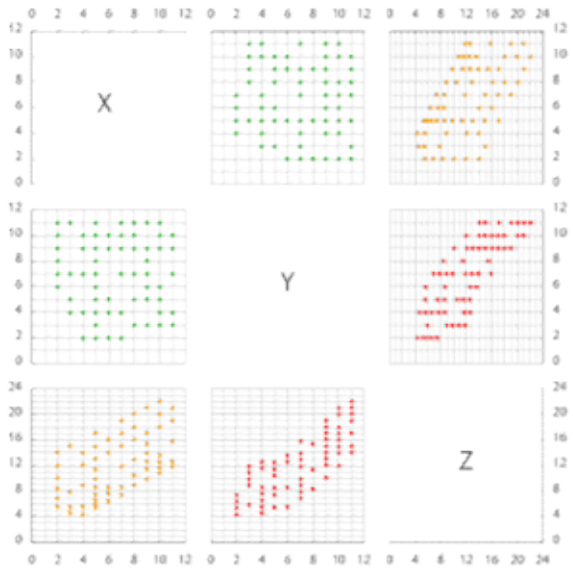
SCATTER PLOT **Scatter plots** are often used to explore the relationship between two or three variables (in 2-D or 2-D visuals). Because they are visual exploration tools, having more than three variables, translating them into more than three dimensions is not easily achievable. Scatter plots are an effective way to explore the existence of trends, concentrations, and outliers. For instance, in a two-variable (two-axis) graph, a scatter plot can be used to illustrate the corelationship between age and weight of heart disease patients or it can illustrate the relationship between the number of customer care representatives and the number of open customer service claims. Often, a trend line is superimposed on a two-dimensional scatter plot to illustrate the nature of the relationship.

Scatter plot

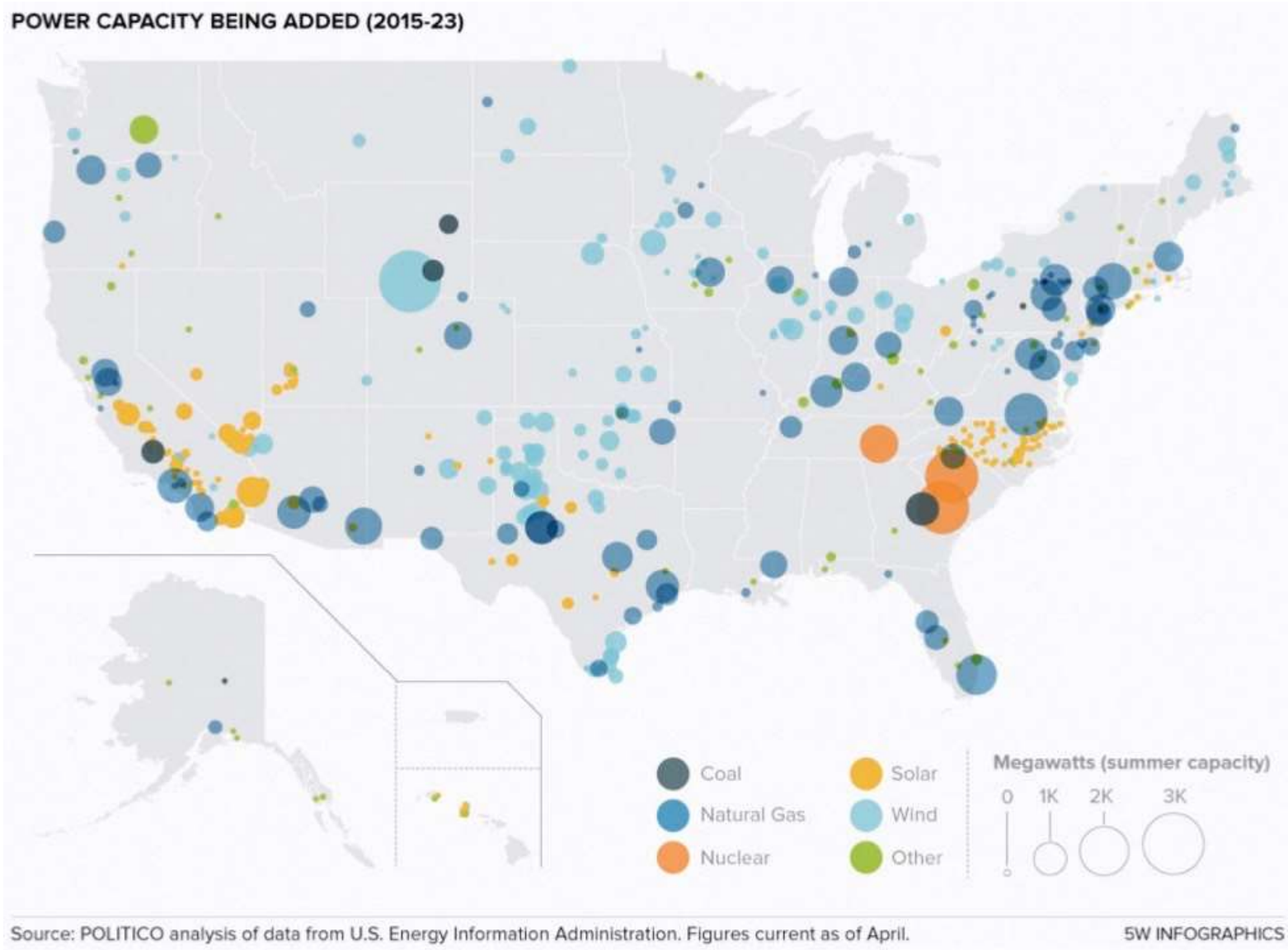


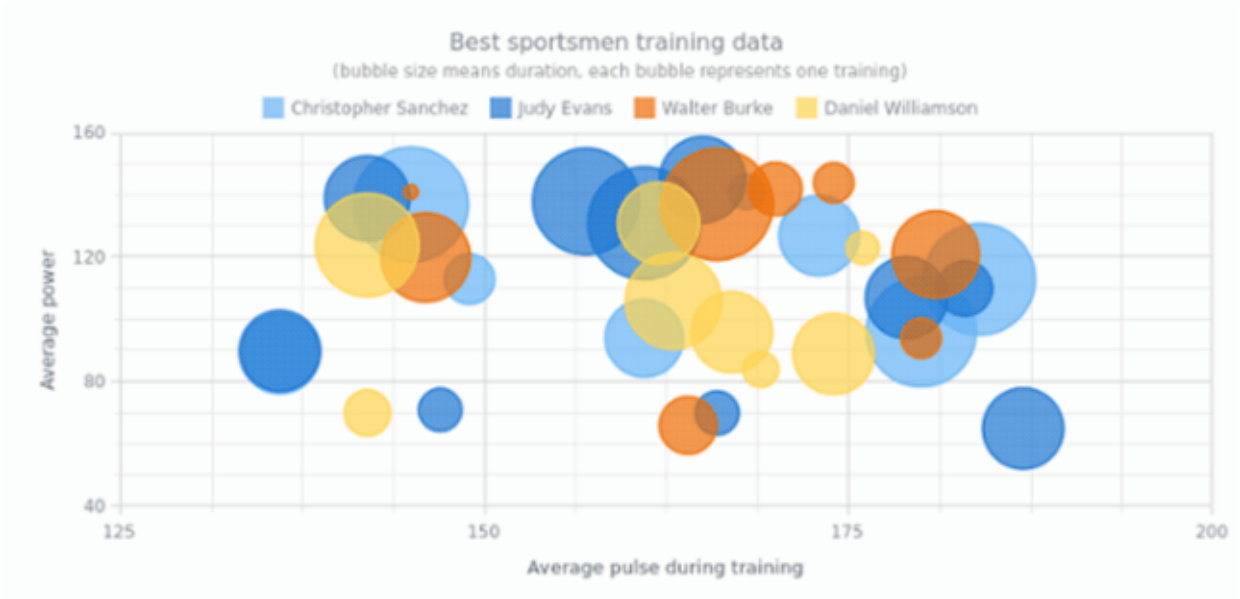






BUBBLE CHART **Bubble charts** are often enhanced versions of scatter plots. Bubble charts, though, are not a new visualization type; instead, they should be viewed as a technique to enrich data illustrated in scatter plots (or even geographic maps). By varying the size and/or color of the circles, one can add additional data dimensions, offering more enriched meaning about the data. For instance, a bubble chart can be used to show a competitive view of college-level class attendance by major and by time of the day, or it can be used to show profit margin by product type and by geographic region.


POWER CAPACITY BEING ADDED (2015-23)
Coal
Natural Gas
Nuclear
Solar
Wind
Other
Megawatts (summer capacity)
0
1K
2K
3K
Source: POLITICO analysis of data from U.S. Energy Information Administration. Figures current as of April.
5W INFOGRAPHICS


Best sportsmen training data
(bubble size means duration, each bubble represents one training)
Christopher Sanchez
Judy Evans
Walter Burke
Daniel Williamson
160
120
80
40
Average power
125
150
175
200
Average pulse during training

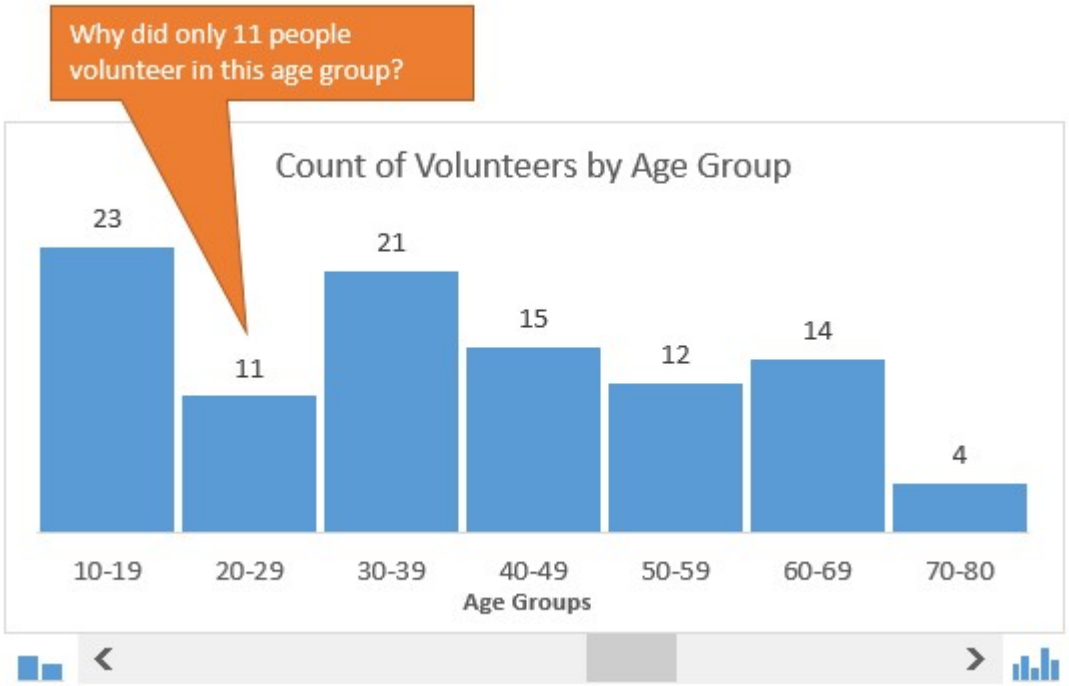
Histogram

Specialized Charts and Graphs

The graphs and charts that we review in this section are either derived from the basic charts as special cases or they are relatively new and are specific to a problem type and/or an application area.

HISTOGRAM Graphically speaking, a **histogram** looks just like a bar chart. The difference between histograms and generic bar charts is the information that is portrayed. Histograms are used to show the frequency distribution of a variable or several variables. In a histogram, the *x*-axis is often used to show the categories or ranges, and the *y*-axis is used to show the measures/values/frequencies. Histograms show the distributional shape of the data. That way, one can visually examine if the data is normally or exponentially distributed. For instance, one can use a histogram to illustrate the exam performance of a class, where distribution of the grades as well as comparative analysis of individual results can be shown, or one can use a histogram to show age distribution of the customer base.

Histogram


Why did only 11 people volunteer in this age group?
Count of Volunteers by Age Group
23
11
21
15
12
14
4
10-19
20-29
30-39
40-49
50-59
60-69
70-80
Age Groups

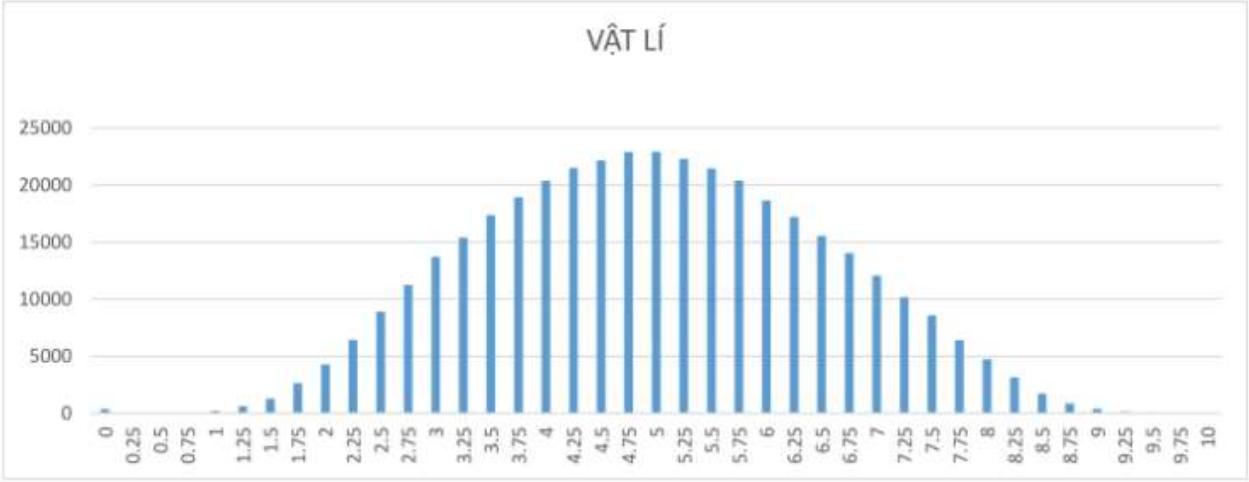
Histogram

PHÂN TÍCH PHỔ ĐIỂM THI CÁC MÔN THI/BÀI THI
KÌ THI TRUNG HỌC PHỔ THÔNG QUỐC GIA NĂM 2018

1. Vật lí

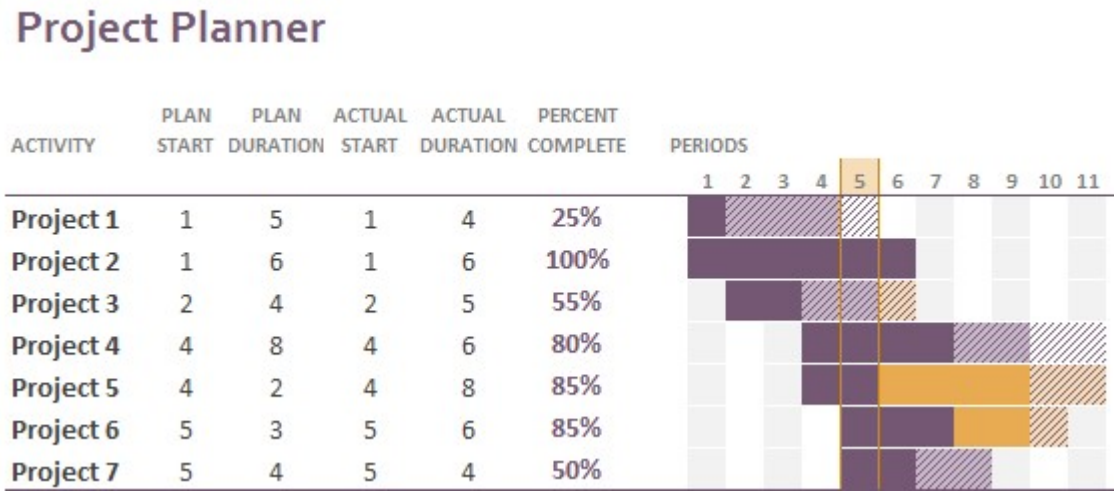
Điểm	0	0.25	0.5	0.75	1	1.25	1.5	1.75	2	2.25	2.5	2.75	3	3.25	3.5	3.75	4	4.25	4.5	4.75	5
Số lượng	372	0	12	50	203	623	1287	2625	4270	6449	8901	11256	13715	15407	17365	18933	20358	21495	22154	22900	22935

Điểm	5.25	5.5	5.75	6	6.25	6.5	6.75	7	7.25	7.5	7.75	8	8.25	8.5	8.75	9	9.25	9.5	9.75	10
Số lượng	22320	21444	20401	18653	17222	15553	14020	12066	10177	8585	6411	4752	3136	1725	849	385	119	71	19	2

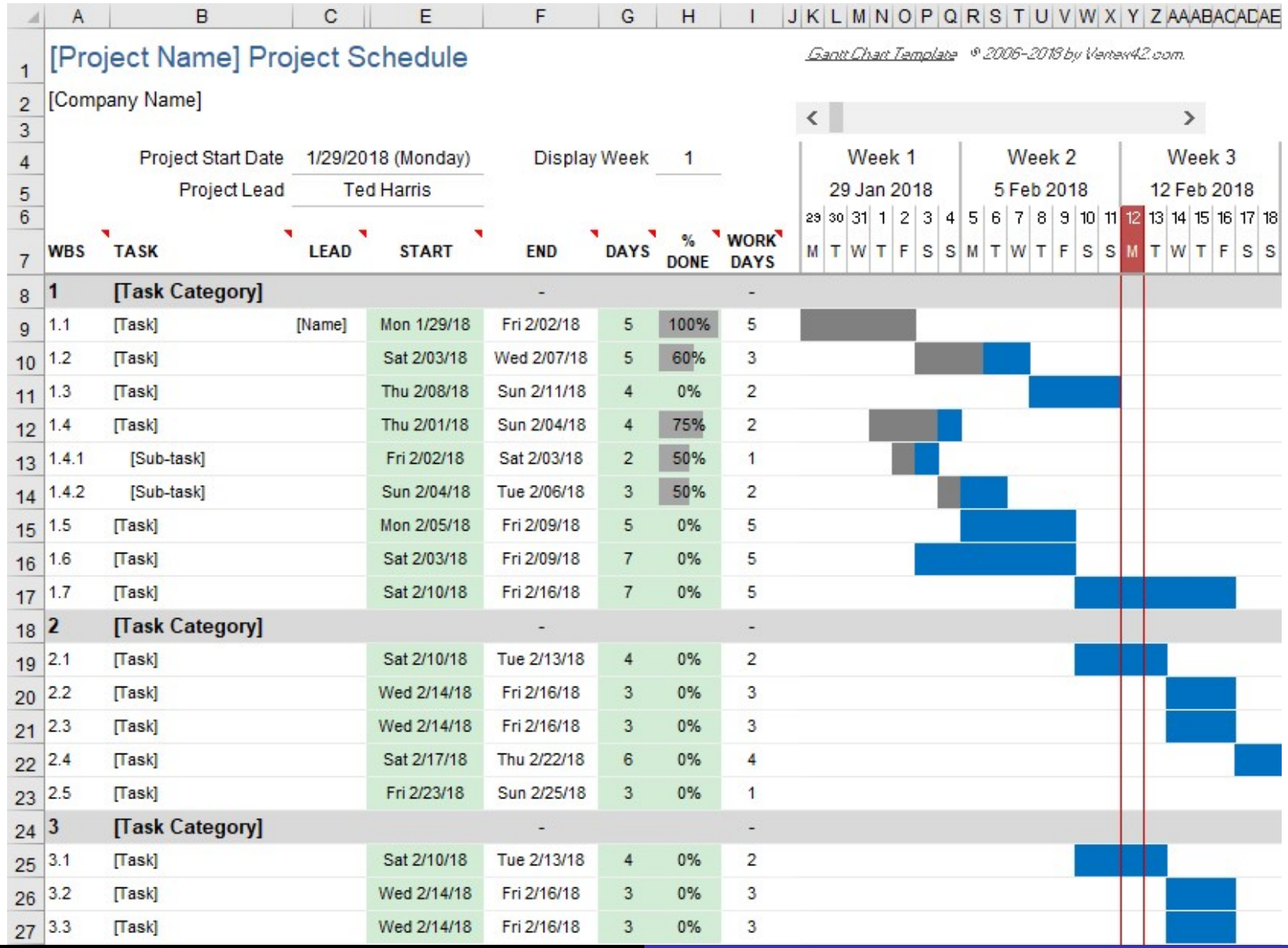


Gantt chart

GANTT CHART Gantt charts are a special case of horizontal bar charts that are used to portray project timelines, project tasks/activity durations, and overlap among the tasks/activities. By showing start and end dates/times of tasks/activities and the overlapping relationships, Gantt charts provide an invaluable aid for management and control of projects. For instance, Gantt charts are often used to show project timelines, task overlaps, relative task completions (a partial bar illustrating the completion percentage inside a bar that shows the actual task duration), resources assigned to each task, milestones, and deliverables.



Gantt chart



[Project Name] Project Schedule

Gantt Chart Template © 2006-2018 by Vertex42.com.

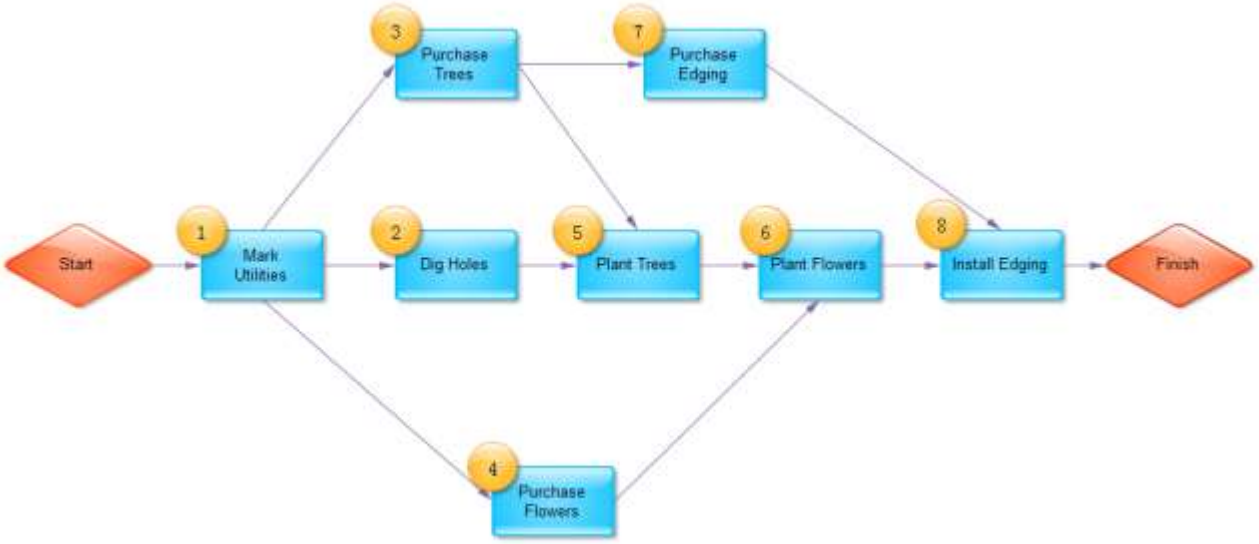
[Company Name]

Project Start Date: 1/29/2018 (Monday) | Display Week: 1 | Project Lead: Ted Harris

WBS	TASK	LEAD	START	END	DAYS	% DONE	WORK DAYS
1	[Task Category]			-			-
1.1	[Task]	[Name]	Mon 1/29/18	Fri 2/02/18	5	100%	5
1.2	[Task]		Sat 2/03/18	Wed 2/07/18	5	60%	3
1.3	[Task]		Thu 2/08/18	Sun 2/11/18	4	0%	2
1.4	[Task]		Thu 2/01/18	Sun 2/04/18	4	75%	2
1.4.1	[Sub-task]		Fri 2/02/18	Sat 2/03/18	2	50%	1
1.4.2	[Sub-task]		Sun 2/04/18	Tue 2/06/18	3	50%	2
1.5	[Task]		Mon 2/05/18	Fri 2/09/18	5	0%	5
1.6	[Task]		Sat 2/03/18	Fri 2/09/18	7	0%	5
1.7	[Task]		Sat 2/10/18	Fri 2/16/18	7	0%	5
2	[Task Category]			-			-
2.1	[Task]		Sat 2/10/18	Tue 2/13/18	4	0%	2
2.2	[Task]		Wed 2/14/18	Fri 2/16/18	3	0%	3
2.3	[Task]		Wed 2/14/18	Fri 2/16/18	3	0%	3
2.4	[Task]		Sat 2/17/18	Thu 2/22/18	6	0%	4
2.5	[Task]		Fri 2/23/18	Sun 2/25/18	3	0%	1
3	[Task Category]			-			-
3.1	[Task]		Sat 2/10/18	Tue 2/13/18	4	0%	2
3.2	[Task]		Wed 2/14/18	Fri 2/16/18	3	0%	3
3.3	[Task]		Wed 2/14/18	Fri 2/16/18	3	0%	3

PERT chart

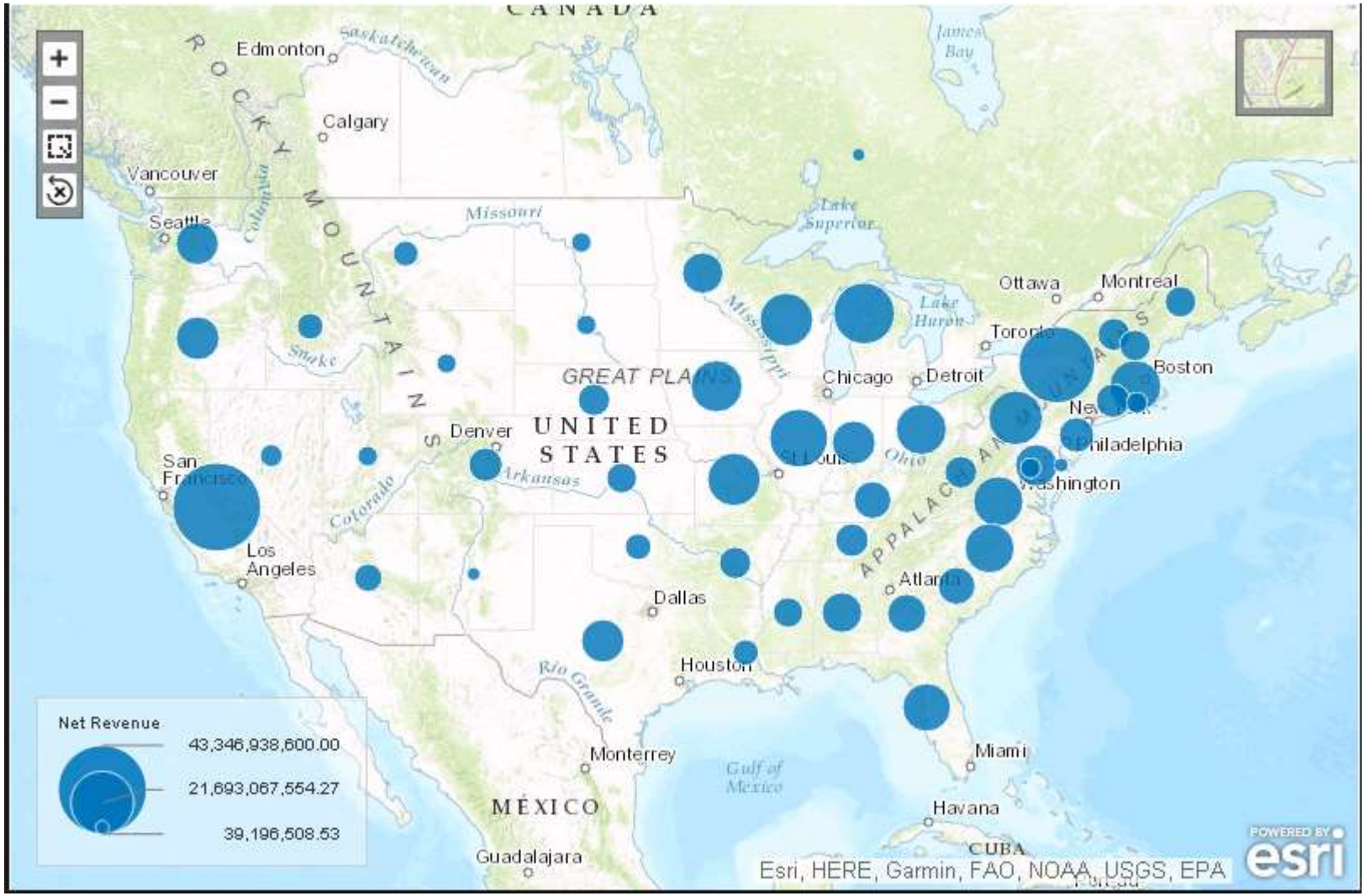
PERT CHART PERT charts (also called network diagrams) are developed primarily to simplify the planning and scheduling of large and complex projects. They show precedence relationships among the project activities/tasks. A PERT chart is composed of nodes (represented as circles or rectangles) and edges (represented with directed arrows). Based on the selected PERT chart convention, either nodes or the edges may be used to represent the project activities/tasks (activity-on-node versus activity-on-arrow representation schema).



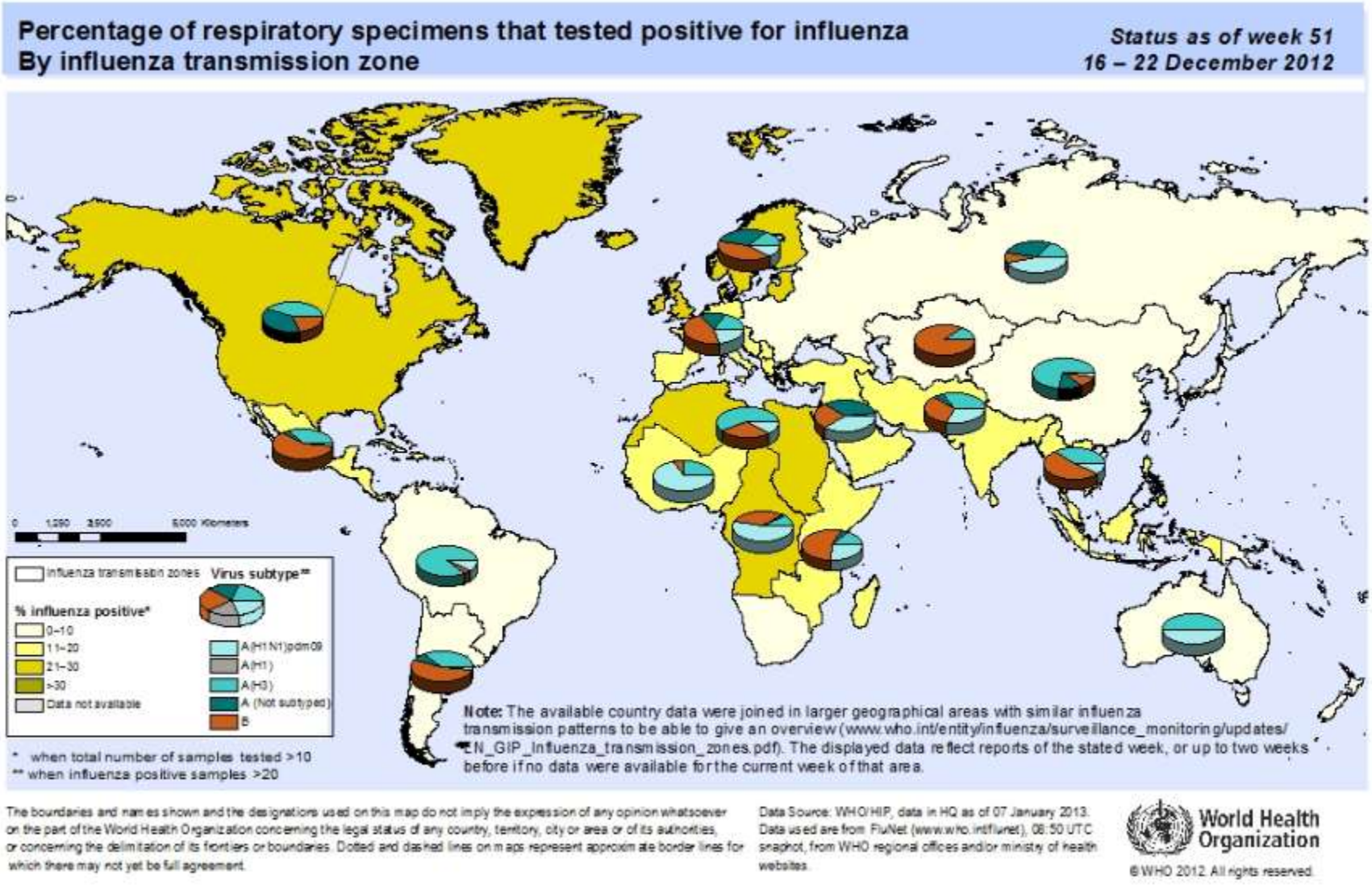
GIS map

GEOGRAPHIC MAP When the data set includes any kind of location data (e.g., physical addresses, postal codes, state names or abbreviations, country names, latitude/longitude, or some type of custom geographic encoding), it is better and more informative to see the data on a map. Maps usually are used in conjunction with other charts and graphs, as opposed to by themselves. For instance, one can use maps to show distribution of customer service requests by product type (depicted in pie charts) by geographic locations. Often a large variety of information (e.g., age distribution, income distribution, education, economic growth, or population changes) can be portrayed in a geographic map to help decide where to open a new restaurant or a new service station. These types of systems are often called geographic information systems (GIS).

GIS map

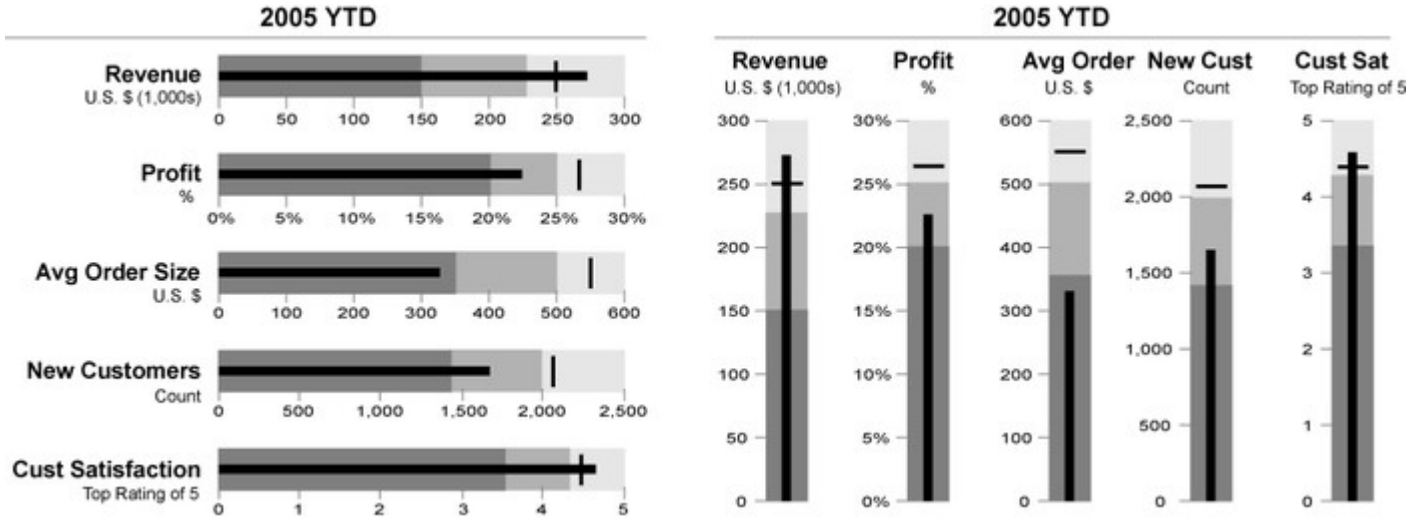

CANADA
Edmonton
Calgary
Vancouver
Seattle
ROCKY MOUNTAINS
Missouri
Saskatchewan
Columbia
Snake
Colorado
Arkansas
Mississippi
Ohio
GREAT PLAINS
UNITED STATES
Denver
San Francisco
Los Angeles
Dallas
Houston
Chicago
Detroit
Ottawa
Montreal
Toronto
Boston
Philadelphia
Washington
Atlanta
Miami
Havana
CUBA
Monterrey
MÉXICO
Guadalajara
Rio Grande
Gulf of Mexico
Lake Superior
Lake Huron
James Bay
APPALACHIANS
Net Revenue
43,346,938,600.00
21,693,067,554.27
39,196,508.53
Esri, HERE, Garmin, FAO, NOAA, USGS, EPA
POWERED BY esri

GIS map

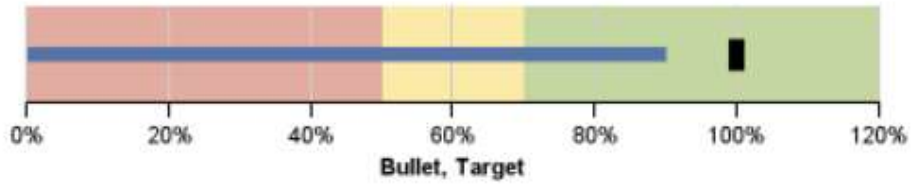
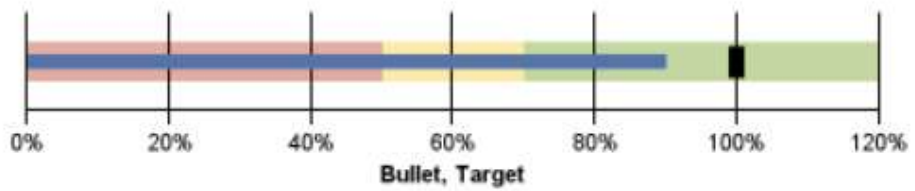

Percentage of respiratory specimens that tested positive for influenza
By influenza transmission zone
Status as of week 51
16 – 22 December 2012
0 1,250 2,500 5,000 Kilometers
Influenza transmission zones
% influenza positive*
0–10
11–20
21–30
>30
Data not available
Virus subtype**
A(H1N1)pdm09
A(H1)
A(H3)
A (Not subtyped)
B
* when total number of samples tested >10
** when influenza positive samples >20
Note: The available country data were joined in larger geographical areas with similar influenza transmission patterns to be able to give an overview (www.who.int/entity/influenza/surveillance_monitoring/updates/TN_GIP_Influenza_transmission_zones.pdf). The displayed data reflect reports of the stated week, or up to two weeks before if no data were available for the current week of that area.
The boundaries and names shown and the designations used on this map do not imply the expression of any opinion whatsoever on the part of the World Health Organization concerning the legal status of any country, territory, city or area or of its authorities, or concerning the delimitation of its frontiers or boundaries. Dotted and dashed lines on maps represent approximate border lines for which there may not yet be full agreement.
Data Source: WHO/HIP, data in HQ as of 07 January 2013.
Data used are from FluNet (www.who.int/flunet), 08:50 UTC snapshot, from WHO regional offices and/or ministry of health websites.
World Health Organization
© WHO 2012. All rights reserved.

Bullet

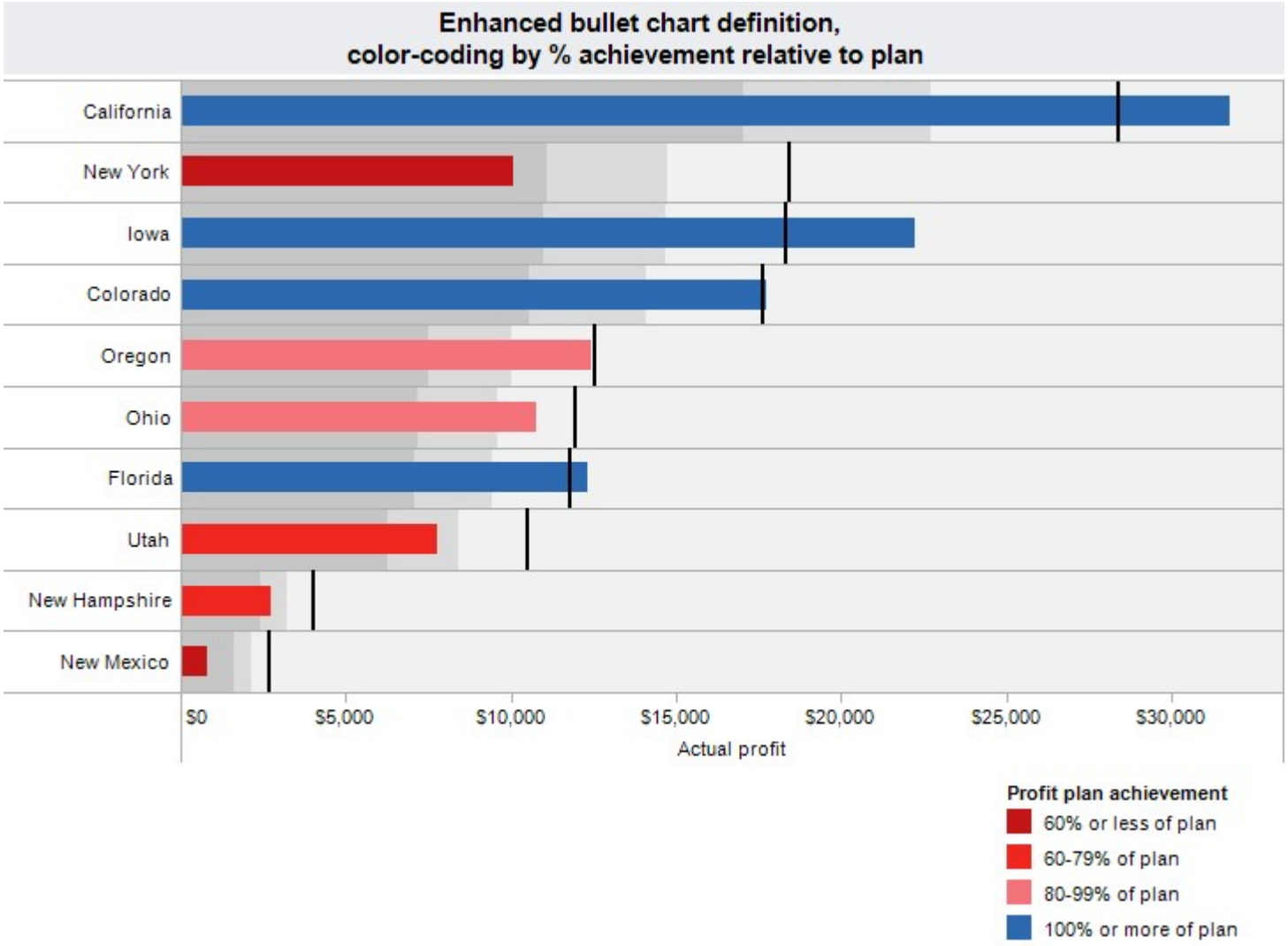
BULLET Bullet graphs are often used to show progress toward a goal. A bullet graph is essentially a variation of a bar chart. Often they are used in place of gauges, meters, and thermometers in a dashboard to more intuitively convey the meaning within a much smaller space. Bullet graphs compare a primary measure (e.g., year-to-date revenue) to one or more other measures (e.g., annual revenue target) and present this in the context of defined performance metrics (e.g., sales quotas). A bullet graph can intuitively illustrate how the primary measure is performing against overall goals (e.g., how close a sales representative is to achieving his/her annual quota).



Bullet

1	bullet measure 0.9 target measure 1 1 start: numberic value: 0.0 end: numberic value: 0.9 2 start: numberic value: 0.9 end: numberic value: 0.95 3 start: numberic value: 0.95end: numberic value: 1.50	
2	bullet measure 0.9 target measure 1 1 start: numberic value: 0.0end: numberic value: 0.5 2 start: numberic value: 0.5end: numberic value: 0.7 3 start: numberic value: 0.7end: numberic value: 0.8	
3	bullet measure 0.9 target measure 1 1 start: numberic value: 0.0 end: numberic value: 0.9 2 start: numberic value: 0.9 end: numberic value: 0.95 3 start: numberic value: 0.95end: Percentage of X %: 100%	
4	bullet measure 0.9 target measure 1 1 start: numberic value: 0.0 end: numberic value: 0.9 2 start: numberic value: 0.9 end: numberic value: 0.95 3 start: numberic value: 0.95 end: Percentage of X %: 100% start: catgory position: 30%End catgory position: 70%	

Bullet


Enhanced bullet chart definition,
color-coding by % achievement relative to plan
California
New York
Iowa
Colorado
Oregon
Ohio
Florida
Utah
New Hampshire
New Mexico
$0
$5,000
$10,000
$15,000
$20,000
$25,000
$30,000
Actual profit
Profit plan achievement
60% or less of plan
60-79% of plan
80-99% of plan
100% or more of plan

Heat map

HEAT MAP Heat maps are great visuals to illustrate the comparison of continuous values across two categories using color. The goal is to help the user quickly see where the intersection of the categories is strongest and weakest in terms of numerical values of the measure being analyzed. For instance, one can use heat maps to show segmentation analysis of target markets where the measure (color gradient would be the purchase amount) and the dimensions would be age and income distribution.

Heat map

Flows into all fund classes (all, including ETFs) – a time series

Figure 15: Heat map* showing the flows as % of total assets into various fund classes (all including ETFs)

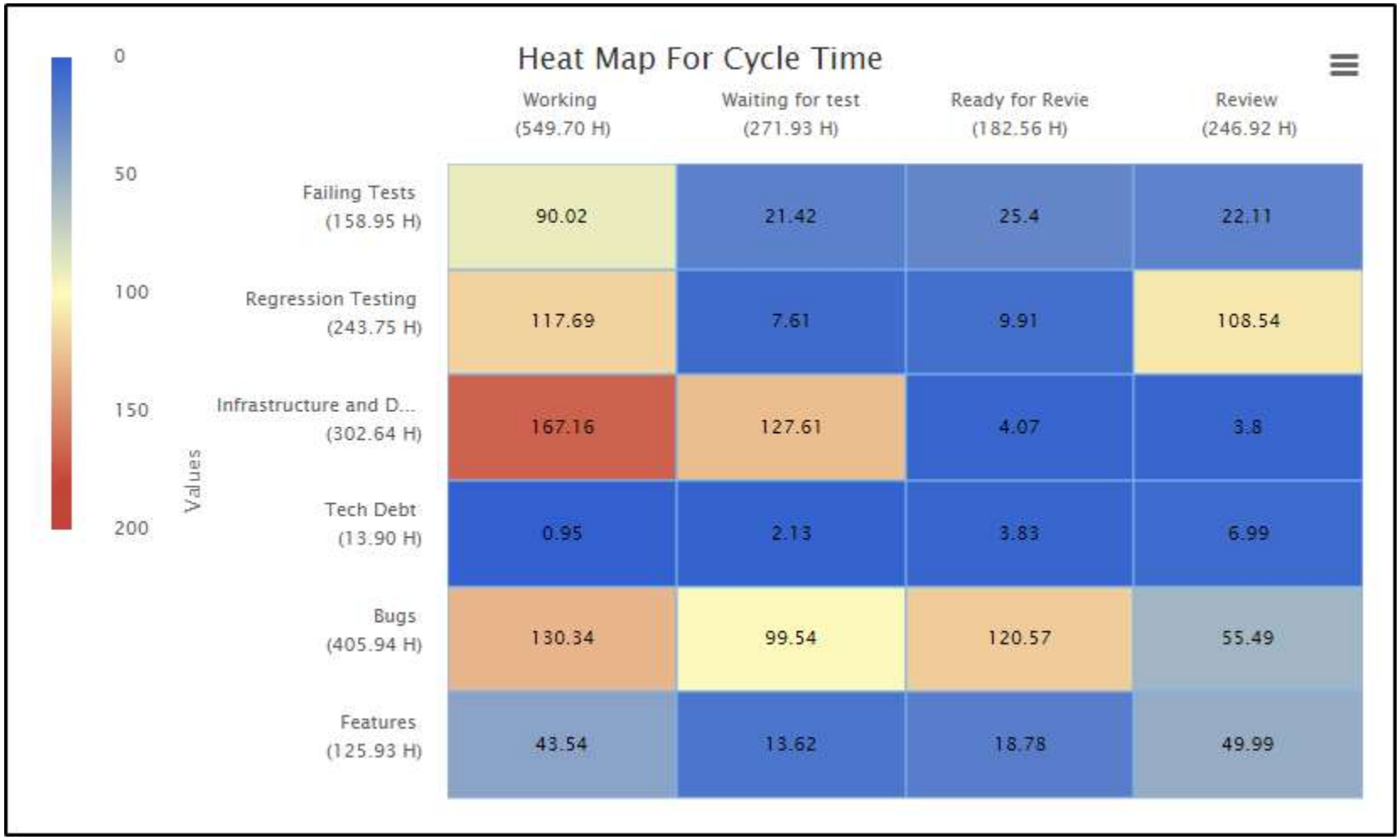
Fund Category	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013 YTD
Total Equity Funds	4%	3%	3%	1%	-3%	2%	2%	-1%	1%	3.4%
Total Developed Market Equity Funds	4%	2%	2%	-1%	-3%	-1%	0%	0%	0%	3.8%
International Equity Funds	8%	6%	7%	6%	-4%	4%	1%	1%	1%	3.8%
US Equity Funds	1%	-1%	-1%	0%	0%	-4%	0%	0%	-1%	3.5%
Western Europe Equity Funds	1%	-1%	7%	-13%	-12%	1%	-3%	-2%	-2%	0.4%
Japan Equity Funds	52%	44%	0%	-27%	-18%	-19%	-3%	5%	10%	24.7%
Pacific Equity Funds	7%	-3%	12%	-1%	-16%	17%	8%	-8%	1%	7.9%
Total Emerging Market Equity Funds	3%	16%	11%	12%	-7%	27%	16%	-5%	7%	0.4%
Global Emerging Market Equity Funds	-10%	3%	4%	10%	-4%	32%	23%	-1%	12%	2.5%
EMEA Equity funds	27%	40%	-6%	-2%	-8%	11%	20%	-11%	-4%	-7.4%
Latin America Equity Funds	10%	81%	27%	46%	-12%	48%	4%	-12%	-1%	-8.5%
Asia Pacific Ex-Japan Funds	21%	22%	27%	14%	-9%	21%	10%	-7%	3%	0.2%
Total Bond Funds	14%	4%	8%	-2%	-10%	24%	16%	4%	11%	1.5%
International Bond Funds	12%	12%	10%	-2%	-24%	25%	23%	3%	6%	1.1%
Corporate High Yield Bond Funds	NA	-18%	-2%	-4%	-5%	40%	15%	4%	18%	1.4%
US Bond Funds	NA	-17%	-9%	4%	-2%	23%	10%	6%	12%	2.2%
Western Europe Bond funds	NA	1%	58%	-8%	-46%	29%	-7%	-28%	2%	-3.4%
Germany Bond funds	NA	NA	NA	NA	NA	NA	29%	25%	-13%	-5.7%
Switzerland Bond funds	NA	NA	NA	NA	NA	NA	-65%	-19%	-2%	-2.0%
United Kingdom Bond funds	NA	22%	-17%	-141%	-26%	64%	8%	-3%	0%	-4.1%
Emerging Markets Debt Funds	12%	24%	18%	9%	-21%	19%	54%	7%	25%	2.4%
Asia ex-Japan Bond funds	NA	4%	3%	16%	-10%	2%	71%	25%	12%	2.2%
Emerging Europe Bond funds	NA	40%	-12%	-18%	-37%	-19%	-8%	-39%	-9%	0.1%
Lat-Am Bond funds	NA	-28%	-22%	-33%	-30%	19%	46%	38%	68%	2.8%
Money Market Funds	NA	NA	NA	NA	31%	-17%	-15%	-4%	-1%	-2.7%

Colour Legend:

13%	8%	3%	-3%	-8%	-13%
>10%	5% to 10%	0% to 5%	-5% to 0%	-10% to -5%	<-10%

Source: EPFR, Deutsche Bank calculations

Heat map


Heat Map For Cycle Time
Working
(549.70 H)
Waiting for test
(271.93 H)
Ready for Revie
(182.56 H)
Review
(246.92 H)
Failing Tests
(158.95 H)
90.02
21.42
25.4
22.11
Regression Testing
(243.75 H)
117.69
7.61
9.91
108.54
Infrastructure and D...
(302.64 H)
167.16
127.61
4.07
3.8
Tech Debt
(13.90 H)
0.95
2.13
3.83
6.99
Bugs
(405.94 H)
130.34
99.54
120.57
55.49
Features
(125.93 H)
43.54
13.62
18.78
49.99
Values
0
50
100
150
200

Heat map

SOMEKA EXCEL TEMPLATES

GERMANY STATES GEOGRAPHIC HEAT MAP

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someka Excel Solutions

POPULATION LEGEND		
Color	Min.	Max.
	0	1.000.000
	1.000.000	2.000.000
	2.000.000	5.000.000
	5.000.000	10.000.000
	10.000.000	300.000.000



DATA		
State	Abbr.	YOUR INFO
Schleswig-Holstein	SH	2.833.000
Hamburg	HH	1.788.000
Mecklenburg-Vorpommern	MV	1.639.000
Niedersachsen	NI	7.914.000
Bremen	HB	661.000
Saxony-Anhalt	ST	2.331.000
Berlin	BE	3.469.000
Brandenburg	BB	2.500.000
Sachsen	SN	4.143.000
Thuringia	TH	2.231.000
Hessen	HE	6.066.000
Nordrhein-Westfalen	NW	17.837.000
Rheinland-Pfalz	RP	4.052.803
Saarland	SL	1.018.000
Baden-Württemberg	BW	10.755.000
Bayern	BY	12.542.000

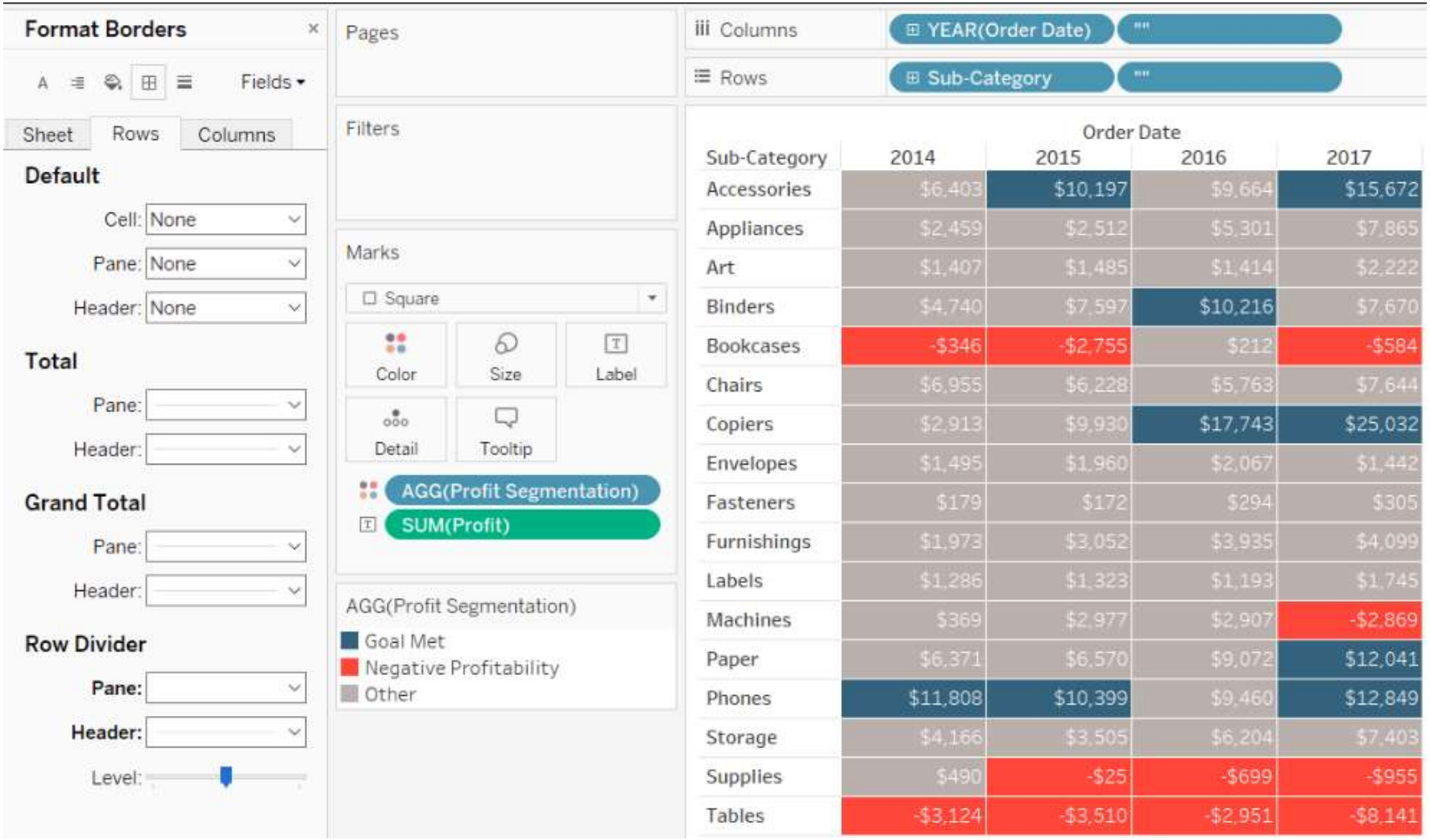
Highlight table

HIGHLIGHT TABLE Highlight tables are intended to take heat maps one step further. In addition to showing how data intersects by using color, highlight tables add a number on top to provide additional detail. That is, they are two-dimensional tables with cells populated with numerical values and gradients of colors. For instance, one can show sales representatives' performance by product type and by sales volume.

SUM(Sales) ($194,268) $1,031,341

		Consumer		Corporate		Home Office		Small Business	
Region	Country / Re..	Furniture	Office Supplies	Furniture	Office Supplies	Furniture	Office Supplies	Furniture	Office Supplies
AsiaPac	Australia		$2,677	$316	$5,753	$43,007	$25,163	$1,976	$52,048
	China	$327,265	$153,923	$520,690	$284,654	$292,090	$198,831	$207,592	$177,013
	India	$169,824	$119,982	$254,423	$336,395	$177,964	$136,734	$141,475	$71,999
	Japan	$79,342	$89,044	$104,936	$122,644	$132,382	$85,140	$42,223	$28,308
EMEA	Germany	$1,350	$1,332	$878	$1,098	$38,522	$27,084	$9,873	$232
	South Africa			$18,122	$8,175	$20,453	$10,940	$24	$4,229
	United Kingdom	$768	$11,250	$2,836	$6,798	$11,143	$32,098	$551	$2,188
Latam	Argentina	$73,676	$20,235	$84,516	$116,670	$148,742	$65,528	$40,353	$93,933
	Brazil	$146,501	$115,550	$271,880	$181,832	$144,187	$115,890	$97,058	$196,796
	Mexico	$19,058	$14,374	$229,770	$140,469	$56,381	$43,140	$29,137	$19,366
North America	Canada	$2,100	$2,734	$6,538	$10,231			$24,142	$10,287
	United States ..	$616,038	$407,799	$1,031,341	$725,000	$663,557	$570,220	$499,786	$509,639

Highlight table

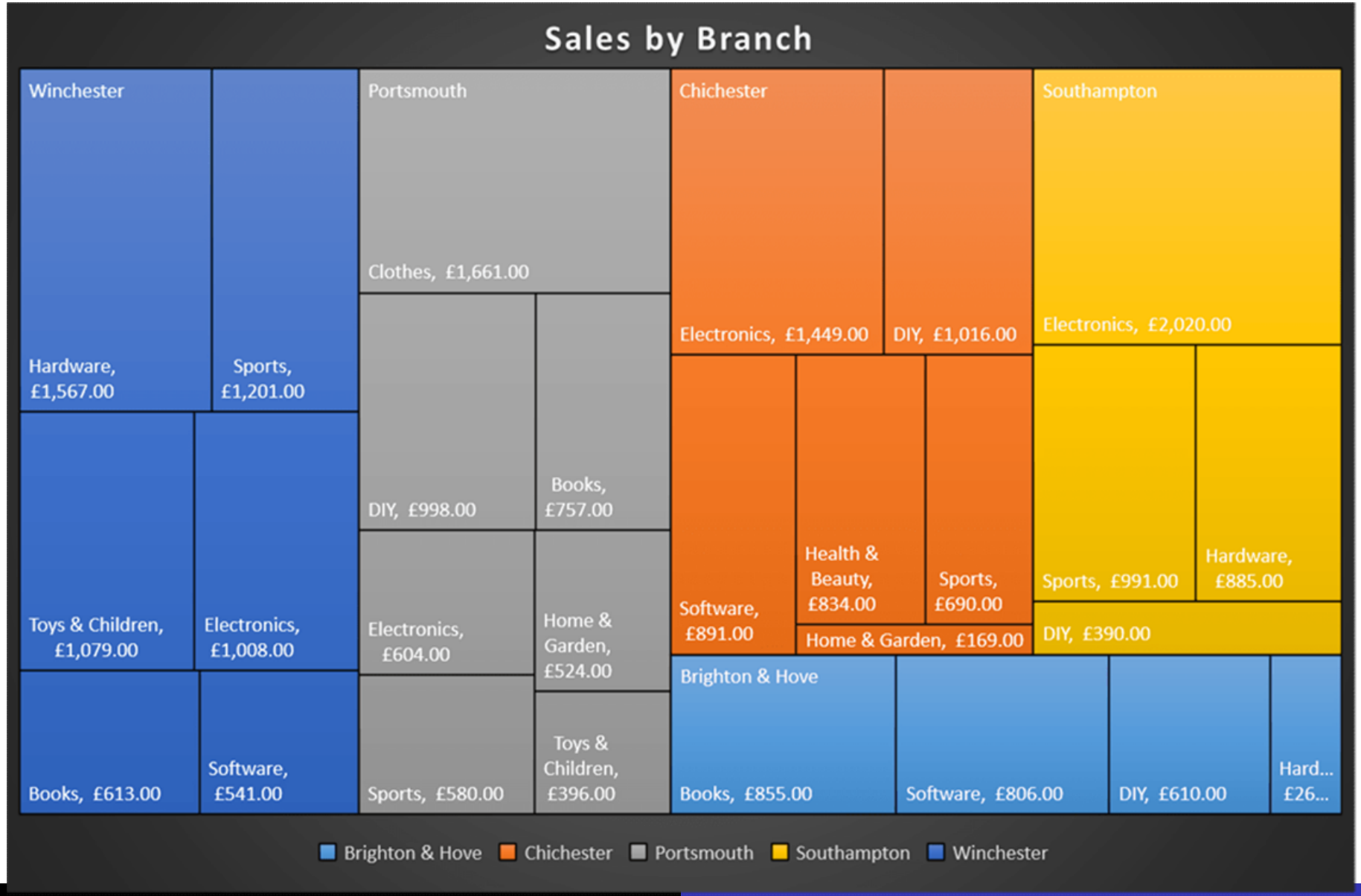


	Order Date			
Sub-Category	2014	2015	2016	2017
Accessories	$6,403	$10,197	$9,664	$15,672
Appliances	$2,459	$2,512	$5,301	$7,865
Art	$1,407	$1,485	$1,414	$2,222
Binders	$4,740	$7,597	$10,216	$7,670
Bookcases	-$346	-$2,755	$212	-$584
Chairs	$6,955	$6,228	$5,763	$7,644
Copiers	$2,913	$9,930	$17,743	$25,032
Envelopes	$1,495	$1,960	$2,067	$1,442
Fasteners	$179	$172	$294	$305
Furnishings	$1,973	$3,052	$3,935	$4,099
Labels	$1,286	$1,323	$1,193	$1,745
Machines	$369	$2,977	$2,907	-$2,869
Paper	$6,371	$6,570	$9,072	$12,041
Phones	$11,808	$10,399	$9,460	$12,849
Storage	$4,166	$3,505	$6,204	$7,403
Supplies	$490	-$25	-$699	-$955
Tables	-$3,124	-$3,510	-$2,951	-$8,141

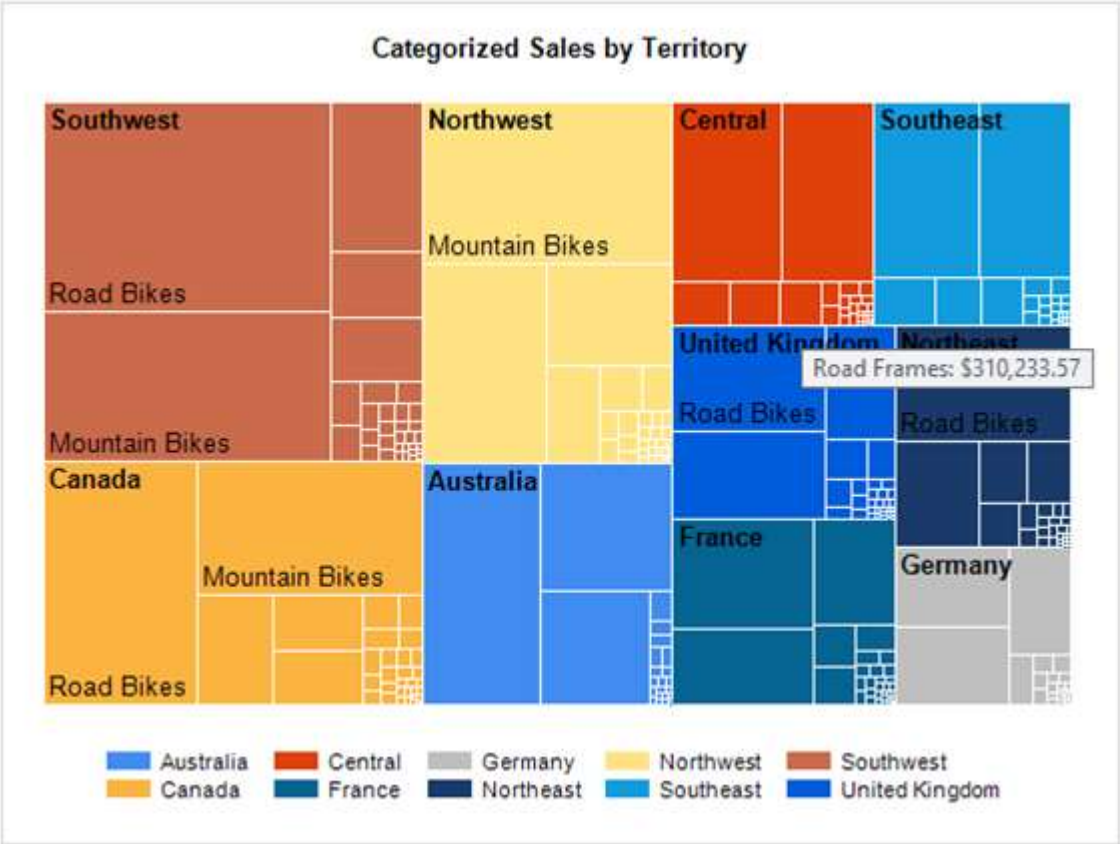
Tree map

TREE MAP Tree maps display hierarchical (tree-structured) data as a set of nested rectangles. Each branch of the tree is given a rectangle, which is then tiled with smaller rectangles representing subbranches. A leaf node's rectangle has an area proportional to a specified dimension on the data. Often the leaf nodes are colored to show a separate dimension of the data. When the color and size dimensions are correlated in some way with the tree structure, one can often easily see patterns that would be difficult to spot in other ways, such as if a certain color is particularly relevant. A second advantage of tree maps is that, by construction, they make efficient use of space. As a result, they can legibly display thousands of items on the screen simultaneously.

Tree map


Sales by Branch
Winchester
Hardware, £1,567.00
Sports, £1,201.00
Toys & Children, £1,079.00
Electronics, £1,008.00
Books, £613.00
Software, £541.00
Portsmouth
Clothes, £1,661.00
DIY, £998.00
Books, £757.00
Electronics, £604.00
Home & Garden, £524.00
Sports, £580.00
Toys & Children, £396.00
Chichester
Electronics, £1,449.00
DIY, £1,016.00
Software, £891.00
Health & Beauty, £834.00
Sports, £690.00
Home & Garden, £169.00
Southampton
Electronics, £2,020.00
Sports, £991.00
Hardware, £885.00
DIY, £390.00
Brighton & Hove
Books, £855.00
Software, £806.00
DIY, £610.00
Hard... £26...
Brighton & Hove
Chichester
Portsmouth
Southampton
Winchester

Tree map


Categorized Sales by Territory
Southwest
Road Bikes
Mountain Bikes
Northwest
Mountain Bikes
Central
Southeast
United Kingdom
Northeast
Road Frames: $310,233.57
Road Bikes
Road Bikes
Canada
Mountain Bikes
Road Bikes
Australia
France
Germany
Australia
Canada
Central
France
Germany
Northeast
Northwest
Southeast
Southwest
United Kingdom

Guideline

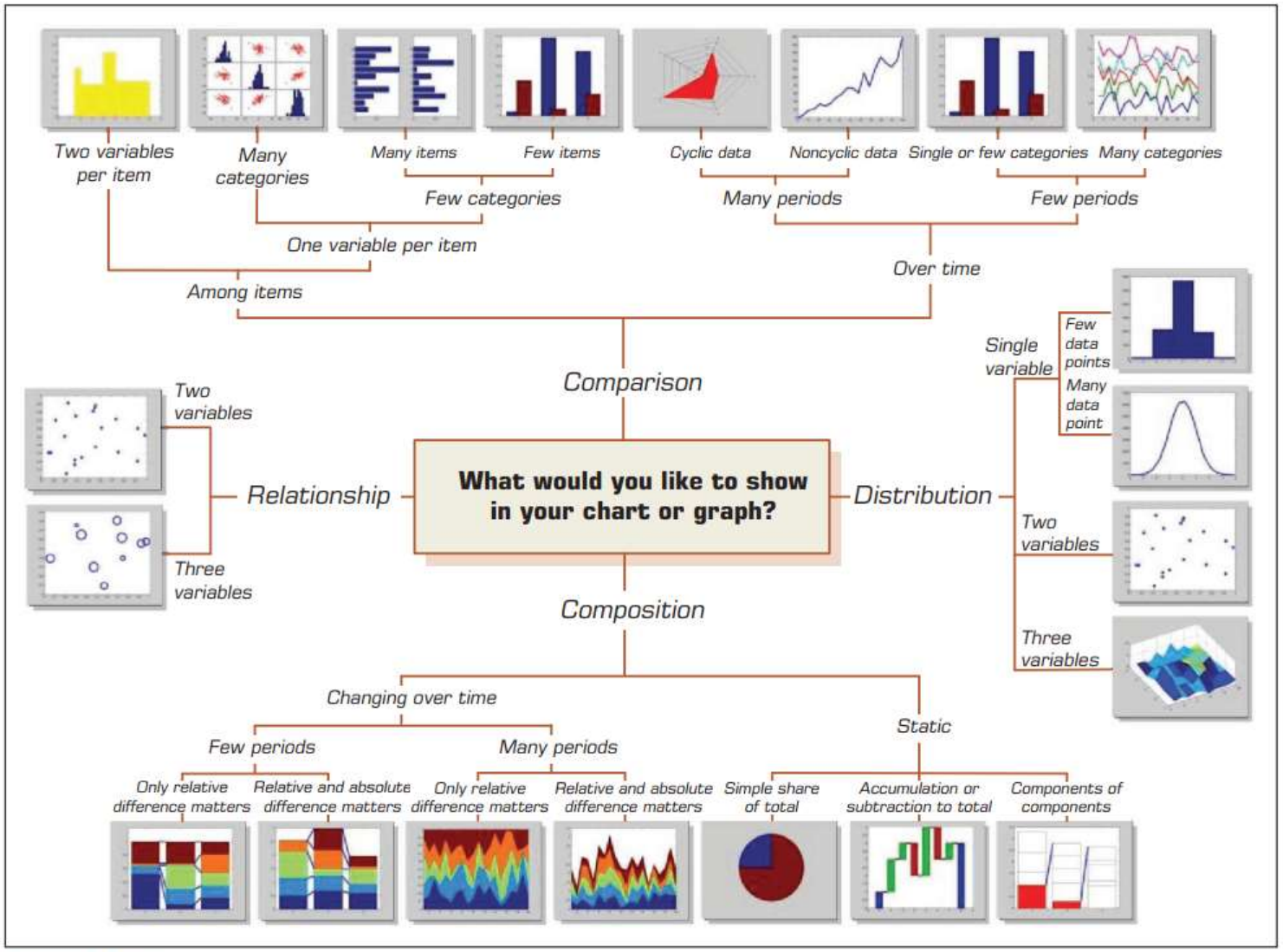
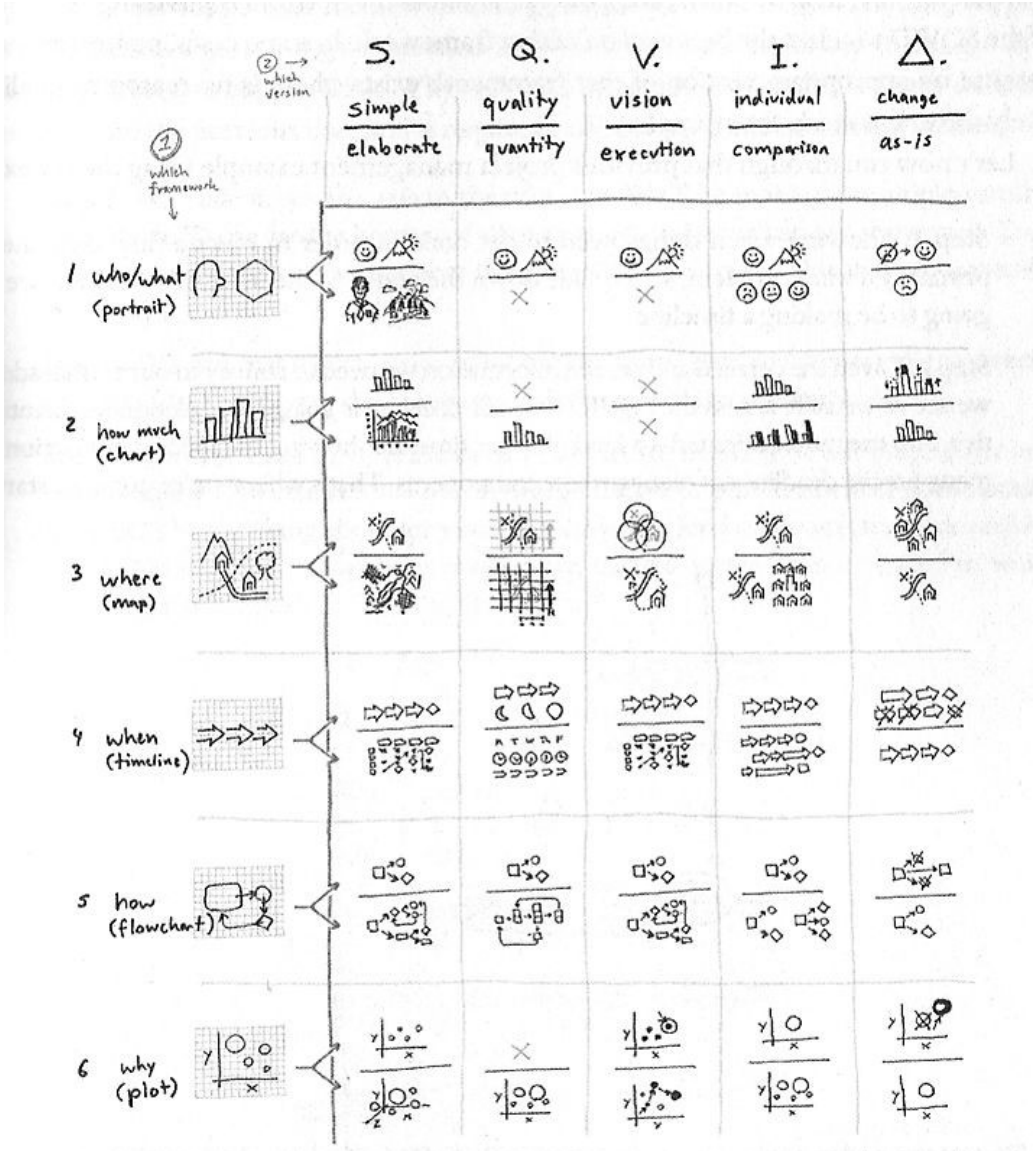
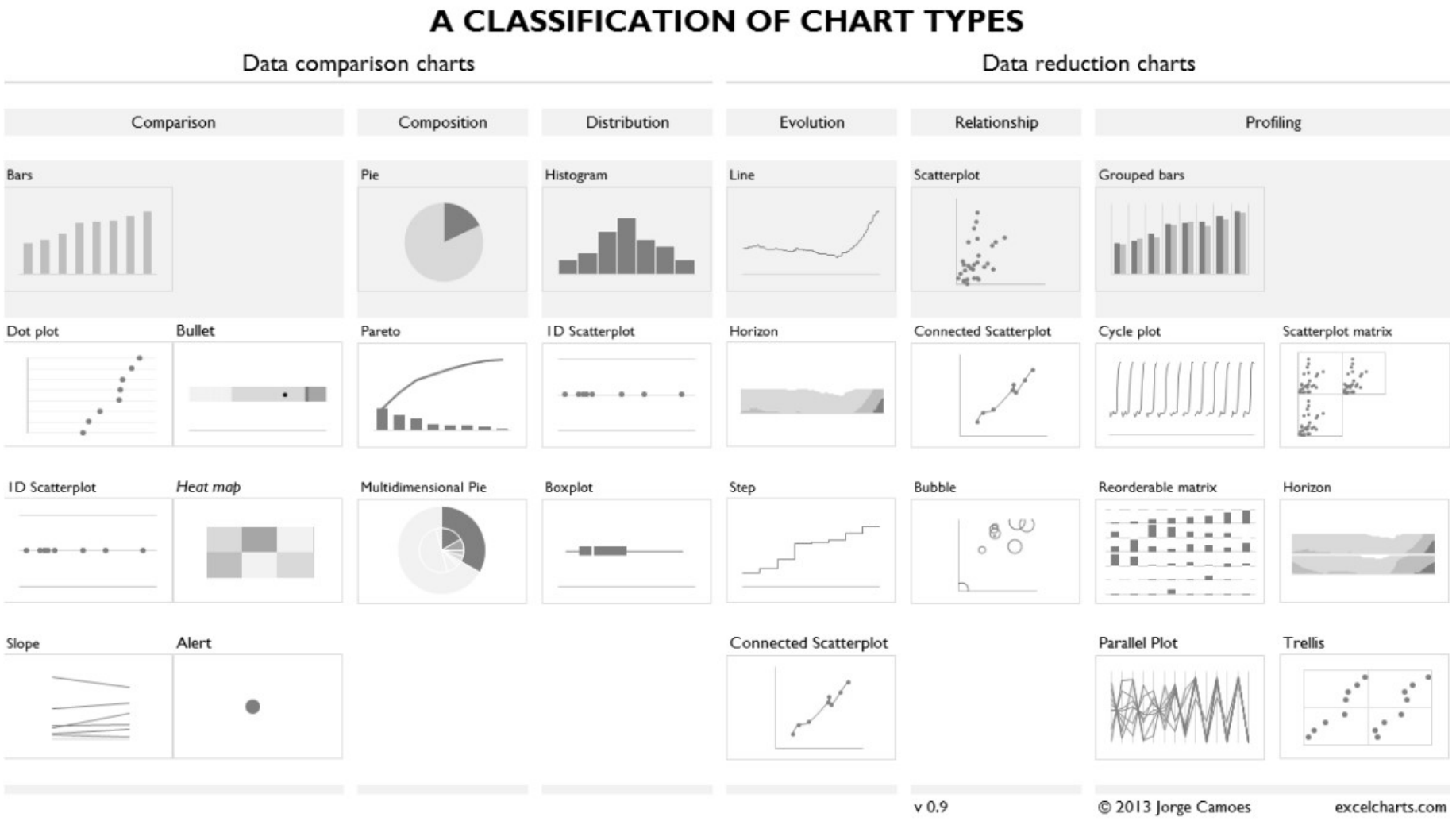


FIGURE 2.21 A Taxonomy of Charts and Graphs. *Source:* Adapted from Abela, A. (2008). Advanced presentations by design: Creating communication that drives action. New York: Wiley.

Guideline


S.
simple
elaborate
Q.
quality
quantity
V.
vision
execution
I.
individual
comparison
Δ.
change
as-is
(2) which version
(1) which framework
1 who/what (portrait)
2 how much (chart)
3 where (map)
4 when (timeline)
5 how (flowchart)
6 why (plot)

Guideline


A CLASSIFICATION OF CHART TYPES
Data comparison charts
Data reduction charts
Comparison
Composition
Distribution
Evolution
Relationship
Profiling
Bars
Pie
Histogram
Line
Scatterplot
Grouped bars
Dot plot
Bullet
Pareto
1D Scatterplot
Horizon
Connected Scatterplot
Cycle plot
Scatterplot matrix
1D Scatterplot
Heat map
Multidimensional Pie
Boxplot
Step
Bubble
Reorderable matrix
Horizon
Slope
Alert
Connected Scatterplot
Parallel Plot
Trellis
v 0.9
© 2013 Jorge Camoes
excelcharts.com

KPI guideline

3.10 Performance Measurement

Underlying BPM is a performance measurement system. According to Simons (2002), **performance measurement systems**:

> Assist managers in tracking the implementations of business strategy by comparing actual results against strategic goals and objectives. A performance measurement system typically comprises systematic methods of setting business goals together with periodic feedback reports that indicate progress against goals. (p. 108)

All measurement is about comparisons. Raw numbers are rarely of value. If you were told that a salesperson completed 50% of the deals he or she was working on within a month, that would have little meaning. Now, suppose you were told that the same salesperson had a monthly close rate of 30% last year. Obviously, the trend is good. What if you were also told that the average close rate for all salespeople at the company was 80%? Obviously, that particular salesperson needs to pick up the pace. As Simons's definition suggests, in performance measurement, the key comparisons revolve around strategies, goals, and objectives. Operational metrics that are used to measure performance are usually called key performance indicators (KPIs).

KPI guideline

Key Performance Indicator (KPI)

There is a difference between a "run of the mill" metric and a "strategically aligned" metric. The term **key performance indicator** (**KPI**) is often used to denote the latter. A KPI represents a strategic objective and measures performance against a goal. According to Eckerson (2009), KPIs are multidimensional. Loosely translated, this means that KPIs have a variety of distinguishing features, including

- ***Strategy.*** KPIs embody a strategic objective.
- ***Targets.*** KPIs measure performance against specific targets. Targets are defined in strategy, planning, or budget sessions and can take different forms (e.g., achievement targets, reduction targets, absolute targets).
- ***Ranges.*** Targets have performance ranges (e.g., above, on, or below target).
- ***Encodings.*** Ranges are encoded in software, enabling the visual display of performance (e.g., green, yellow, red). Encodings can be based on percentages or more complex rules.
- ***Time frames.*** Targets are assigned time frames by which they must be accomplished. A time frame is often divided into smaller intervals to provide performance mileposts.
- ***Benchmarks.*** Targets are measured against a baseline or benchmark. The previous year's results often serve as a benchmark, but arbitrary numbers or external benchmarks may also be used.

Dashboard guideline

What to Look for in a Dashboard

Although performance dashboards and other information visualization frameworks differ, they all do share some common design characteristics. First, they all fit within the larger BI and/or performance measurement system. This means that their underlying architecture is the BI or performance management architecture of the larger system. Second, all well-designed dashboard and other information visualizations possess the following characteristics (Novell, 2009):

- They use visual components (e.g., charts, performance bars, sparklines, gauges, meters, stoplights) to highlight, at a glance, the data and exceptions that require action.
- They are transparent to the user, meaning that they require minimal training and are extremely easy to use.
- They combine data from a variety of systems into a single, summarized, unified view of the business.
- They enable drill-down or drill-through to underlying data sources or reports, providing more detail about the underlying comparative and evaluative context.
- They present a dynamic, real-world view with timely data refreshes, enabling the end u up to date with any recent changes in the business.
- They require little, if any, customized coding to implement, deploy, and maintain.

Visual analytics

Visual Analytics

Visual analytics is a recently coined term that is often used loosely to mean nothing more than information visualization. What is meant by **visual analytics** is the combination of visualization and predictive analytics. Whereas information visualization is aimed at answering, "What happened?" and "What is happening?" and is closely associated with BI (routine reports, scorecards, and dashboards), visual analytics is aimed at answering, "Why is it happening?" "What is more likely to happen?" and is usually associated with business analytics (forecasting, segmentation, correlation analysis). Many of the information visualization vendors are adding the capabilities to call themselves visual analytics solution providers. One of the top, long-time analytics solution providers, SAS Institute, is approaching it from another direction. They are embedding their analytics capabilities into a high-performance data visualization environment that they call visual analytics.

Visual or not visual, automated or manual, online or paper based, business reporting is not much different than telling a story. Technology Insights 2.3 provides a different, unorthodox viewpoint to better business reporting.

Visual analytics

TECHNOLOGY INSIGHTS 2.3
Telling Great Stories with Data and Visualization

Everyone who has data to analyze has stories to tell, whether it's diagnosing the reasons for manufacturing defects, selling a new idea in a way that captures the imagination of your target audience, or informing colleagues about a particular customer service improvement program. And when it's telling the story behind a big strategic choice so that you and your senior management team can make a solid decision, providing a fact-based story can be especially challenging. In all cases, it's a big job. You want to be interesting and memorable; you know you need to keep it simple for your busy executives and colleagues. Yet you also know you have to be factual, detail oriented, and data driven, especially in today's metric-centric world.

It's tempting to present just the data and facts, but when colleagues and senior management are overwhelmed by data and facts without context, you lose. We have all experienced presentations with large slide decks, only to find that the audience is so overwhelmed with data that they don't know what to think, or they are so completely tuned out that they take away only a fraction of the key points.

Start engaging your executive team and explaining your strategies and results more powerfully by approaching your assignment as a story. You will need the "what" of your story (the facts and data) but you also need the "Who?" "How?" "Why?" and the often-missed "So what?" It's these story elements that will make your data relevant and tangible for your audience. Creating a good story can aid you and senior management in focusing on what is important.

Visual analytics

Why Story?

Stories bring life to data and facts. They can help you make sense and order out of a disparate collection of facts. They make it easier to remember key points and can paint a vivid picture of what the future can look like. Stories also create interactivity—people put themselves into stories and can relate to the situation.

Cultures have long used **storytelling** to pass on knowledge and content. In some cultures, storytelling is critical to their identity. For example, in New Zealand, some of the Maori people tattoo their faces with *mokus*. A *moku* is a facial tattoo containing a story about ancestors—the family tribe. A man may have a tattoo design on his face that shows features of a hammerhead to highlight unique qualities about his lineage. The design he chooses signifies what is part of his "true self" and his ancestral home.

Likewise, when we are trying to understand a story, the storyteller navigates to finding the "true north." If senior management is looking to discuss how they will respond to a competitive change, a good story can make sense and order out of a lot of noise. For example, you may have facts and data from two studies, one including results from an advertising study and one from a product satisfaction study. Developing a story for what you measured across both studies can help people see the whole where there were disparate parts. For rallying your distributors around a new product, you can employ a story to give vision to what the future can look like. Most important, storytelling is interactive—typically the presenter uses words and pictures that audience members can put themselves into. As a result, they become more engaged and better understand the information.

Visual analytics

Think of Your Analysis as a Story—Use a Story Structure

When crafting a data-rich story, the first objective is to find the story. Who are the characters? What is the drama or challenge? What hurdles have to be overcome? And at the end of your story, what do you want your audience to do as a result?

Once you know the core story, craft your other story elements: define your characters, understand the challenge, identify the hurdles, and crystallize the outcome or decision question. Make sure you are clear with what you want people to do as a result. This will shape how your audience will recall your story. With the story elements in place, write out the storyboard, which represents the structure and form of your story. Although it's tempting to skip this step, it is better first to understand the story you are telling and then to focus on the presentation structure and form. Once the storyboard is in place, the other elements will fall into place. The storyboard will help you to think about the best analogies or metaphors, to clearly set up challenge or opportunity, and to finally see the flow and transitions needed. The storyboard also helps you focus on key visuals (graphs, charts, and graphics) that you need your executives to recall. Figure 2.24 shows a storyline for the impact of small loans in a worldwide view within the Tableau visual analytics environment.

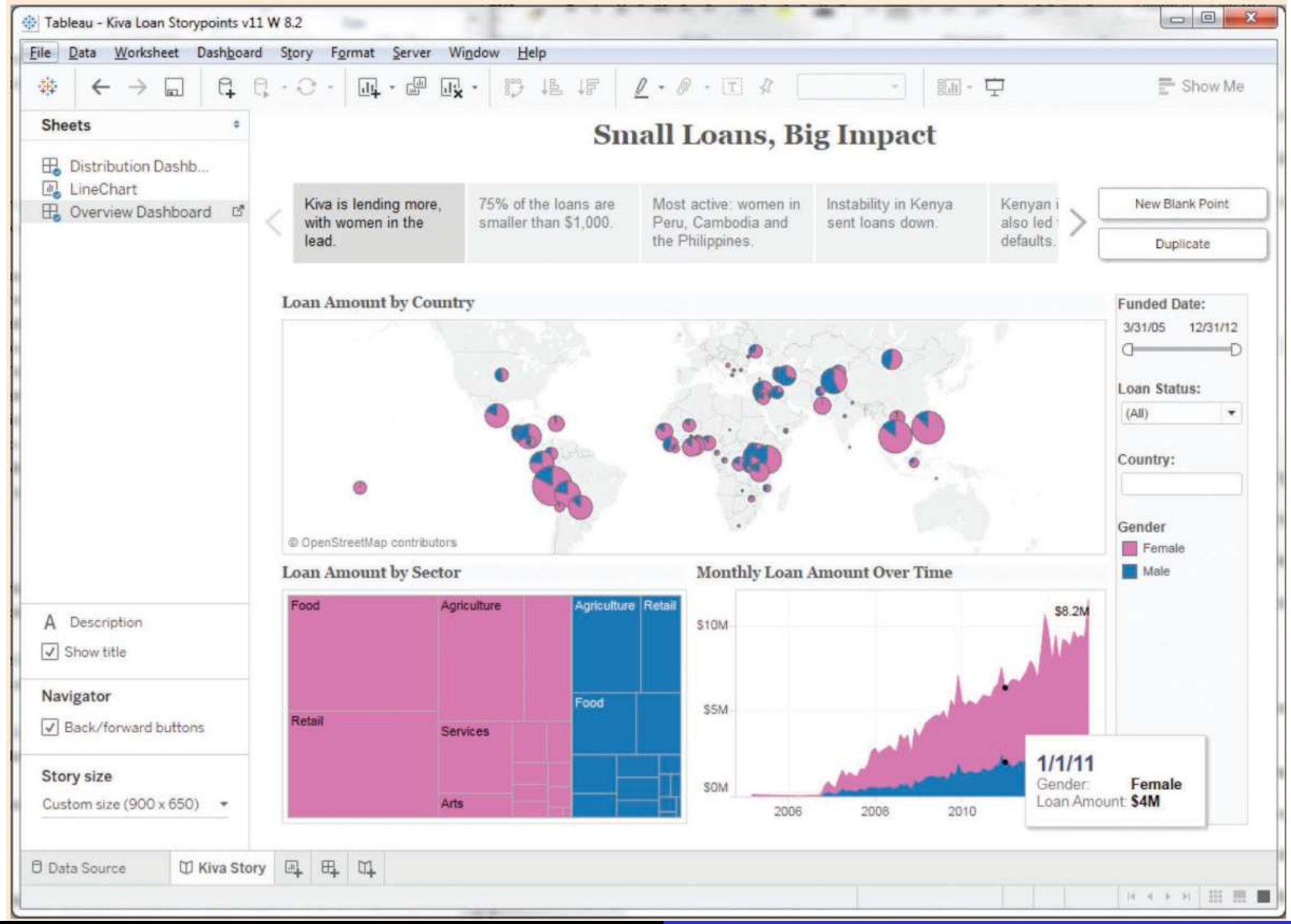
Visual analytics

In summary, don't be afraid to use data to tell great stories. Being factual, detail oriented, and data driven is critical in today's metric-centric world, but it does not have to mean being boring and lengthy. In fact, by finding the real stories in your data and following the best practices, you can get people to focus on your message—and thus on what's important. Here are those best practices:

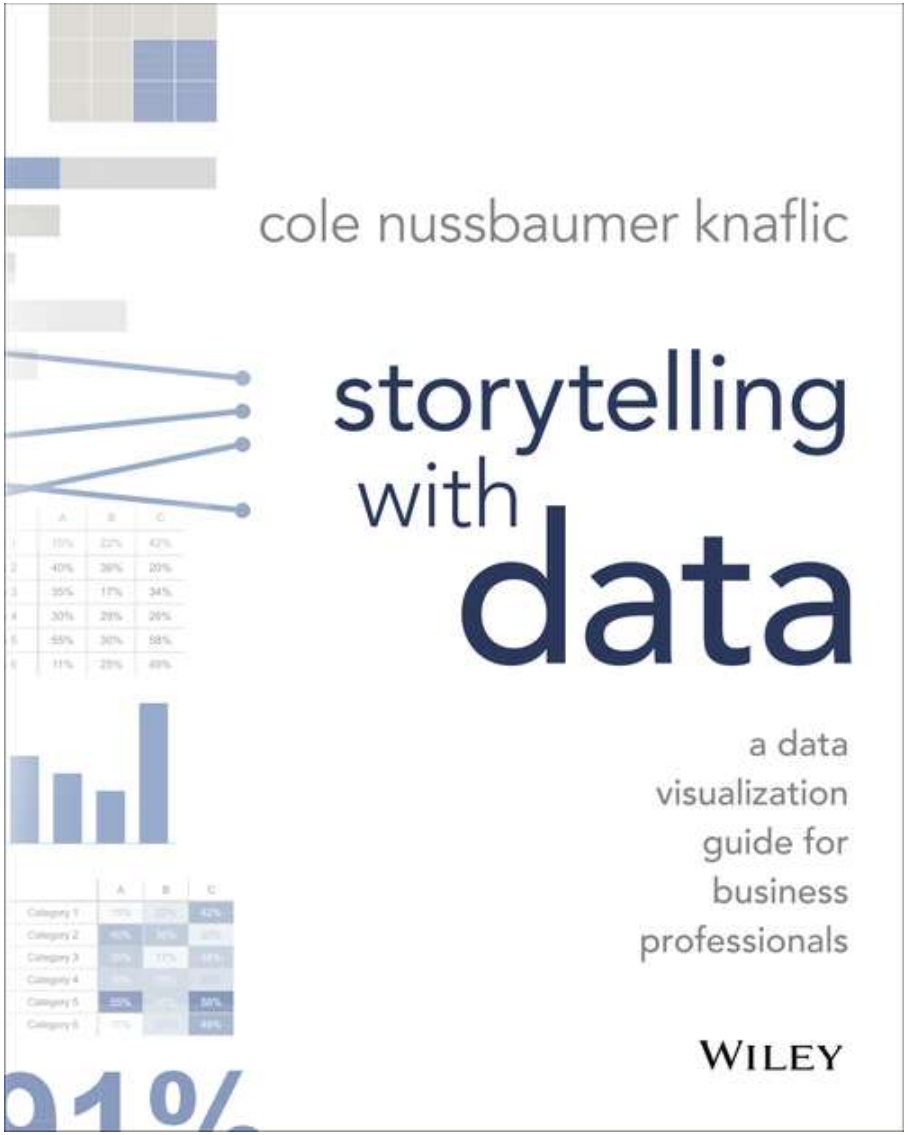
1. Think of your analysis as a story—use a story structure.
2. Be authentic—your story will flow.
3. Be visual—think of yourself as a film editor.
4. Make it easy for your audience and you.
5. Invite and direct discussion.

Source: Fink, E., & Moore, S. J. (2012). Five best practices for telling great stories with data. White paper by Tableau Software, Inc., www.tableau.com/whitepapers/telling-data-stories (accessed May 2016).

Visual analytics



Book


cole nussbaumer knaflic
storytelling with data
a data visualization guide for business professionals
WILEY

Visual analytics

2.11 Information Dashboards

Information dashboards are common components of most, if not all, BI or business analytics platforms, business performance management systems, and performance measurement software suites. **Dashboards** provide visual displays of important information that is consolidated and arranged on a single screen so that information can be digested at a single glance and easily drilled in and further explored. A typical dashboard is shown in Figure 2.27. This particular executive dashboard displays a variety of KPIs for a hypothetical software company called Sonatica (selling audio tools). This executive dashboard shows a high-level view of the different functional groups surrounding the products, starting from a general overview to the marketing efforts, sales, finance, and support departments. All of this is intended to give executive decision makers a quick and accurate idea of what is going on within the organization. On the left side of the dashboard, we can see (in a time series fashion) the quarterly changes in revenues, expenses, and margins, as well as the comparison of those figures to previous years' monthly numbers. On the upper-right side we see two dials with color-coded regions showing the amount of

Visual analytics



FIGURE 2.27 A Sample Executive Dashboard. *Source:* dundas.com.

Visual analytics

Dashboard Design

Dashboards are not a new concept. Their roots can be traced at least to the executive information system of the 1980s. Today, dashboards are ubiquitous. For example, a few years back, Forrester Research estimated that over 40% of the largest 2,000 companies in the world used the technology (Ante & McGregor, 2006). Since then, one can safely assume that this number has gone up quite significantly. In fact, nowadays it would be rather unusual to see a large company using a BI system that does not employ some sort of performance dashboards. The Dashboard Spy Web site (dashboardspy.com/about) provides further evidence of their ubiquity. The site contains descriptions and screenshots of thousands of BI dashboards, scorecards, and BI interfaces used by businesses of all sizes and industries, nonprofits, and government agencies.

According to Eckerson (2006), a well-known expert on BI in general and dashboards in particular, the most distinctive feature of a dashboard is its three layers of information:

1. ***Monitoring:*** Graphical, abstracted data to monitor key performance metrics.
2. ***Analysis:*** Summarized dimensional data to analyze the root cause of problems.
3. ***Management:*** Detailed operational data that identify what actions to take to resolve a problem.

Visual analytics

Because of these layers, dashboards pack a lot of information into a single screen. According to Few (2005), "The fundamental challenge of dashboard design is to display all the required information on a single screen, clearly and without distraction, in a manner that can be assimilated quickly." To speed assimilation of the numbers, the numbers need to be placed in context. This can be done by comparing the numbers of interest to other baseline or target numbers, by indicating whether the numbers are good or bad, by denoting whether a trend is better or worse, and by using specialized display widgets or components to set the comparative and evaluative context. Some of the common comparisons that are typically made in BI systems include comparisons against past values, forecasted values, targeted values, benchmark or average values, multiple instances of the same measure, and the values of other measures (e.g., revenues versus costs).

Even with comparative measures, it is important to specifically point out whether a particular number is good or bad and whether it is trending in the right direction. Without these types of evaluative designations, it can be time consuming to determine the status of a particular number or result. Typically, either specialized visual objects (e.g., traffic lights, dials, and gauges) or visual attributes (e.g., color coding) are used to set the evaluative context. An interactive dashboard-driven reporting data exploration solution built by an energy company is featured in Application Case 2.8.

Best practices in dashboard design

Benchmark Key Performance Indicators with Industry Standards

Many customers, at some point in time, want to know if the metrics they are measuring are the right metrics to monitor. Sometimes customers have found that the metrics they are tracking are not the right ones to track. Doing a gap assessment with industry benchmarks aligns you with industry best practices.

Wrap the Dashboard Metrics with Contextual Metadata

Often when a report or a visual dashboard/scorecard is presented to business users, questions remain unanswered. The following are some examples:

- Where did you source this data from?
- While loading the data warehouse, what percentage of the data got rejected/encountered data quality problems?
- Is the dashboard presenting "fresh" information or "stale" information?
- When was the data warehouse last refreshed?
- When is it going to be refreshed next?
- Were any high-value transactions that would skew the overall trends rejected as a part of the loading process?

Best practices in dashboard design

Validate the Dashboard Design by a Usability Specialist

In most dashboard environments, the dashboard is designed by a tool specialist without giving consideration to usability principles. Even though it's a well-engineered data warehouse that can perform well, many business users do not use the dashboard, as it is perceived as not being user friendly, leading to poor adoption of the infrastructure and change management issues. Up-front validation of the dashboard design by a usability specialist can mitigate this risk.

Prioritize and Rank Alerts/Exceptions Streamed to the Dashboard

Because there are tons of raw data, it is important to have a mechanism by which important exceptions/behaviors are proactively pushed to the information consumers. A business rule can be codified, which detects the alert pattern of interest. It can be coded into a program, using database-stored procedures, which can crawl through the fact tables and detect patterns that need immediate attention. This way, information finds the business user as opposed to the business user polling the fact tables for the occurrence of critical patterns.

Best practices in dashboard design

Enrich the Dashboard with Business-User Comments

When the same dashboard information is presented to multiple business users, a small text box can be provided that can capture the comments from an end-user's perspective. This can often be tagged to the dashboard to put the information in context, adding perspective to the structured KPIs being rendered.

Present Information in Three Different Levels

Information can be presented in three layers depending on the granularity of the information: the visual dashboard level, the static report level, and the self-service cube level. When a user navigates the dashboard, a simple set of 8 to 12 KPIs can be presented, which would give a sense of what is going well and what is not.

Best practices in dashboard design

Pick the Right Visual Construct Using Dashboard Design Principles

In presenting information in a dashboard, some information is presented best with bar charts, some with time series line graphs, and when presenting correlations, a scatter plot is useful. Sometimes merely rendering it as simple tables is effective. Once the dashboard design principles are explicitly documented, all the developers working on the front end can adhere to the same principles while rendering the reports and dashboard.

Provide for Guided Analytics

In a typical organization, business users can be at various levels of analytical maturity. The capability of the dashboard can be used to guide the "average" business user to access the same navigational path as that of an analytically savvy business user.

Best practices in dashboard design

Radha, R. (2008). Eight best practices in dashboard design. *Information Management*. www.information-management.com/news/columns/-10001129-1.html (accessed July 2016).

Thank you for your attention!

QUIZES AND DICUSSIONS!